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MedDiscover: A Domain-Specific Retrieval-Augmented Generation Framework for Evidence-Grounded Knowledge Extraction in Metabolomics

--Manuscript Draft--

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Abstract:	Retrieval-augmented generation (RAG) can improve biomedical question answering, but claims about domain-specific retrievers require transparent, reproducible evaluation. We present MedDiscover, an open-source RAG implementation and benchmark instantiated for metabolomics and metabolic-disorder literature, with a two-tier dataset design: a Gold set of 10 papers and 30 expert-curated question-answer pairs with traceable references, and a Silver set of 100 papers curated using ICD-10 metabolic disorder codes E70-E88 with ~600 synthetic QA pairs and retrievability metadata (Ada n=300, MedCPT n=300). On the Gold set, retrieval augmentation improves grounding compared with a non-RAG baseline. On the Silver benchmark, MedCPT and Ada show comparable retrieval-centric performance (faithfulness p=0.85, context recall p=0.50, context precision p=0.49, answer relevancy p=0.82), while MedCPT yields higher answer correctness (Mann-Whitney U p=5.72 x 10 ⁻¹¹ , Cliff's delta=0.303; common-language effect=0.651). We release code, evaluation scripts, and document lists to support reproducibility.
Opposed Reviewers:	
Response to Reviewers:	Review 2 (22 Feb 2026) Reviewer #3 R3-K1: This revised submission presents a domain specific approach using LLM for information retrieval in metabolomics and the biosciences. This is viewed as a very valuable approach, despite not being the final model version. Any progress towards the goal of better (and faster) answering of biomedical questions is welcome. The authors have done a very good job at responding the queries from the first two reviewers. The most critical one being the size of the training set. In this new, revised version, the training set has been largely expanded, and it is now seen as a more innovative approach. The size of the dataset is still not as large as it should be, with the metabolomics literature having thousands of papers and coverage of many more diseases, so the authors are encouraged to continue pursuing this work and expanding to the whole body of literature. This type of tool, when made accessible to the whole metabolomics community, could save enormous amounts of time, which are currently devoted to digging into the literature to find the biological interpretation of the data generated. Note that the citations seem to be missing, at least from the pdf document I was provided. Response: we acknowledge the comments and have submitted and attached the complimentary reference.bib file along with the latex missed out from our previous submission. Review 1 (18 Jan 2026) Reviewer #1

R1-O1: In this article, the author proposes a RAG-based pipeline named MedDiscover for metabolomics question answering. The author claims that SemiDevKG supports question answering through a GraphRAG pipeline. However, the paper suffers from major issues in methodological novelty and dataset size (only 10 articles are considered, limited to 3 metabolic disorders). Moreover, the proposed RAG application could be applied to any dataset; it does not showcase features specific to metabolomics.

Response: We agree the pipeline is not metabolomics-unique as an architecture. We therefore reposition MedDiscover as a metabolomics-instantiated benchmark and evaluation pipeline, and we define metabolomics-specificity in terms of the corpus, question construction, and evaluation constraints rather than algorithmic novelty.

Manuscript: Retrieval-augmented generation (RAG) can improve biomedical question answering, but claims about domain-specific retrievers require transparent, reproducible evaluation. We present MedDiscover, an open-source RAG implementation and benchmark instantiated for metabolomics and metabolic-disorder literature, with a two-tier dataset design: a Gold set of 10 papers and 30 expert-curated question-answer pairs with traceable references, and a Silver set of 100 papers curated using ICD-10 metabolic disorder codes E70-E88 with approx. 600 synthetic QA pairs and retrievability metadata (Ada n=300, MedCPT n=300). On the Gold set, retrieval augmentation improves grounding compared with a non-RAG baseline. On the Silver benchmark, MedCPT and Ada show comparable retrieval-centric performance (faithfulness p=0.85, context recall p=0.50, context precision p=0.49, answer relevancy p=0.82), while MedCPT yields higher answer correctness (Mann-Whitney U p=5.72x10⁻¹¹, Cliff's delta=0.303; common-language effect =0.651). We release code, evaluation scripts, and document lists to support reproducibility.

R1-O2: Overall, the contribution appears weak and not suitable for publication in a scientific journal. Therefore, I recommend rejection of the paper.

Response: We respectfully note that the revision substantially changes the study scope and evaluation rigor. The updated manuscript addresses novelty, scale, and fairness concerns and provides a clearer, evidence-grounded contribution.

Manuscript: We implement MedDiscover as an open-source RAG evaluation pipeline and apply it to metabolomics/metabolic-disorder literature using a two-tier dataset (Gold and Silver). The primary comparisons fix the decoder (GPT-4o) to isolate retrieval effects, while decoder upgrades (GPT-4.1-nano/mini) are reported as sensitivity analyses. NotebookLM is included as a black-box comparator for answer style, with grounding metrics reported only where retrieved contexts are available.

Major Comments

R1-M1: Graphical abstract: Too simplistic. It is unclear how the retrieval and evidence-based answering steps are implemented (e.g., prompts are missing).

Response: We agree and have revised the pipeline figure to show retrieval, evidence grounding, and the Gold/Silver split. We now also provide the exact prompt templates in a dedicated appendix (Prompt Templates) to make the retrieval and evidence-based answering steps fully transparent.

Manuscript: MedDiscover pipeline and benchmark design. A metabolomics/metabolic-disorder corpus (ICD-10 E70-E88) is ingested, cleaned, chunked (500 tokens with 50-token overlap), and indexed with two embedding configurations (MedCPT and Ada-002) using FAISS. Evaluation is organized into a Gold set (expert-curated QA with traceable references) and a Silver set (synthetic QA with retrievability metadata). RAG inference retrieves top-k contexts and uses an evidence-grounded prompt to produce answers with explicit supporting context; evaluation reports reference-based metrics on Gold and retrieval-grounding metrics plus retrievability statistics on Silver.

Prompt templates

Evidence-grounded answering prompt (MedDiscover RAG; GPT-4o)

[SYSTEM]

You are a biomedical assistant. Answer the question using ONLY the provided context. If the context does not contain the answer, reply: "Insufficient evidence in the provided context."

[USER]

Question: {question}

Context:

{top_k_context_chunks}

Instructions:

- Provide a concise answer (max 25 tokens).
- Prefer specific entities (metabolites, diseases, study descriptors).
- Do not add unsupported claims.
- If multiple findings exist, report the one directly supported by the context.

Synthetic QA generation prompt (Silver set; gpt-4.1-mini)

[SYSTEM]

You generate biomedical questions and short answers from scientific text excerpts.

[USER]

Given the excerpt below, generate exactly 3 diverse questions that are answerable from the excerpt. Prefer non-trivial questions when possible.

Excerpt:

{chunk_or_grouped_context}

Output format (one per line):

Q: ...

A: ...

R1-M2: Title/Name: MedDiscover is a misleading name since the pipeline is not specifically tailored to the metabolomics domain.

Response: We revised the title and framing to avoid claiming a metabolomics-only architecture. The paper now presents MedDiscover as a reproducible benchmark and evaluation pipeline instantiated for metabolomics/metabolic-disorder literature.

Manuscript: MedDiscover: Two-tier benchmark for retrieval-augmented metabolomics question answering in metabolic disorders

This paper introduces MedDiscover, an open-source RAG implementation and evaluation benchmark instantiated for metabolomics literature. Rather than proposing a new retrieval architecture, we focus on reproducible assessment of retrieval augmentation, embedding choice, and decoder sensitivity within a metabolomics/metabolic-disorder corpus. Metabolomics, pivotal for biomarker discovery, disease diagnosis, and personalized medicine, faces significant literature management challenges due to its rapidly expanding knowledge base.

R1-M3: Model choice: Why was gpt4o used instead of other LLMs?

Response: We clarified the model-choice rationale and fixed GPT-4o as the primary decoder for fair comparisons; GPT-4.1 variants are treated as sensitivity analyses.

Manuscript: We implement MedDiscover as an open-source RAG evaluation pipeline and apply it to metabolomics/metabolic-disorder literature using a two-tier dataset (Gold and Silver). The primary comparisons fix the decoder (GPT-4o) to isolate retrieval effects, while decoder upgrades (GPT-4.1-nano/mini) are reported as sensitivity analyses. NotebookLM is included as a black-box comparator for answer style, with grounding metrics reported only where retrieved contexts are available. The evaluation compares three primary strategies: (1) domain-specific embeddings (MedCPT) with a GPT-4o decoder, (2) general-purpose embeddings (OpenAI Ada-002) with a GPT-4o decoder, and (3) a non-RAG baseline (GPT-4o without retrieval). The choice of MedCPT, Ada, and GPT-4o was guided by an extensive literature review, which identified MedCPT as the only published, publicly available domain-specific embedding model with validated superior performance. Other alternatives (e.g., BioGPT, SciPhi) were either not available for standalone use or did not surpass MedCPT in comparative studies. Ada and GPT-4o represent widely used general-purpose embedding and generation baselines, enabling a controlled comparison.

R1-M4: Section 2.3: The chunking phase as described in the article seems domain-specific, not something general to RA.

Response: We revised the chunking description to emphasize standard RAG hyperparameters (500-token windows, 50-token overlap) rather than a domain-specific procedure.

Manuscript: The chunking process uses overlapping chunks of 500 tokens with a 50-token overlap, a standard RAG setting selected based on prevailing practices in recent biomedical studies (e.g., RDGuru, AsthmaBot, EARA) and to preserve entity co-

occurrence across chunk boundaries. This chunk size is compatible with the context windows of current LLMs and balances retrieval precision with contextual integrity. For documents exceeding the LLM token limit, multiple overlapping chunks were generated and, during retrieval, the top-ranked segments were concatenated or batched as input context for the decoder LLM.

R1-M5: Figure 1: Comparative approaches should not be injected into this figure; the focus should remain on the proposed approach.

Response: We separated the architecture figure from the experimental design. Figure 1 now focuses only on the MedDiscover pipeline and dataset design; comparative evaluations are described in the study-design figure and Results.

Manuscript: MedDiscover pipeline and benchmark design. A metabolomics/metabolic-disorder corpus (ICD-10 E70-E88) is ingested, cleaned, chunked (500 tokens with 50-token overlap), and indexed with two embedding configurations (MedCPT and Ada-002) using FAISS. Evaluation is organized into a Gold set (expert-curated QA with traceable references) and a Silver set (synthetic QA with retrievability metadata). RAG inference retrieves top-k contexts and uses an evidence-grounded prompt to produce answers with explicit supporting context; evaluation reports reference-based metrics on Gold and retrieval-grounding metrics plus retrievability statistics on Silver.

Experimental design overview. Primary tests fix the decoder (GPT-4o) to isolate retrieval effects (RAG vs non-RAG on the Gold set; MedCPT vs Ada at scale on the Silver benchmark). Secondary analyses include decoder sensitivity (GPT-4o vs GPT-4.1 variants on Gold) and retrievability quality-control analyses on Silver (hit-rate and rank-based ablations).

R1-M6: Section 3.1: It is unclear whether the focus is on metabolomics in general or specifically on metabolic diseases/disorders.

Response: We clarified the scope as metabolomics literature in the context of metabolic disorders, explicitly referencing ICD-10 E70-E88 in the Introduction and Data Collection sections.

Manuscript: We focus on metabolomics literature in the context of metabolic disorders (ICD-10 E70-E88) because these conditions are metabolite-centric and representative of biomarker-driven metabolomics workflows.

We assembled a metabolomics/metabolic-disorder corpus of 100 papers published between 2014 and 2024 from PubMed and Google Scholar, curated using ICD-10 metabolic disorder codes E70-E88. Text extraction and cleaning were performed by extracting raw text from PDFs using PyPDF2 and removing headers, footers, and non-informative formatting. Numeric and scientifically relevant data (metabolite concentrations, p-values) were preserved, and each processed article retained original paragraph sequences with metadata identifiers mapping back to the source PDF.

R1-M7: Section 3.6: This section should be moved earlier, and the ROUGE metric should be formally defined. What does E represent in the formulae?

Response: We revised the Evaluation Metrics section to define ROUGE formally and to define the embedding function used in semantic metrics, removing ambiguity around notation.

Manuscript: Performance evaluation incorporated both generation-centric and retrieval-centric metrics, applied to MedDiscover configurations and, where appropriate, to NotebookLM. Let $\phi(\cdot)$ denote the embedding function used for semantic similarity metrics.

Generation Metrics: Accuracy, Precision, Recall, and F1 Score quantitatively assessed the correctness and comprehensiveness of generated responses. ROUGE-1/2/L measure unigram overlap, bigram overlap, and longest-common-subsequence overlap between candidate and reference answers. BERTScore captures semantic similarity: $F1\ Score = 2 \times (Precision \times Recall / (Precision + Recall))$

$BERTScore = CosSim(\phi(\text{generated}), \phi(\text{reference}))$

Retrieval Metrics (Primarily for RAG systems like MedDiscover): Context Precision and Context Recall assessed the relevance and completeness of retrieved information, while Faithfulness measured factual grounding of generated content:

$Context\ Precision@K = (Relevant\ Retrieved\ Contexts / K)$

$Context\ Recall = (Relevant\ Retrieved\ Contexts / Total\ Relevant\ Contexts)$

$Faithfulness = (Supported\ Claims\ in\ Generated\ Answer / Total\ Claims\ in\ Generated\ Answer)$

Additionally, Answer Relevancy quantified semantic alignment between queries and

responses:

Answer Relevancy = $(1 / N) \sum_{i=1}^N \cos(\phi(\text{answer}_i), \phi(\text{query}_i))$

We compute RAGAS metrics (faithfulness, answer correctness, context recall, context precision, answer relevancy) using the RAGAS framework and report these where retrieved contexts are available. ROUGE and BLEU are reported only for the Gold set because the Silver set lacks human reference answers. For NotebookLM, only metrics not dependent on explicit retrieved context (like Answer Relevancy and generation metrics) are fully comparable.

R1-M8: Results: It seems that NotebookLM performs best in terms of accuracy. Why should we prefer your approach? This is not convincingly discussed.

Response: We clarified that NotebookLM does not expose retrieved contexts, so grounding metrics are not comparable. The Discussion now explains why transparent evidence grounding is preferred for biomedical use cases.

Manuscript: NotebookLM comparison

NotebookLM shows strong lexical overlap, especially when prompted for minimal output, but it does not expose retrieved contexts. As a result, grounding metrics (context precision/recall, faithfulness) cannot be computed for NotebookLM, limiting its comparability in evidence-critical settings.

R1-M9: Figure 2: please improve readability.

Response: We improved readability by replacing the dense multi-model plot in the main text with a clear study-design figure and moving the full multi-model plot to the Supplementary section; Silver results are summarized in a table with CIs and effect sizes.

Manuscript: Experimental design overview. Primary tests fix the decoder (GPT-4o) to isolate retrieval effects (RAG vs non-RAG on the Gold set; MedCPT vs Ada at scale on the Silver benchmark). Secondary analyses include decoder sensitivity (GPT-4o vs GPT-4.1 variants on Gold) and retrievability quality-control analyses on Silver (hit-rate and rank-based ablations).

Supplementary Gold-set results across additional configurations (decoder sensitivity variants and black-box comparator). This figure is provided for completeness; the primary comparisons in the main text restrict to fixed-decoder (GPT-4o) evaluations to avoid conflating retriever effects with decoder capacity. Note that NotebookLM does not expose retrieved contexts; context-dependent metrics are therefore not comparable and should be interpreted as not available rather than as zero performance.

R1-M10: Section 4.4: The last paragraph is unclear and difficult to read.

Response: We rewrote the Discussion for clarity and structure, focusing on three defensible claims aligned with the revised benchmark framing.

Manuscript: This study evaluates MedDiscover as a reproducible RAG benchmark with two datasets: a Gold set (expert-curated) and a Silver set (retrievability-filtered synthetic). On the Gold set, retrieval augmentation improves grounding relative to a non-RAG baseline when the decoder is held constant. When the decoder is fixed at GPT-4o, differences between MedCPT and Ada are modest and not consistently significant across metrics.

On the Silver benchmark, embedding choice produces comparable retrieval-centric metrics, while MedCPT yields a statistically significant improvement in answer correctness (Table). This indicates that domain-specific embeddings can provide a targeted gain in answer quality, but not a universal advantage across all grounding metrics.

R1-M11: Section 5.3: The discussion is verbose and lacks clarity.

Response: Addressed via the same Discussion rewrite, with reduced verbosity and clearer separation of primary vs sensitivity analyses.

Manuscript: NotebookLM provides strong lexical alignment under minimal-output prompting, but its retrieval mechanism is opaque. Because retrieved contexts are not exposed, grounding metrics cannot be computed for NotebookLM, limiting its suitability for evidence-critical biomedical workflows.

All MedDiscover runs restricted GPT-4o to 25 generation tokens to control verbosity and encourage precision. This design choice likely depressed length-sensitive scores such as ROUGE and BLEU relative to longer responses, underscoring the need to interpret lexical overlap metrics cautiously in short-answer biomedical settings.

R1-M12: Dataset size: Only 10 articles and 3 diseases are included, which is insufficient for a scientific journal publication.
Response: We expanded the dataset using a 100-paper Silver benchmark with ~600 synthetic QA pairs while retaining the Gold expert-curated set; the dataset summary table and Silver results address scale and generalization.
Manuscript: Gold QA set (expert-curated)
The Gold set consists of ten metabolomics papers focusing on Gaucher disease (Type 1), Nonalcoholic Fatty Liver Disease (NAFLD), and Type 1 diabetes, with 30 expert-curated questions and traceable reference answers. This set is used to validate answer quality against human references. The full Gold set contains 30 expert-curated QA pairs. For cross-system comparisons that require outputs from every evaluated configuration (including black-box comparators), we report results on the subset of 20 Gold questions for which all model outputs are available (evaluation subset; see Section).

Silver QA set (synthetic, retrievability-filtered)
The Silver set uses the 100-paper corpus and approx. 600 synthetic QA pairs generated with retrievability metadata (Ada n=300, MedCPT n=300). This set is used to stress-test retrieval grounding at scale rather than to provide human reference answers. We refer to this collection as a Silver set because the questions and reference answers are machine-generated and only partially filtered: each QA pair is retained based on lightweight quality controls (Section), including a retrievability check, but the content is not exhaustively expert-validated. Accordingly, Silver-set results are interpreted primarily as evidence about retrieval grounding and robustness at scale, rather than as definitive clinical correctness.

Minor Comments
R1-m1: "RAG-LLM" is redundant; use simply "RAG."
Response: Implemented. We use the term RAG consistently throughout the manuscript.
Manuscript: Background: RAG in Metabolomics
Metabolomics Research: Relevance and Challenges

R1-m2: Inappropriate use of the opening quotation mark (").
Response: Corrected in the revised manuscript; we performed a pass to normalize quotation marks.
Manuscript: RAG's core mechanism involves encoding a user query—e.g., "What metabolites are associated with early-stage Alzheimer's disease?"—into a dense vector embedding using models like MedCPT or BioBERT. This embedding, capturing semantic intent, queries a vector database of pre-processed documents (e.g., PubMed articles). Retrieved text chunks augment the query, enabling the LLM to generate evidence-based responses. Embeddings, built from tokenized text (as discussed in the Background section), are indexed in vector databases like FAISS or Qdrant, using algorithms like HNSW (Hierarchical Navigable Small Worlds) for efficient similarity searches.

R1-m3: Missing spaces after several periods.
Response: Corrected with targeted copy edits and spacing fixes; we will do a final proofreading pass on the clean manuscript.
Manuscript: Detailed methodologies for data collection, preparation, the RAG pipeline, and evaluation metrics are described in the following subsections. Further experimental setups and extensive supplementary results can be found in the Supplementary Material, ensuring transparency, reproducibility, and facilitating further exploration of our methodology.

R1-m4: References should be ordered numerically, e.g., [12,19] --> [19,12].
Response: Implemented by enabling numeric sorting/compression in the bibliography options.
Manuscript: biboptions{sort&compress}

Reviewer #2
R2-C1: This is a well-written manuscript that represents a novel contribution in the area of LLM RAG systems for biomedical (in particular, metabolomics) research; it introduces a new domain-specific LLM RAG framework and compares it to multiple

other related frameworks on a curated metabolomics query set. The study also contributes to general understanding of the benefits of RAG (whether general or domain-specific) in biomedical question answering.

Response: Thank you for the positive assessment.

Manuscript: Our contributions are threefold:

- A two-tier benchmark: a Gold set with expert-curated QA pairs and a Silver set with retrievability-filtered synthetic QA pairs at scale.
- Transparent RAG evaluation with retrieval-centric metrics (RAGAS) and statistical testing that reports effect sizes and confidence intervals.
- An open-source reproducibility package (code, prompts, document lists, and evaluation pipeline) that enables independent re-analysis.

R2-C2: That said, I think some of the claims may be misleading - specifically, in the Abstract and elsewhere, the authors report that MedDiscover with a GPT-4.1 decoder outperformed the general-purpose RAG comparator (which was Ada with a GPT-4o decoder). While this is true, it is misleading to present as the primary finding given that - when holding the decoder constant and using GPT-4o for both MedDiscover and Ada - MedDiscover did not significantly outperform Ada. And using a GPT-4.1 decoder provided a big bump in performance for MedDiscover (suggesting GPT-4.1 may be much better suited than GPT-4o for the questions tested), yet was not evaluated for the comparators, making it impossible to determine whether performance gains stem from the domain-specific embeddings or simply from using a more capable decoder. Given this, I don't think the results are sufficient to conclude that "domain-adapted embeddings significantly improve retrieval precision, factual grounding, and answer correctness".

Response: We agree. The revised manuscript avoids decoder-confounded claims: primary comparisons fix GPT-4o, while GPT-4.1 variants are explicitly labeled as sensitivity analyses, and the Abstract/Conclusion were updated accordingly. Primary conclusion statements are now based only on fixed-decoder (GPT-4o) comparisons; decoder upgrades are treated strictly as sensitivity analyses and are not used to support embedding superiority claims.

Manuscript: We implement MedDiscover as an open-source RAG evaluation pipeline and apply it to metabolomics/metabolic-disorder literature using a two-tier dataset (Gold and Silver). The primary comparisons fix the decoder (GPT-4o) to isolate retrieval effects, while decoder upgrades (GPT-4.1-nano/mini) are reported as sensitivity analyses. NotebookLM is included as a black-box comparator for answer style, with grounding metrics reported only where retrieved contexts are available. Results on the Gold set (expert curated)

We evaluated an expert-curated Gold set of 30 QA pairs; results reported here use the 20-question evaluation subset for which outputs are available across all compared configurations. Primary comparisons fix the decoder (GPT-4o) to isolate retrieval effects across the non-RAG baseline, Ada + GPT-4o, and MedCPT + GPT-4o. Supplementary Figure summarizes LLM-specific and retrieval-specific metrics for these models, along with NotebookLM and the MedCPT decoder sensitivity variants.

R2-C3: While it would be ideal if the authors evaluated the comparators with GPT-4.1 as well, I don't think it's required for this manuscript to still be a novel contribution to research. However, if not doing further experiments, the authors should be much more careful with their presentation of results and claims. In addition to revising claims/conclusions reported throughout (e.g., Abstract lines 11-16 and 19-22; Conclusion lines 2-7 and 13-15), I would suggest excluding the MedDiscover GPT-4.1 results from the primary results figure (Figure 2), and then only later in the Results section including a second version of that figure that includes the MedDiscover GPT-4.1 results, clearly noting that it was a supplementary analysis without a fair comparator.

Response: Implemented. GPT-4.1 results were moved to the Supplementary section and are not used for primary conclusions. Primary conclusions are based only on fixed-decoder (GPT-4o) comparisons; decoder upgrades are treated strictly as sensitivity analyses and do not support embedding-superiority claims.

Manuscript: Experimental design overview. Primary tests fix the decoder (GPT-4o) to isolate retrieval effects (RAG vs non-RAG on the Gold set; MedCPT vs Ada at scale on the Silver benchmark). Secondary analyses include decoder sensitivity (GPT-4o vs GPT-4.1 variants on Gold) and retrievability quality-control analyses on Silver (hit-rate and rank-based ablations).

Supplementary Gold-set results across additional configurations (decoder sensitivity variants and black-box comparator). This figure is provided for completeness; the primary comparisons in the main text restrict to fixed-decoder (GPT-4o) evaluations to avoid conflating retriever effects with decoder capacity. Note that NotebookLM does not expose retrieved contexts; context-dependent metrics are therefore not comparable and should be interpreted as not available rather than as zero performance.

R2-C4: One final minor comment - on page 23 of the PDF, in the sentence "By contrast, upgrading the decoder within the MedCPT pipeline from GPT-4 to GPT-4.1-nano and further to GPT-4.1-mini..." - I'd suggest changing the first "GPT-4" to "GPT-4o" for clarity.

Response: Corrected throughout the manuscript.

Manuscript: We implement MedDiscover as an open-source RAG evaluation pipeline and apply it to metabolomics/metabolic-disorder literature using a two-tier dataset (Gold and Silver). The primary comparisons fix the decoder (GPT-4o) to isolate retrieval effects, while decoder upgrades (GPT-4.1-nano/mini) are reported as sensitivity analyses. NotebookLM is included as a black-box comparator for answer style, with grounding metrics reported only where retrieved contexts are available.

Reviewer #3

R3-O1: This manuscript presents MedDiscover, a Retrieval-Augmented Generation (RAG) framework tailored to metabolomics knowledge extraction. However, there are methodological and reporting issues that should be addressed before acceptance.

Response: Thank you for the careful review. We revised the manuscript to address the methodological and reporting issues and to tighten the study framing and evaluation rigor.

Manuscript: We implement MedDiscover as an open-source RAG evaluation pipeline and apply it to metabolomics/metabolic-disorder literature using a two-tier dataset (Gold and Silver). The primary comparisons fix the decoder (GPT-4o) to isolate retrieval effects, while decoder upgrades (GPT-4.1-nano/mini) are reported as sensitivity analyses. NotebookLM is included as a black-box comparator for answer style, with grounding metrics reported only where retrieved contexts are available.

Results on the Gold set (expert curated)

We evaluated an expert-curated Gold set of 30 QA pairs; results reported here use the 20-question evaluation subset for which outputs are available across all compared configurations. Primary comparisons fix the decoder (GPT-4o) to isolate retrieval effects across the non-RAG baseline, Ada + GPT-4o, and MedCPT + GPT-4o.

Supplementary Figure summarizes LLM-specific and retrieval-specific metrics for these models, along with NotebookLM and the MedCPT decoder sensitivity variants.

Major Comments

R3-M1: 1. Experimental design and dataset limitations

The study evaluates MedDiscover on only 30 question-answer pairs derived from 10 metabolomics papers across three diseases. This sample is too small for robust statistical inference in high-variance LLM evaluations. The paper acknowledges this but should:

Quantify statistical power and confidence intervals;

Clarify the manual curation process and whether inter-annotator agreement or expert validation of the reference answers was performed;

Discuss possible selection bias due to the limited corpus and disease scope.

Suggestion: Expand the dataset or validate the model on an external corpus to support generalization claims.

Response: We expanded the evaluation with a 100-paper Silver benchmark and kept the Gold set for reference-based validation. We added bootstrap confidence intervals and effect sizes for the Silver results and clarified selection bias and scope limitations in the Limitations section.

Manuscript: The Gold set consists of ten metabolomics papers focusing on Gaucher disease (Type 1), Nonalcoholic Fatty Liver Disease (NAFLD), and Type 1 diabetes, with 30 expert-curated questions and traceable reference answers. This set is used to validate answer quality against human references. The full Gold set contains 30 expert-curated QA pairs. For cross-system comparisons that require outputs from every evaluated configuration (including black-box comparators), we report results on the subset of 20 Gold questions for which all model outputs are available (evaluation

subset; see Section).

Silver QA set (synthetic, retrievability-filtered)

The Silver set uses the 100-paper corpus and approx. 600 synthetic QA pairs generated with retrievability metadata (Ada n=300, MedCPT n=300). This set is used to stress-test retrieval grounding at scale rather than to provide human reference answers. We refer to this collection as a Silver set because the questions and reference answers are machine-generated and only partially filtered: each QA pair is retained based on lightweight quality controls (Section), including a retrievability check, but the content is not exhaustively expert-validated. Accordingly, Silver-set results are interpreted primarily as evidence about retrieval grounding and robustness at scale, rather than as definitive clinical correctness.

This study demonstrates the potential of RAG frameworks for metabolomic biomarker discovery, but several important limitations must be acknowledged. The Gold set is limited to ten English articles spanning three diseases (Gaucher disease Type 1, NAFLD, and Type 1 diabetes), which constrains the strength of reference-based conclusions. The Silver set expands coverage to 100 papers, but the scope remains restricted to metabolic-disorder literature (ICD-10 E70-E88), limiting generalizability to other biomedical domains. A larger and more diverse corpus would provide a more representative evaluation of the framework's scalability and applicability across a wider range of conditions.

Gold and Silver sets differ in both scope and construction: the Gold set is expert-curated over three disease-focused topics, whereas the Silver set spans ICD-10 E70-E88 metabolic-disorder literature and uses machine-generated QA with partial filtering. This introduces selection and distribution shift between tiers; consequently, we do not interpret absolute metric values as directly comparable across Gold and Silver, and we use each tier to support different validity claims (reference-based validation vs retrieval-grounding stress testing).

R3-M2: 2. Evaluation methodology and metric validity

The use of ROUGE/BLEU for concise biomedical answers (≤ 25 tokens) may underestimate semantic quality and exaggerate differences in output style. Metrics like Faithfulness and Context Precision rely on author-defined definitions rather than standardized formulations.

Suggestion: Include semantic similarity or LLM-as-a-Judge evaluation calibrated by biomedical experts. Report confidence intervals or bootstrap variance for all metrics.

Response: We now limit ROUGE/BLEU reporting to the Gold set and use standard RAGAS metrics for the Silver benchmark. For Silver, we describe answer correctness as a silver-standard answer-correctness score (i.e., an answerability/grounding proxy rather than expert-validated clinical correctness), and we report confidence intervals for the Silver results.

Manuscript: We compute RAGAS metrics (faithfulness, answer correctness, context recall, context precision, answer relevancy) using the RAGAS framework and report these where retrieved contexts are available. ROUGE and BLEU are reported only for the Gold set because the Silver set lacks human reference answers. For NotebookLM, only metrics not dependent on explicit retrieved context (like Answer Relevancy and generation metrics) are fully comparable.

Because the Silver benchmark uses synthetic reference answers that are identical to the responses in the evaluation files, answer-quality metrics should be interpreted as proxy measures of answerability and grounding rather than as expert-validated clinical correctness.

R3-M3: 3. Statistical analysis

The authors correctly apply non-parametric tests (Kruskal-Wallis + Dunn), but there is no correction for multiple testing across metrics, beyond Bonferroni within metric. Small-sample effects may inflate p-values or significance of large differences.

Suggestion: Include effect sizes alongside p-values to quantify practical importance. Add a summary table of test statistics.

Response: We added Holm-Bonferroni correction across the primary metric family, and report Mann-Whitney U with Cliff's delta and bootstrap CIs in the Silver results table.

Manuscript: Unless otherwise stated, we treat the retrieval-centric RAGAS metrics (faithfulness, answer correctness, context recall, context precision, answer relevancy) as a primary metric family and control family-wise error across this family using Holm-Bonferroni adjustment. Metrics outside this family are treated as secondary and interpreted cautiously.

All model-by-metric score distributions were subjected to the Shapiro-Wilk test; almost every distribution departed from normality ($p < 0.05$). A subsequent Levene test revealed unequal variances for several evaluation metrics. In view of these results, groupwise comparisons were carried out with rank-based, non-parametric methods. For the Gold set, a Kruskal-Wallis one-way ANOVA on ranks was computed across the model configurations, followed by Dunn's post-hoc tests with Ada + GPT-4o as the reference group. Holm-Bonferroni correction was applied across the primary metric family (faithfulness, answer correctness, context recall, context precision, answer relevancy). Supplementary Figure reports adjusted contrasts. For the Silver benchmark, we report Mann-Whitney U tests for Ada vs MedCPT on each RAGAS metric, along with Cliff's delta effect sizes and bootstrap 95% confidence intervals for mean scores (2,000 resamples; Table). This isolates embedding effects under a fixed decoder and provides a scale-appropriate estimate of practical impact.

R3-M4: 4. Novelty and comparison to prior RAG frameworks

While the paper situates MedDiscover relative to BioRAG, DRAGON, and Self-BioRAG, it does not clearly define what is technically novel beyond domain restriction (metabolomics) and embedding choice.

Suggestion: Clarify whether MedDiscover introduces a new retrieval strategy, interface design, or evaluation framework, and distinguish these from prior MedCPT-based pipelines.

Response: We clarified that the contribution is a two-tier benchmark and evaluation pipeline rather than a new retrieval algorithm, and positioned MedDiscover accordingly in the Introduction and Methods.

Manuscript: This paper introduces MedDiscover, an open-source RAG implementation and evaluation benchmark instantiated for metabolomics literature. Rather than proposing a new retrieval architecture, we focus on reproducible assessment of retrieval augmentation, embedding choice, and decoder sensitivity within a metabolomics/metabolic-disorder corpus. Metabolomics, pivotal for biomarker discovery, disease diagnosis, and personalized medicine, faces significant literature management challenges due to its rapidly expanding knowledge base.

RAG frameworks were introduced to bridge this gap by injecting up-to-date evidence at inference time. Systems such as BIORAG, RAG², and Self-BioRAG have since added multi-stage retrieval pipelines, self-evaluation loops, and knowledge-graph conditioning to cope with the scale and dynamism of biomedical knowledge.

MedDiscover follows this lineage but narrows the focus to metabolomics, a sub-discipline that has received comparatively little attention in prior RAG work. By anchoring retrieval to experimentally validated metabolite literature rather than broader biomedical text, our study illustrates how increased domain granularity can sharpen factual grounding.

Retrieval Architectures: Similarities and Divergences

Compared with earlier frameworks, MedDiscover adopts a deliberately lightweight retrieval architecture: a single-stage dense lookup over FAISS indices produced by MedCPT or Ada-002 embeddings. DRAGON and RAG², by contrast, implement multi-hop or multi-stage retrieval. DRAGON fuses knowledge graphs with text through a cross-modal encoder, while RAG² first rewrites the query into a rationale, then balances evidence across PubMed, PMC, textbooks and guidelines, and finally reranks with a cross-encoder. BioRAG extends the concept still further by coupling MeSH-filtered hierarchical look-ups with external search-engine calls and a self-evaluation module that decides when additional retrieval is needed.

Although these more elaborate pipelines improve performance on broad medical QA benchmarks, our results indicate that a streamlined, single-stage design can suffice for narrowly scoped, high-precision tasks. MedCPT-based retrieval delivers competitive context precision and faithfulness on the Gold set while maintaining lower latency and a smaller memory footprint. This finding implies that retrieval complexity should be matched to the breadth of the information need: dense, domain-specific embeddings may already capture enough local structure when the target corpus is well circumscribed.

R3-M5: 5. Transparency, reproducibility, and open science

The GitHub repository is cited, but the paper should specify:

What data and scripts are available (PDF text, embeddings, evaluation code);

Licensing and reproducibility (random seeds, hardware, API versions);

Whether PubMed content redistribution complies with copyright.

Suggestion: Add a “Reproducibility Checklist” table (dataset availability, model weights, code, and statistical analysis scripts).

Response: We added a reproducibility checklist table that specifies available data, scripts, model/config versions, and evaluation artifacts, and we clarified data availability in the Data and Code Availability section.

Manuscript: Reproducibility checklist (artifacts and where to find them).

Data and Code Availability

The MedDiscover framework is available as open-source software at <https://github.com/VatsalPatel18/MedDiscover>. The analysis pipeline, evaluation outputs, and figure-generation scripts used in this manuscript are available at <https://github.com/VatsalPatel18/MedDiscover-Analysis>.

Due to copyright and licensing constraints, we do not redistribute publisher PDFs or extracted full-text derived from closed-access articles. Instead, we release the list of PubMed IDs/DOIs used to assemble the corpus and provide scripts to reproduce preprocessing and indexing from documents that the user can legally access (e.g., open-access sources). Released derived artifacts (Gold QA pairs, Silver QA pairs, retrieval metadata, and evaluation tables) are provided without redistributing publisher PDF files.

R3-M6: 6. Clinical and ethical considerations

The discussion briefly mentions regulatory readiness (GDPR, CE/FDA) but lacks concrete exploration of bias, interpretability, or safety in biomedical deployment.

Suggestion: Expand Section 5.4 to include ethical implications (traceability, hallucination risk, domain shift) and planned safeguards.

Response: We expanded the Limitations and Discussion to address traceability, hallucination risk, domain shift, and deployment constraints for biomedical use.

Manuscript: The indexing and retrieval processes used in this study were static, with no mechanisms for continuous updates or adaptive refinement. Biomedical research is a rapidly evolving field, and the inability of the framework to incorporate new findings or adjust to user feedback limits its relevance and longevity. Furthermore, this study did not sufficiently explore real-world usability concerns, such as providing confidence scores, highlighting retrieved documents, or offering transparent explanations of how generated responses were synthesized. These features are critical for building trust and fostering adoption among clinicians and researchers who depend on reliability and interpretability.

From an engineering perspective, computational efficiency and cost were not rigorously profiled. The study relied on commercial GPT-4o decoding and two embedding models without accounting for the latency or financial burden that would accrue in large-scale or real-time deployments. For institutions with limited computational budgets—or for privacy-sensitive environments that cannot rely on third-party APIs—these cost and infrastructure questions are pivotal.

In addition to the previously discussed limitations, several further aspects warrant critical consideration. First, the current evaluation strategy relied exclusively on automated metrics (e.g., ROUGE, BLEU, BERTScore, Faithfulness), without incorporating structured human-in-the-loop assessments or inter-rater agreement. This absence of qualitative expert judgment may obscure important nuances in factual accuracy and biomedical relevance, especially in domains where minor terminological shifts (e.g., “reduced” vs. “unchanged”) can meaningfully alter interpretation. Second, all generative outputs were produced using variants of GPT-4 (GPT-4o or GPT-4.1-mini), thereby limiting insights into decoder variability. Given that different LLM architectures exhibit distinct generation behaviors, a broader decoder comparison, including open-source biomedical models, would be essential for generalizing the framework’s applicability across institutional settings. Third, the study did not develop a formal taxonomy of observed error types. While macro-level metrics suggested performance differences between domain-specific and general-purpose pipelines, it remains unclear whether these stemmed from hallucinations, factual omissions, semantic contradiction, or ambiguous phrasing. A structured error analysis would help isolate these causes and guide future improvements. Fourth, while the QA dataset was intentionally designed for clinical specificity and scientific rigor, it does not capture the breadth or ambiguity of real-world information-seeking behavior by clinicians, biologists, or translational researchers. Incorporating user-centric evaluations, log analyses, or exploratory query types could improve the system’s ecological validity. Lastly, although the framework holds promise for biomedical support tasks, its ethical, regulatory, and operational readiness for clinical deployment remains unassessed.

Issues such as GDPR compliance, CE/FDA documentation traceability, and robust audit trails for generated content were beyond the current scope but are indispensable for translation into regulated medical environments.

R3-M7: 7. Typographical issues

Grammatical inconsistencies (“nove” instead of “novel” on p. 34).

I would like to offer a strong recommendation to the authors to take steps towards enhancing the clarity of their manuscript and making revisions to improve the overall quality of the language and sentence structure. In order to achieve this, it is highly advisable that the authors consider arranging for a comprehensive proofreading of the manuscript. This will not only contribute to a more polished and refined document but also ensure that the manuscript effectively conveys its intended message to its readers.

Response: We corrected typos and performed language cleanup, and we will complete a final proofreading pass before resubmission.

Manuscript: Coupling this capability with iterative re-ranking strategies and user-driven feedback loops would enable the system to retire outdated passages and elevate novel, high-quality evidence, ensuring that clinicians and researchers receive the most current insights available.

R3-M8: 8. Discussion

Consider discussing and properly citing related works such as:

<https://doi.org/10.1016/j.csbj.2024.10.017>

<https://doi.org/10.1016/j.csbj.2024.03.016>

<https://doi.org/10.1016/j.csbj.2024.08.016>

<https://doi.org/10.1016/j.csbj.2024.05.004>

<https://doi.org/10.1016/j.csbj.2023.11.011>

Response: We added and discussed the requested CSBJ-related works in the Background/Related Work sections.

Manuscript: Metabolomics Research: Relevance and Challenges

Metabolomics connects genetic and environmental influences with phenotypic outcomes through metabolites, acting as substrates, intermediates, and end products within biological pathways, thereby reflecting immediate physiological states and adaptations. Researchers often struggle with the complexity, terminology, and rapidly expanding literature, complicating efficient knowledge extraction and practical hypothesis generation. Traditional metabolomics databases require structured queries, limiting intuitive exploration, especially for researchers with limited domain expertise. Beyond the rapid growth of the metabolomics literature, metabolomics datasets themselves are often high-dimensional and sparse (i.e., $p \gg n$), and can exhibit class- or feature-level imbalance, which complicates robust patient classification and biomarker discovery. Recent benchmarking work in metabolomics biomedical data systematically evaluated feature extraction and feature selection pipelines and reported that supervised feature selection can improve downstream classification performance, providing practical guidance on when to apply these techniques. In parallel, predictive biomedicine increasingly combines multiple data modalities (e.g., multi-omics measurements with imaging or genomic variation), further increasing the complexity of evidence synthesis and interpretation. Existing biomedical literature-mining tools such as BioTextQuest v2.0 support term-driven clustering and named-entity extraction from PubMed abstracts, but remain primarily query-and-browse systems rather than evidence-grounded natural-language question answering interfaces.

In biomedical NLP, these modelling choices must also be evaluated through the lens of transparency and interpretability. A recent scoping review in healthcare NLP highlights persistent gaps in best practices and systematic benchmarks for explainable and interpretable deep learning, reinforcing the need for designs that support auditability and evidence traceability in high-stakes settings.

Knowledge-graph resources can further complement text retrieval by encoding metabolite-gene-disease relationships and enabling multi-hop evidence traversal. However, constructing and maintaining biomedical knowledge graphs at scale remains challenging. BioKGrapher provides an example of automated KG construction from PubMed IDs using NER/NEL with UMLS normalization, concept weighting and re-ranking, and ontology-driven relation building (e.g., via terminologies such as SNOMED CT and NCI), illustrating a practical pathway for producing condition-specific graphs that could be coupled to graph-based retrieval in biomedical RAG pipelines.

Minor Comments

R3-m1: How do the researchers/clinicians can benefit from the work of this manuscript?

Response: We clarified the intended benefit as a reproducible, evidence-grounded benchmark and evaluation pipeline for metabolomics/metabolic-disorder QA.

Manuscript: Our contributions are threefold:

- A two-tier benchmark: a Gold set with expert-curated QA pairs and a Silver set with retrievability-filtered synthetic QA pairs at scale.
- Transparent RAG evaluation with retrieval-centric metrics (RAGAS) and statistical testing that reports effect sizes and confidence intervals.
- An open-source reproducibility package (code, prompts, document lists, and evaluation pipeline) that enables independent re-analysis.

This study ...

Revised Cover Letter

Dear Editor,

On behalf of the authors, I am pleased to submit the revised manuscript titled “**MedDiscover: A Domain-Specific Retrieval-Augmented Generation Framework for Evidence-Grounded Knowledge Extraction in Metabolomics**” for consideration in the *Computational and Structural Biotechnology Journal (CSBJ) Special Issue: “AI and Data Science for Precision Medicine and Healthcare Innovation.”*

In response to the constructive feedback from the previous review round, we have significantly overhauled the manuscript to move beyond a simple benchmark. We now present **MedDiscover** as a functional, open-source framework designed to solve the "black-box" and hallucination challenges of using LLMs in specialized metabolomics research.

Key enhancements in this revision include:

- **Significant Dataset Expansion:** We have expanded the corpus from **10 to 110 peer-reviewed papers** curated using ICD-10 metabolic disorder codes (E70–E88), addressing concerns regarding statistical power and domain scope.
- **Two-Tier Evaluation Strategy:** We introduce a robust dual-testing design: a **Gold set** (30 expert-curated QA with traceable references) and a **Silver set** (600 synthetic QA pairs with retrievability metadata) to stress-test the system at scale.
- **Transparency & Open Science:** To support the CSBJ community, we have released a full **reproducibility package**, including the MedDiscover code, prompts, and document lists, ensuring our findings can be independently verified.
- **Refined Methodological Focus:** We have isolated retrieval effects by fixing the decoder (GPT-4o) and providing sensitivity analyses for advanced models (GPT-4.1), ensuring the results clearly distinguish between embedding quality and LLM capacity.

We believe MedDiscover is now a perfect fit for this Special Issue. It provides a practical, interpretable tool that bridges the gap between massive unstructured metabolomics literature and actionable insights, a core requirement for the next generation of precision medicine and healthcare innovation.

All authors have approved this submission. This manuscript is original, not previously published, and not under consideration elsewhere.

Thank you for your continued consideration of our work.

Sincerely,

Anne Schwerk

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Juri-Gagarin-Ring 152, 99084 Erfurt, Germany

Review 2 (22 Feb 2026)

Reviewer #3

R3-K1: This revised submission presents a domain specific approach using LLM for information retrieval in metabolomics and the biosciences. This is viewed as a very valuable approach, despite not being the final model version. Any progress towards the goal of better (and faster) answering of biomedical questions is welcome. The authors have done a very good job at responding the queries from the first two reviewers. The most critical one being the size of the training set. In this new, revised version, the training set has been largely expanded, and it is now seen as a more innovative approach. The size of the dataset is still not as large as it should be, with the metabolomics literature having thousands of papers and coverage of many more diseases, so the authors are encouraged to continue pursuing this work and expanding to the whole body of literature. This type of tool, when made accessible to the whole metabolomics community, could save enormous amounts of time, which are currently devoted to digging into the literature to find the biological interpretation of the data generated.

Note that the citations seem to be missing, at least from the pdf document I was provided.

Response: we acknowledge the comments and have submitted and attached the complimentary reference.bib file along with the latex missed out from our previous submission.

Review 1 (18 Jan 2026)

Reviewer #1

R1-O1: In this article, the author proposes a RAG-based pipeline named MedDiscover for metabolomics question answering. The author claims that SemiDevKG supports question answering through a GraphRAG pipeline. However, the paper suffers from major issues in methodological novelty and dataset size (only 10 articles are considered, limited to 3 metabolic disorders). Moreover, the proposed RAG application could be applied to any dataset; it does not showcase features specific to metabolomics.

Response: We agree the pipeline is not metabolomics-unique as an architecture. We therefore reposition MedDiscover as a metabolomics-instantiated benchmark and evaluation pipeline, and we define metabolomics-specificity in terms of the corpus, question construction, and evaluation constraints rather than algorithmic novelty.

Manuscript: Retrieval-augmented generation (RAG) can improve biomedical question answering, but claims about domain-specific retrievers require transparent, reproducible evaluation. We present MedDiscover, an open-source RAG implementation and benchmark instantiated for metabolomics and metabolic-disorder literature, with a two-tier dataset

design: a Gold set of 10 papers and 30 expert-curated question-answer pairs with traceable references, and a Silver set of 100 papers curated using ICD-10 metabolic disorder codes E70-E88 with approx. 600 synthetic QA pairs and retrievability metadata (Ada n=300, MedCPT n=300). On the Gold set, retrieval augmentation improves grounding compared with a non-RAG baseline. On the Silver benchmark, MedCPT and Ada show comparable retrieval-centric performance (faithfulness p=0.85, context recall p=0.50, context precision p=0.49, answer relevancy p=0.82), while MedCPT yields higher answer correctness (Mann-Whitney U p=5.72x10⁻¹¹, Cliff's delta=0.303; common-language effect =0.651). We release code, evaluation scripts, and document lists to support reproducibility.

R1-O2: Overall, the contribution appears weak and not suitable for publication in a scientific journal. Therefore, I recommend rejection of the paper.

Response: We respectfully note that the revision substantially changes the study scope and evaluation rigor. The updated manuscript addresses novelty, scale, and fairness concerns and provides a clearer, evidence-grounded contribution.

Manuscript: We implement MedDiscover as an open-source RAG evaluation pipeline and apply it to metabolomics/metabolic-disorder literature using a two-tier dataset (Gold and Silver). The primary comparisons fix the decoder (GPT-4o) to isolate retrieval effects, while decoder upgrades (GPT-4.1-nano/mini) are reported as sensitivity analyses. NotebookLM is included as a black-box comparator for answer style, with grounding metrics reported only where retrieved contexts are available.

Major Comments

R1-M1: Graphical abstract: Too simplistic. It is unclear how the retrieval and evidence-based answering steps are implemented (e.g., prompts are missing).

Response: We agree and have revised the pipeline figure to show retrieval, evidence grounding, and the Gold/Silver split. We now also provide the exact prompt templates in a dedicated appendix (Prompt Templates) to make the retrieval and evidence-based answering steps fully transparent.

Manuscript: MedDiscover pipeline and benchmark design. A metabolomics/metabolic-disorder corpus (ICD-10 E70-E88) is ingested, cleaned, chunked (500 tokens with 50-token overlap), and indexed with two embedding configurations (MedCPT and Ada-002) using FAISS. Evaluation is organized into a Gold set (expert-curated QA with traceable references) and a Silver set (synthetic QA with retrievability metadata). RAG inference retrieves top-k contexts and uses an evidence-grounded prompt to produce answers with explicit supporting context; evaluation reports reference-based metrics on Gold and retrieval-grounding metrics plus retrievability statistics on Silver.

Prompt templates

Evidence-grounded answering prompt (MedDiscover RAG; GPT-4o)

[SYSTEM]

You are a biomedical assistant. Answer the question using ONLY the provided context. If the context does not contain the answer, reply: "Insufficient evidence in the provided context."

[USER]

Question: {question}

Context:

{top_k_context_chunks}

Instructions:

- Provide a concise answer (max 25 tokens).
- Prefer specific entities (metabolites, diseases, study descriptors).
- Do not add unsupported claims.
- If multiple findings exist, report the one directly supported by the context.

Synthetic QA generation prompt (Silver set; gpt-4.1-mini)

[SYSTEM]

You generate biomedical questions and short answers from scientific text excerpts.

[USER]

Given the excerpt below, generate exactly 3 diverse questions that are answerable from the excerpt. Prefer non-trivial questions when possible.

Excerpt:

{chunk_or_grouped_context}

Output format (one per line):

Q: ...

A: ...

R1-M2: Title/Name: MedDiscover is a misleading name since the pipeline is not specifically tailored to the metabolomics domain.

Response: We revised the title and framing to avoid claiming a metabolomics-only architecture. The paper now presents MedDiscover as a reproducible benchmark and evaluation pipeline instantiated for metabolomics/metabolic-disorder literature.

Manuscript: [MedDiscover: Two-tier benchmark for retrieval-augmented metabolomics question answering in metabolic disorders](#)

This paper introduces MedDiscover, an open-source RAG implementation and evaluation

benchmark instantiated for metabolomics literature. Rather than proposing a new retrieval architecture, we focus on reproducible assessment of retrieval augmentation, embedding choice, and decoder sensitivity within a metabolomics/metabolic-disorder corpus. Metabolomics, pivotal for biomarker discovery, disease diagnosis, and personalized medicine, faces significant literature management challenges due to its rapidly expanding knowledge base.

R1-M3: Model choice: Why was gpt4o used instead of other LLMs?

Response: We clarified the model-choice rationale and fixed GPT-4o as the primary decoder for fair comparisons; GPT-4.1 variants are treated as sensitivity analyses.

Manuscript: We implement MedDiscover as an open-source RAG evaluation pipeline and apply it to metabolomics/metabolic-disorder literature using a two-tier dataset (Gold and Silver). The primary comparisons fix the decoder (GPT-4o) to isolate retrieval effects, while decoder upgrades (GPT-4.1-nano/mini) are reported as sensitivity analyses. NotebookLM is included as a black-box comparator for answer style, with grounding metrics reported only where retrieved contexts are available.

The evaluation compares three primary strategies: (1) domain-specific embeddings (MedCPT) with a GPT-4o decoder, (2) general-purpose embeddings (OpenAI Ada-002) with a GPT-4o decoder, and (3) a non-RAG baseline (GPT-4o without retrieval). The choice of MedCPT, Ada, and GPT-4o was guided by an extensive literature review, which identified MedCPT as the only published, publicly available domain-specific embedding model with validated superior performance. Other alternatives (e.g., BioGPT, SciPhi) were either not available for standalone use or did not surpass MedCPT in comparative studies. Ada and GPT-4o represent widely used general-purpose embedding and generation baselines, enabling a controlled comparison.

R1-M4: Section 2.3: The chunking phase as described in the article seems domain-specific, not something general to RA.

Response: We revised the chunking description to emphasize standard RAG hyperparameters (500-token windows, 50-token overlap) rather than a domain-specific procedure.

Manuscript: The chunking process uses overlapping chunks of 500 tokens with a 50-token overlap, a standard RAG setting selected based on prevailing practices in recent biomedical studies (e.g., RDGuru, AsthmaBot, EARA) and to preserve entity co-occurrence across chunk boundaries. This chunk size is compatible with the context windows of current LLMs and balances retrieval precision with contextual integrity. For documents exceeding the LLM token limit, multiple overlapping chunks were generated and, during retrieval, the top-ranked segments were concatenated or batched as input context for the decoder LLM.

R1-M5: Figure 1: Comparative approaches should not be injected into this figure; the focus should remain on the proposed approach.

Response: We separated the architecture figure from the experimental design. Figure 1 now focuses only on the MedDiscover pipeline and dataset design; comparative evaluations are described in the study-design figure and Results.

Manuscript: [MedDiscover pipeline and benchmark design](#). A metabolomics/metabolic-disorder corpus (ICD-10 E70-E88) is ingested, cleaned, chunked (500 tokens with 50-token overlap), and indexed with two embedding configurations (MedCPT and Ada-002) using FAISS. Evaluation is organized into a Gold set (expert-curated QA with traceable references) and a Silver set (synthetic QA with retrievability metadata). RAG inference retrieves top-k contexts and uses an evidence-grounded prompt to produce answers with explicit supporting context; evaluation reports reference-based metrics on Gold and retrieval-grounding metrics plus retrievability statistics on Silver. Experimental design overview. Primary tests fix the decoder (GPT-4o) to isolate retrieval effects (RAG vs non-RAG on the Gold set; MedCPT vs Ada at scale on the Silver benchmark). Secondary analyses include decoder sensitivity (GPT-4o vs GPT-4.1 variants on Gold) and retrievability quality-control analyses on Silver (hit-rate and rank-based ablations).

R1-M6: Section 3.1: It is unclear whether the focus is on metabolomics in general or specifically on metabolic diseases/disorders.

Response: We clarified the scope as metabolomics literature in the context of metabolic disorders, explicitly referencing ICD-10 E70-E88 in the Introduction and Data Collection sections.

Manuscript: [We focus on metabolomics literature in the context of metabolic disorders \(ICD-10 E70-E88\) because these conditions are metabolite-centric and representative of biomarker-driven metabolomics workflows](#). We assembled a metabolomics/metabolic-disorder corpus of 100 papers published between 2014 and 2024 from PubMed and Google Scholar, curated using ICD-10 metabolic disorder codes E70-E88. Text extraction and cleaning were performed by extracting raw text from PDFs using PyPDF2 and removing headers, footers, and non-informative formatting. Numeric and scientifically relevant data (metabolite concentrations, p-values) were preserved, and each processed article retained original paragraph sequences with metadata identifiers mapping back to the source PDF.

R1-M7: Section 3.6: This section should be moved earlier, and the ROUGE metric should be formally defined. What does E represent in the formulae?

Response: We revised the Evaluation Metrics section to define ROUGE formally and to define the embedding function used in semantic metrics, removing ambiguity around notation.

Manuscript: Performance evaluation incorporated both generation-centric and retrieval-centric metrics, applied to MedDiscover configurations and, where appropriate, to NotebookLM. Let $\phi(\cdot)$ denote the embedding function used for semantic similarity metrics. Generation Metrics: Accuracy, Precision, Recall, and F1 Score quantitatively assessed the correctness and comprehensiveness of generated responses. ROUGE-1/2/L measure unigram overlap, bigram overlap, and longest-common-subsequence overlap between candidate and reference answers. BERTScore captures semantic similarity:
$$F1\ Score = 2 \times (Precision \times Recall / (Precision + Recall))$$
$$BERTScore = CosSim(\phi(\text{generated}), \phi(\text{reference}))$$
Retrieval Metrics (Primarily for RAG systems like MedDiscover): Context Precision and Context Recall assessed the relevance and completeness of retrieved information, while Faithfulness measured factual grounding of generated content:
$$Context\ Precision@K = (\text{Relevant Retrieved Contexts} / K)$$
$$Context\ Recall = (\text{Relevant Retrieved Contexts} / \text{Total Relevant Contexts})$$
$$Faithfulness = (\text{Supported Claims in Generated Answer} / \text{Total Claims in Generated Answer})$$
Additionally, Answer Relevancy quantified semantic alignment between queries and responses:
$$Answer\ Relevancy = (1 / N) \sum_{i=1}^N \cos(\phi(\text{answer}_i), \phi(\text{query}_i))$$
We compute RAGAS metrics (faithfulness, answer correctness, context recall, context precision, answer relevancy) using the RAGAS framework and report these where retrieved contexts are available. ROUGE and BLEU are reported only for the Gold set because the Silver set lacks human reference answers. For NotebookLM, only metrics not dependent on explicit retrieved context (like Answer Relevancy and generation metrics) are fully comparable.

R1-M8: Results: It seems that NotebookLM performs best in terms of accuracy. Why should we prefer your approach? This is not convincingly discussed.

Response: We clarified that NotebookLM does not expose retrieved contexts, so grounding metrics are not comparable. The Discussion now explains why transparent evidence grounding is preferred for biomedical use cases.

Manuscript: [NotebookLM comparison](#)
NotebookLM shows strong lexical overlap, especially when prompted for minimal output, but it does not expose retrieved contexts. As a result, grounding metrics (context precision/recall, faithfulness) cannot be computed for NotebookLM, limiting its comparability in evidence-critical settings.

R1-M9: Figure 2: please improve readability.

Response: We improved readability by replacing the dense multi-model plot in the main text with a clear study-design figure and moving the full multi-model plot to the Supplementary section; Silver results are summarized in a table with CIs and effect sizes.

Manuscript: Experimental design overview. Primary tests fix the decoder (GPT-4o) to isolate retrieval effects (RAG vs non-RAG on the Gold set; MedCPT vs Ada at scale on the Silver benchmark). Secondary analyses include decoder sensitivity (GPT-4o vs GPT-4.1 variants on Gold) and retrievability quality-control analyses on Silver (hit-rate and rank-based ablations).

Supplementary Gold-set results across additional configurations (decoder sensitivity variants and black-box comparator). This figure is provided for completeness; the primary comparisons in the main text restrict to fixed-decoder (GPT-4o) evaluations to avoid conflating retriever effects with decoder capacity. Note that NotebookLM does not expose retrieved contexts; context-dependent metrics are therefore not comparable and should be interpreted as not available rather than as zero performance.

R1-M10: Section 4.4: The last paragraph is unclear and difficult to read.

Response: We rewrote the Discussion for clarity and structure, focusing on three defensible claims aligned with the revised benchmark framing.

Manuscript: This study evaluates MedDiscover as a reproducible RAG benchmark with two datasets: a Gold set (expert-curated) and a Silver set (retrievability-filtered synthetic). On the Gold set, retrieval augmentation improves grounding relative to a non-RAG baseline when the decoder is held constant. When the decoder is fixed at GPT-4o, differences between MedCPT and Ada are modest and not consistently significant across metrics. On the Silver benchmark, embedding choice produces comparable retrieval-centric metrics, while MedCPT yields a statistically significant improvement in answer correctness (Table). This indicates that domain-specific embeddings can provide a targeted gain in answer quality, but not a universal advantage across all grounding metrics.

R1-M11: Section 5.3: The discussion is verbose and lacks clarity.

Response: Addressed via the same Discussion rewrite, with reduced verbosity and clearer separation of primary vs sensitivity analyses.

Manuscript: NotebookLM provides strong lexical alignment under minimal-output prompting, but its retrieval mechanism is opaque. Because retrieved contexts are not exposed, grounding metrics cannot be computed for NotebookLM, limiting its suitability for evidence-critical biomedical workflows.

All MedDiscover runs restricted GPT-4o to 25 generation tokens to control verbosity and encourage precision. This design choice likely depressed length-sensitive scores such as

ROUGE and BLEU relative to longer responses, underscoring the need to interpret lexical overlap metrics cautiously in short-answer biomedical settings.

R1-M12: Dataset size: Only 10 articles and 3 diseases are included, which is insufficient for a scientific journal publication.

Response: We expanded the dataset using a 100-paper Silver benchmark with ~600 synthetic QA pairs while retaining the Gold expert-curated set; the dataset summary table and Silver results address scale and generalization.

Manuscript: [Gold QA set \(expert-curated\)](#)

The Gold set consists of ten metabolomics papers focusing on Gaucher disease (Type 1), Nonalcoholic Fatty Liver Disease (NAFLD), and Type 1 diabetes, with 30 expert-curated questions and traceable reference answers. This set is used to validate answer quality against human references. The full Gold set contains 30 expert-curated QA pairs. For cross-system comparisons that require outputs from every evaluated configuration (including black-box comparators), we report results on the subset of 20 Gold questions for which all model outputs are available (evaluation subset; see Section).

Silver QA set (synthetic, retrievability-filtered)

The Silver set uses the 100-paper corpus and approx. 600 synthetic QA pairs generated with retrievability metadata (Ada n=300, MedCPT n=300). This set is used to stress-test retrieval grounding at scale rather than to provide human reference answers. We refer to this collection as a Silver set because the questions and reference answers are machine-generated and only partially filtered: each QA pair is retained based on lightweight quality controls (Section), including a retrievability check, but the content is not exhaustively expert-validated. Accordingly, Silver-set results are interpreted primarily as evidence about retrieval grounding and robustness at scale, rather than as definitive clinical correctness.

Minor Comments

R1-m1: "RAG-LLM" is redundant; use simply "RAG."

Response: Implemented. We use the term RAG consistently throughout the manuscript.

Manuscript: [Background: RAG in Metabolomics](#)
[Metabolomics Research: Relevance and Challenges](#)

R1-m2: Inappropriate use of the opening quotation mark (").

Response: Corrected in the revised manuscript; we performed a pass to normalize quotation marks.

Manuscript: RAG's core mechanism involves encoding a user query—e.g., "What metabolites are associated with early-stage Alzheimer's disease?"—into a dense vector embedding using models like MedCPT or BioBERT. This embedding, capturing semantic intent, queries a vector database of pre-processed documents (e.g., PubMed articles). Retrieved text chunks augment the query, enabling the LLM to generate evidence-based responses. Embeddings, built from tokenized text (as discussed in the Background section), are indexed in vector databases like FAISS or Qdrant, using algorithms like HNSW (Hierarchical Navigable Small Worlds) for efficient similarity searches.

R1-m3: Missing spaces after several periods.

Response: Corrected with targeted copy edits and spacing fixes; we will do a final proofreading pass on the clean manuscript.

Manuscript: Detailed methodologies for data collection, preparation, the RAG pipeline, and evaluation metrics are described in the following subsections. Further experimental setups and extensive supplementary results can be found in the Supplementary Material, ensuring transparency, reproducibility, and facilitating further exploration of our methodology.

R1-m4: References should be ordered numerically, e.g., [12,19] --> [19,12].

Response: Implemented by enabling numeric sorting/compression in the bibliography options.

Manuscript: `biboptions{sort&compress}`

Reviewer #2

R2-C1: This is a well-written manuscript that represents a novel contribution in the area of LLM RAG systems for biomedical (in particular, metabolomics) research; it introduces a new domain-specific LLM RAG framework and compares it to multiple other related frameworks on a curated metabolomics query set. The study also contributes to general understanding of the benefits of RAG (whether general or domain-specific) in biomedical question answering.

Response: Thank you for the positive assessment.

Manuscript: Our contributions are threefold:

- A two-tier benchmark: a Gold set with expert-curated QA pairs and a Silver set with

retrievability-filtered synthetic QA pairs at scale.

- Transparent RAG evaluation with retrieval-centric metrics (RAGAS) and statistical testing that reports effect sizes and confidence intervals.
- An open-source reproducibility package (code, prompts, document lists, and evaluation pipeline) that enables independent re-analysis.

R2-C2: That said, I think some of the claims may be misleading - specifically, in the Abstract and elsewhere, the authors report that MedDiscover with a GPT-4.1 decoder outperformed the general-purpose RAG comparator (which was Ada with a GPT-4o decoder). While this is true, it is misleading to present as the primary finding given that - when holding the decoder constant and using GPT-4o for both MedDiscover and Ada - MedDiscover did not significantly outperform Ada. And using a GPT-4.1 decoder provided a big bump in performance for MedDiscover (suggesting GPT-4.1 may be much better suited than GPT-4o for the questions tested), yet was not evaluated for the comparators, making it impossible to determine whether performance gains stem from the domain-specific embeddings or simply from using a more capable decoder. Given this, I don't think the results are sufficient to conclude that "domain-adapted embeddings significantly improve retrieval precision, factual grounding, and answer correctness".

Response: We agree. The revised manuscript avoids decoder-confounded claims: primary comparisons fix GPT-4o, while GPT-4.1 variants are explicitly labeled as sensitivity analyses, and the Abstract/Conclusion were updated accordingly. Primary conclusion statements are now based only on fixed-decoder (GPT-4o) comparisons; decoder upgrades are treated strictly as sensitivity analyses and are not used to support embedding superiority claims.

Manuscript: We implement MedDiscover as an open-source RAG evaluation pipeline and apply it to metabolomics/metabolic-disorder literature using a two-tier dataset (Gold and Silver). The primary comparisons fix the decoder (GPT-4o) to isolate retrieval effects, while decoder upgrades (GPT-4.1-nano/mini) are reported as sensitivity analyses. NotebookLM is included as a black-box comparator for answer style, with grounding metrics reported only where retrieved contexts are available.

Results on the Gold set (expert curated)

We evaluated an expert-curated Gold set of 30 QA pairs; results reported here use the 20-question evaluation subset for which outputs are available across all compared configurations. Primary comparisons fix the decoder (GPT-4o) to isolate retrieval effects across the non-RAG baseline, Ada + GPT-4o, and MedCPT + GPT-4o. Supplementary Figure summarizes LLM-specific and retrieval-specific metrics for these models, along with NotebookLM and the MedCPT decoder sensitivity variants.

R2-C3: While it would be ideal if the authors evaluated the comparators with GPT-4.1 as well, I don't think it's required for this manuscript to still be a novel contribution to

research. However, if not doing further experiments, the authors should be much more careful with their presentation of results and claims. In addition to revising claims/conclusions reported throughout (e.g., Abstract lines 11-16 and 19-22; Conclusion lines 2-7 and 13-15), I would suggest excluding the MedDiscover GPT-4.1 results from the primary results figure (Figure 2), and then only later in the Results section including a second version of that figure that includes the MedDiscover GPT-4.1 results, clearly noting that it was a supplementary analysis without a fair comparator.

Response: Implemented. GPT-4.1 results were moved to the Supplementary section and are not used for primary conclusions. Primary conclusions are based only on fixed-decoder (GPT-4o) comparisons; decoder upgrades are treated strictly as sensitivity analyses and do not support embedding-superiority claims.

Manuscript: [Experimental design overview](#). Primary tests fix the decoder (GPT-4o) to isolate retrieval effects (RAG vs non-RAG on the Gold set; MedCPT vs Ada at scale on the Silver benchmark). Secondary analyses include decoder sensitivity (GPT-4o vs GPT-4.1 variants on Gold) and retrievability quality-control analyses on Silver (hit-rate and rank-based ablations).

[Supplementary Gold-set results across additional configurations \(decoder sensitivity variants and black-box comparator\)](#). This figure is provided for completeness; the primary comparisons in the main text restrict to fixed-decoder (GPT-4o) evaluations to avoid conflating retriever effects with decoder capacity. Note that NotebookLM does not expose retrieved contexts; context-dependent metrics are therefore not comparable and should be interpreted as not available rather than as zero performance.

R2-C4: One final minor comment - on page 23 of the PDF, in the sentence "By contrast, upgrading the decoder within the MedCPT pipeline from GPT-4 to GPT-4.1-nano and further to GPT-4.1-mini..." - I'd suggest changing the first "GPT-4" to "GPT-4o" for clarity.

Response: Corrected throughout the manuscript.

Manuscript: [We implement MedDiscover as an open-source RAG evaluation pipeline and apply it to metabolomics/metabolic-disorder literature using a two-tier dataset \(Gold and Silver\)](#). The primary comparisons fix the decoder (GPT-4o) to isolate retrieval effects, while decoder upgrades (GPT-4.1-nano/mini) are reported as sensitivity analyses. NotebookLM is included as a black-box comparator for answer style, with grounding metrics reported only where retrieved contexts are available.

Reviewer #3

R3-O1: This manuscript presents MedDiscover, a Retrieval-Augmented Generation (RAG) framework tailored to metabolomics knowledge extraction. However, there are methodological and reporting issues that should be addressed before acceptance.

Response: Thank you for the careful review. We revised the manuscript to address the methodological and reporting issues and to tighten the study framing and evaluation rigor.

Manuscript: We implement MedDiscover as an open-source RAG evaluation pipeline and apply it to metabolomics/metabolic-disorder literature using a two-tier dataset (Gold and Silver). The primary comparisons fix the decoder (GPT-4o) to isolate retrieval effects, while decoder upgrades (GPT-4.1-nano/mini) are reported as sensitivity analyses. NotebookLM is included as a black-box comparator for answer style, with grounding metrics reported only where retrieved contexts are available.

Results on the Gold set (expert curated)

We evaluated an expert-curated Gold set of 30 QA pairs; results reported here use the 20-question evaluation subset for which outputs are available across all compared configurations. Primary comparisons fix the decoder (GPT-4o) to isolate retrieval effects across the non-RAG baseline, Ada + GPT-4o, and MedCPT + GPT-4o. Supplementary Figure summarizes LLM-specific and retrieval-specific metrics for these models, along with NotebookLM and the MedCPT decoder sensitivity variants.

Major Comments

R3-M1: 1. Experimental design and dataset limitations

The study evaluates MedDiscover on only 30 question–answer pairs derived from 10 metabolomics papers across three diseases. This sample is too small for robust statistical inference in high-variance LLM evaluations. The paper acknowledges this but should:

Quantify statistical power and confidence intervals;

Clarify the manual curation process and whether inter-annotator agreement or expert validation of the reference answers was performed;

Discuss possible selection bias due to the limited corpus and disease scope.

Suggestion: Expand the dataset or validate the model on an external corpus to support generalization claims.

Response: We expanded the evaluation with a 100-paper Silver benchmark and kept the Gold set for reference-based validation. We added bootstrap confidence intervals and effect sizes for the Silver results and clarified selection bias and scope limitations in the Limitations section.

Manuscript: The Gold set consists of ten metabolomics papers focusing on Gaucher disease (Type 1), Nonalcoholic Fatty Liver Disease (NAFLD), and Type 1 diabetes, with 30 expert-curated questions and traceable reference answers. This set is used to validate answer quality against human references. The full Gold set contains 30 expert-curated QA pairs. For cross-system comparisons that require outputs from every evaluated

configuration (including black-box comparators), we report results on the subset of 20 Gold questions for which all model outputs are available (evaluation subset; see Section).

Silver QA set (synthetic, retrievability-filtered)

The Silver set uses the 100-paper corpus and approx. 600 synthetic QA pairs generated with retrievability metadata (Ada n=300, MedCPT n=300). This set is used to stress-test retrieval grounding at scale rather than to provide human reference answers. We refer to this collection as a Silver set because the questions and reference answers are machine-generated and only partially filtered: each QA pair is retained based on lightweight quality controls (Section), including a retrievability check, but the content is not exhaustively expert-validated. Accordingly, Silver-set results are interpreted primarily as evidence about retrieval grounding and robustness at scale, rather than as definitive clinical correctness.

This study demonstrates the potential of RAG frameworks for metabolomic biomarker discovery, but several important limitations must be acknowledged. The Gold set is limited to ten English articles spanning three diseases (Gaucher disease Type 1, NAFLD, and Type 1 diabetes), which constrains the strength of reference-based conclusions. The Silver set expands coverage to 100 papers, but the scope remains restricted to metabolic-disorder literature (ICD-10 E70-E88), limiting generalizability to other biomedical domains. A larger and more diverse corpus would provide a more representative evaluation of the framework's scalability and applicability across a wider range of conditions.

Gold and Silver sets differ in both scope and construction: the Gold set is expert-curated over three disease-focused topics, whereas the Silver set spans ICD-10 E70-E88 metabolic-disorder literature and uses machine-generated QA with partial filtering. This introduces selection and distribution shift between tiers; consequently, we do not interpret absolute metric values as directly comparable across Gold and Silver, and we use each tier to support different validity claims (reference-based validation vs retrieval-grounding stress testing).

R3-M2: 2. Evaluation methodology and metric validity

The use of ROUGE/BLEU for concise biomedical answers (≤ 25 tokens) may underestimate semantic quality and exaggerate differences in output style.

Metrics like Faithfulness and Context Precision rely on author-defined definitions rather than standardized formulations.

Suggestion: Include semantic similarity or LLM-as-a-Judge evaluation calibrated by biomedical experts. Report confidence intervals or bootstrap variance for all metrics.

Response: We now limit ROUGE/BLEU reporting to the Gold set and use standard RAGAS metrics for the Silver benchmark. For Silver, we describe answer correctness as a silver-standard answer-correctness score (i.e., an answerability/grounding proxy rather than expert-validated clinical correctness), and we report confidence intervals for the Silver results.

Manuscript: We compute RAGAS metrics (faithfulness, answer correctness, context recall, context precision, answer relevancy) using the RAGAS framework and report these where retrieved contexts are available. ROUGE and BLEU are reported only for the Gold set because the Silver set lacks human reference answers. For NotebookLM, only metrics not dependent on explicit retrieved context (like Answer Relevancy and generation metrics) are fully comparable.

Because the Silver benchmark uses synthetic reference answers that are identical to the responses in the evaluation files, answer-quality metrics should be interpreted as proxy measures of answerability and grounding rather than as expert-validated clinical correctness.

R3-M3: 3. Statistical analysis

The authors correctly apply non-parametric tests (Kruskal-Wallis + Dunn), but there is no correction for multiple testing across metrics, beyond Bonferroni within metric.

Small-sample effects may inflate p-values or significance of large differences.

Suggestion: Include effect sizes alongside p-values to quantify practical importance. Add a summary table of test statistics.

Response: We added Holm-Bonferroni correction across the primary metric family, and report Mann-Whitney U with Cliff's delta and bootstrap CIs in the Silver results table.

Manuscript: Unless otherwise stated, we treat the retrieval-centric RAGAS metrics (faithfulness, answer correctness, context recall, context precision, answer relevancy) as a primary metric family and control family-wise error across this family using Holm-Bonferroni adjustment. Metrics outside this family are treated as secondary and interpreted cautiously.

All model-by-metric score distributions were subjected to the Shapiro-Wilk test; almost every distribution departed from normality ($p < 0.05$). A subsequent Levene test revealed unequal variances for several evaluation metrics. In view of these results, groupwise comparisons were carried out with rank-based, non-parametric methods.

For the Gold set, a Kruskal-Wallis one-way ANOVA on ranks was computed across the model configurations, followed by Dunn's post-hoc tests with Ada + GPT-4o as the reference group. Holm-Bonferroni correction was applied across the primary metric family (faithfulness, answer correctness, context recall, context precision, answer relevancy). Supplementary Figure reports adjusted contrasts.

For the Silver benchmark, we report Mann-Whitney U tests for Ada vs MedCPT on each RAGAS metric, along with Cliff's delta effect sizes and bootstrap 95% confidence intervals for mean scores (2,000 resamples; Table). This isolates embedding effects under a fixed decoder and provides a scale-appropriate estimate of practical impact.

R3-M4: 4. Novelty and comparison to prior RAG frameworks

While the paper situates MedDiscover relative to BioRAG, DRAGON, and Self-BioRAG, it does not clearly define what is technically novel beyond domain restriction (metabolomics) and embedding choice.

Suggestion: Clarify whether MedDiscover introduces a new retrieval strategy, interface design, or evaluation framework, and distinguish these from prior MedCPT-based pipelines.

Response: We clarified that the contribution is a two-tier benchmark and evaluation pipeline rather than a new retrieval algorithm, and positioned MedDiscover accordingly in the Introduction and Methods.

Manuscript: This paper introduces MedDiscover, an open-source RAG implementation and evaluation benchmark instantiated for metabolomics literature. Rather than proposing a new retrieval architecture, we focus on reproducible assessment of retrieval augmentation, embedding choice, and decoder sensitivity within a metabolomics/metabolic-disorder corpus. Metabolomics, pivotal for biomarker discovery, disease diagnosis, and personalized medicine, faces significant literature management challenges due to its rapidly expanding knowledge base.

RAG frameworks were introduced to bridge this gap by injecting up-to-date evidence at inference time. Systems such as BIORAG, RAG², and Self-BioRAG have since added multi-stage retrieval pipelines, self-evaluation loops, and knowledge-graph conditioning to cope with the scale and dynamism of biomedical knowledge. MedDiscover follows this lineage but narrows the focus to metabolomics, a sub-discipline that has received comparatively little attention in prior RAG work. By anchoring retrieval to experimentally validated metabolite literature rather than broader biomedical text, our study illustrates how increased domain granularity can sharpen factual grounding.

Retrieval Architectures: Similarities and Divergences

Compared with earlier frameworks, MedDiscover adopts a deliberately lightweight retrieval architecture: a single-stage dense lookup over FAISS indices produced by MedCPT or Ada-002 embeddings. DRAGON and RAG², by contrast, implement multi-hop or multi-stage retrieval. DRAGON fuses knowledge graphs with text through a cross-modal encoder, while RAG² first rewrites the query into a rationale, then balances evidence across PubMed, PMC, textbooks and guidelines, and finally reranks with a cross-encoder. BioRAG extends the concept still further by coupling MeSH-filtered hierarchical look-ups with external search-engine calls and a self-evaluation module that decides when additional retrieval is needed.

Although these more elaborate pipelines improve performance on broad medical QA benchmarks, our results indicate that a streamlined, single-stage design can suffice for narrowly scoped, high-precision tasks. MedCPT-based retrieval delivers competitive context precision and faithfulness on the Gold set while maintaining lower latency and a smaller memory footprint. This finding implies that retrieval complexity should be matched to the breadth of the information need: dense, domain-specific embeddings may already capture enough local structure when the target corpus is well circumscribed.

R3-M5: 5. Transparency, reproducibility, and open science

The GitHub repository is cited, but the paper should specify:

What data and scripts are available (PDF text, embeddings, evaluation code);

Licensing and reproducibility (random seeds, hardware, API versions);

Whether PubMed content redistribution complies with copyright.

Suggestion: Add a “Reproducibility Checklist” table (dataset availability, model weights, code, and statistical analysis scripts).

Response: We added a reproducibility checklist table that specifies available data, scripts, model/config versions, and evaluation artifacts, and we clarified data availability in the Data and Code Availability section.

Manuscript: [Reproducibility checklist \(artifacts and where to find them\)](#).

[Data and Code Availability](#)

The MedDiscover framework is available as open-source software at

<https://github.com/VatsalPatel18/MedDiscover>. The analysis pipeline, evaluation outputs, and figure-generation scripts used in this manuscript are available at

<https://github.com/VatsalPatel18/MedDiscover-Analysis>.

Due to copyright and licensing constraints, we do not redistribute publisher PDFs or extracted full-text derived from closed-access articles. Instead, we release the list of PubMed IDs/DOIs used to assemble the corpus and provide scripts to reproduce preprocessing and indexing from documents that the user can legally access (e.g., open-access sources).

Released derived artifacts (Gold QA pairs, Silver QA pairs, retrieval metadata, and evaluation tables) are provided without redistributing publisher PDF files.

R3-M6: 6. Clinical and ethical considerations

The discussion briefly mentions regulatory readiness (GDPR, CE/FDA) but lacks concrete exploration of bias, interpretability, or safety in biomedical deployment.

Suggestion: Expand Section 5.4 to include ethical implications (traceability, hallucination risk, domain shift) and planned safeguards.

Response: We expanded the Limitations and Discussion to address traceability, hallucination risk, domain shift, and deployment constraints for biomedical use.

Manuscript: [The indexing and retrieval processes used in this study were static, with no mechanisms for continuous updates or adaptive refinement. Biomedical research is a rapidly evolving field, and the inability of the framework to incorporate new findings or adjust to user feedback limits its relevance and longevity. Furthermore, this study did not sufficiently explore real-world usability concerns, such as providing confidence scores, highlighting retrieved documents, or offering transparent explanations of how generated responses were synthesized. These features are critical for building trust and fostering adoption among clinicians and researchers who depend on reliability and interpretability.](#)

From an engineering perspective, computational efficiency and cost were not rigorously profiled. The study relied on commercial GPT-4o decoding and two embedding models without accounting for the latency or financial burden that would accrue in large-scale or real-time deployments. For institutions with limited computational budgets—or for privacy-sensitive environments that cannot rely on third-party APIs—these cost and infrastructure questions are pivotal.

In addition to the previously discussed limitations, several further aspects warrant critical consideration. First, the current evaluation strategy relied exclusively on automated metrics (e.g., ROUGE, BLEU, BERTScore, Faithfulness), without incorporating structured human-in-the-loop assessments or inter-rater agreement. This absence of qualitative expert judgment may obscure important nuances in factual accuracy and biomedical relevance, especially in domains where minor terminological shifts (e.g., “reduced” vs. “unchanged”) can meaningfully alter interpretation. Second, all generative outputs were produced using variants of GPT-4 (GPT-4o or GPT-4.1-mini), thereby limiting insights into decoder variability. Given that different LLM architectures exhibit distinct generation behaviors, a broader decoder comparison, including open-source biomedical models, would be essential for generalizing the framework's applicability across institutional settings. Third, the study did not develop a formal taxonomy of observed error types. While macro-level metrics suggested performance differences between domain-specific and general-purpose pipelines, it remains unclear whether these stemmed from hallucinations, factual omissions, semantic contradiction, or ambiguous phrasing. A structured error analysis would help isolate these causes and guide future improvements. Fourth, while the QA dataset was intentionally designed for clinical specificity and scientific rigor, it does not capture the breadth or ambiguity of real-world information-seeking behavior by clinicians, biologists, or translational researchers. Incorporating user-centric evaluations, log analyses, or exploratory query types could improve the system's ecological validity. Lastly, although the framework holds promise for biomedical support tasks, its ethical, regulatory, and operational readiness for clinical deployment remains unassessed. Issues such as GDPR compliance, CE/FDA documentation traceability, and robust audit trails for generated content were beyond the current scope but are indispensable for translation into regulated medical environments.

R3-M7: 7. Typographical issues

Grammatical inconsistencies (“nove” instead of “novel” on p. 34).

I would like to offer a strong recommendation to the authors to take steps towards enhancing the clarity of their manuscript and making revisions to improve the overall quality of the language and sentence structure. In order to achieve this, it is highly advisable that the authors consider arranging for a comprehensive proofreading of the manuscript. This will not only contribute to a more polished and refined document but also ensure that the manuscript effectively conveys its intended message to its readers.

Response: We corrected typos and performed language cleanup, and we will complete a final proofreading pass before resubmission.

Manuscript: Coupling this capability with iterative re-ranking strategies and user-driven feedback loops would enable the system to retire outdated passages and elevate novel, high-quality evidence, ensuring that clinicians and researchers receive the most current insights available.

R3-M8: 8. Discussion

Consider discussing and properly citing related works such as:

<https://doi.org/10.1016/j.csbj.2024.10.017>

<https://doi.org/10.1016/j.csbj.2024.03.016>

<https://doi.org/10.1016/j.csbj.2024.08.016>

<https://doi.org/10.1016/j.csbj.2024.05.004>

<https://doi.org/10.1016/j.csbj.2023.11.011>

Response: We added and discussed the requested CSBJ-related works in the Background/Related Work sections.

Manuscript: Metabolomics Research: Relevance and Challenges

Metabolomics connects genetic and environmental influences with phenotypic outcomes through metabolites, acting as substrates, intermediates, and end products within biological pathways, thereby reflecting immediate physiological states and adaptations. Researchers often struggle with the complexity, terminology, and rapidly expanding literature, complicating efficient knowledge extraction and practical hypothesis generation. Traditional metabolomics databases require structured queries, limiting intuitive exploration, especially for researchers with limited domain expertise.

Beyond the rapid growth of the metabolomics literature, metabolomics datasets themselves are often high-dimensional and sparse (i.e., $p \gg n$), and can exhibit class- or feature-level imbalance, which complicates robust patient classification and biomarker discovery. Recent benchmarking work in metabolomics biomedical data systematically evaluated feature extraction and feature selection pipelines and reported that supervised feature selection can improve downstream classification performance, providing practical guidance on when to apply these techniques. In parallel, predictive biomedicine increasingly combines multiple data modalities (e.g., multi-omics measurements with imaging or genomic variation), further increasing the complexity of evidence synthesis and interpretation. Existing biomedical literature-mining tools such as BioTextQuest v2.0 support term-driven clustering and named-entity extraction from PubMed abstracts, but remain primarily query-and-browse systems rather than evidence-grounded natural-language question answering interfaces.

In biomedical NLP, these modelling choices must also be evaluated through the lens of transparency and interpretability. A recent scoping review in healthcare NLP highlights persistent gaps in best practices and systematic benchmarks for explainable and

interpretable deep learning, reinforcing the need for designs that support auditability and evidence traceability in high-stakes settings.

Knowledge-graph resources can further complement text retrieval by encoding metabolite-gene-disease relationships and enabling multi-hop evidence traversal. However, constructing and maintaining biomedical knowledge graphs at scale remains challenging. BioKGrapher provides an example of automated KG construction from PubMed IDs using NER/NEL with UMLS normalization, concept weighting and re-ranking, and ontology-driven relation building (e.g., via terminologies such as SNOMED CT and NCIt), illustrating a practical pathway for producing condition-specific graphs that could be coupled to graph-based retrieval in biomedical RAG pipelines.

Minor Comments

R3-m1: How do the researchers/clinicians can benefit from the work of this manuscript?

Response: We clarified the intended benefit as a reproducible, evidence-grounded benchmark and evaluation pipeline for metabolomics/metabolic-disorder QA.

Manuscript: Our contributions are threefold:

- A two-tier benchmark: a Gold set with expert-curated QA pairs and a Silver set with retrievability-filtered synthetic QA pairs at scale.
- Transparent RAG evaluation with retrieval-centric metrics (RAGAS) and statistical testing that reports effect sizes and confidence intervals.
- An open-source reproducibility package (code, prompts, document lists, and evaluation pipeline) that enables independent re-analysis.

This study presents MedDiscover as a reproducible RAG evaluation benchmark for metabolomics and metabolic-disorder literature. Using a two-tier design (Gold expert-curated QA and Silver retrievability-filtered synthetic QA), we show that retrieval augmentation improves grounding relative to non-RAG baselines, and that MedCPT vs Ada yields comparable retrieval metrics with a modest but significant gain in answer correctness on the Silver benchmark. The open-source pipeline and document lists support reproducibility and provide a foundation for expansion into broader biomedical domains and modalities.

R3-m2: For maximal benefit, the authors are encouraged to develop a user-friendly webserver or tool based on this work. Without such resources, this manuscript risks becoming part of the growing body of bioinformatics research that remains underutilized.

Response: We note user-facing tooling as future work and acknowledge that a webserver would improve accessibility and adoption.

Manuscript: Transparency is key to gain researchers' trust. Users should be able to inspect why a particular passage was retrieved and how it influenced the generated answer. Future work will therefore explore techniques for displaying source information with similarity scores alongside the retrieved text. Visualising token-level saliency could provide experts a deeper understanding of the logic of both the retriever and the generator.

MedDiscover: A Domain-Specific RAG Framework

Evidence-Grounded Knowledge Extraction in Metabolomics



Key Scientific Contribution

Validated that domain-specific MedCPT embeddings significantly outperform general-purpose retrievers in metabolite-disease association correctness through a novel two-tier benchmark evaluation.

OPEN SOURCE

[GitHub Repository](#)
[Benchmarking Code](#)

MedDiscover: A Domain-Specific Retrieval-Augmented Generation Framework for Evidence-Grounded Knowledge Extraction in Metabolomics

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Abstract

Retrieval-augmented generation (RAG) can improve biomedical question answering, but claims about domain-specific retrievers require transparent, reproducible evaluation. We present *MedDiscover*, an open-source RAG implementation and benchmark instantiated for metabolomics and metabolic-disorder literature, with a two-tier dataset design: a Gold set of 10 papers and 30 expert-curated question-answer pairs with traceable references, and a Silver set of 100 papers curated using ICD-10 metabolic disorder codes E70–E88 with ~600 synthetic QA pairs and retrievability metadata (Ada n=300, MedCPT n=300). On the Gold set, retrieval augmentation improves grounding compared with a non-RAG baseline. On the Silver benchmark, MedCPT and Ada show comparable retrieval-centric performance (faithfulness $p = 0.85$, context recall $p = 0.50$, context precision $p = 0.49$, answer relevancy $p = 0.82$), while MedCPT yields higher answer correctness (Mann–Whitney U $p = 5.72 \times 10^{-11}$, Cliff’s $\delta = 0.303$; common-language effect = 0.651). We release code, evaluation scripts, and document lists to support reproducibility.

Keywords: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Biomedical Research, Metabolomics, Knowledge Extraction

1. Introduction

Large Language Models (LLMs), such as GPT-3 and GPT-4, have significantly transformed natural language processing (NLP), showcasing advanced capabilities in tasks including summarization, dialogue systems, reasoning, and text generation [1]. Their capacity to handle vast amounts of unstructured textual data, generating contextually relevant and coherent responses, presents significant opportunities in biomedical research, especially for navigating the extensive and complex biomedical literature available in repositories like PubMed [2]. Nonetheless, deploying these general-purpose LLMs in specialized biomedical subfields, such as metabolomics, presents substantial limitations, notably the occurrence of hallucinations, outputs that seem plausible but are factually incorrect, and difficulty in accurately interpreting complex biomedical terminologies [3–5]. Furthermore, traditional LLMs are limited by static knowledge acquired during training, lacking dynamic integration of emerging, domain-specific data necessary in rapidly evolving fields like precision medicine and metabolomics.

Retrieval-Augmented Generation (RAG) methodologies have emerged to address these limitations by dynamically retrieving external knowledge and integrating it into the generative process. RAG significantly improves factual accuracy, transparency, and context-specificity of responses generated by LLMs [6]. Despite these advantages, optimizing RAG pipelines for domain-specific applications remains challenging. The quality and relevance of retrieved documents, efficiency of document ranking and filtering mechanisms, and the effective integration of retrieval signals into generative processes critically influence the overall performance of RAG-enhanced systems

[7]. Moreover, evaluating the reliability and interpretability of responses generated by RAG systems remains essential, particularly in biomedical contexts where decision-making depends on accurate, reliable, and well-sourced information.

This paper introduces *MedDiscover*, an open-source RAG implementation and evaluation benchmark instantiated for metabolomics literature. Rather than proposing a new retrieval architecture, we focus on reproducible assessment of retrieval augmentation, embedding choice, and decoder sensitivity within a metabolomics/metabolic-disorder corpus (see Appendix A). Metabolomics, pivotal for biomarker discovery, disease diagnosis, and personalized medicine, faces significant literature management challenges due to its rapidly expanding knowledge base [8, 9].

We focus on metabolomics literature in the context of metabolic disorders (ICD-10 E70–E88) because these conditions are metabolite-centric and representative of biomarker-driven metabolomics workflows.

Our contributions are threefold:

- A two-tier benchmark: a Gold set with expert-curated QA pairs and a Silver set with retrievability-filtered synthetic QA pairs at scale.
- Transparent RAG evaluation with retrieval-centric metrics (RAGAS) and statistical testing that reports effect sizes and confidence intervals.
- An open-source reproducibility package (code, prompts, document lists, and evaluation pipeline) that enables independent re-analysis (see Appendix A).

2. Background: RAG in Metabolomics

2.1. Metabolomics Research: Relevance and Challenges

Metabolomics connects genetic and environmental influences with phenotypic outcomes through metabolites, acting as substrates, intermediates, and end products within biological pathways, thereby reflecting immediate physiological states and adaptations [10, 11]. Researchers often struggle with the complexity, terminology, and rapidly expanding literature, complicating efficient knowledge extraction and practical hypothesis generation. Traditional metabolomics databases require structured queries, limiting intuitive exploration, especially for researchers with limited domain expertise. Beyond the rapid growth of the metabolomics literature, metabolomics datasets themselves are often high-dimensional and sparse (i.e., $p \gg n$), and can exhibit class- or feature-level imbalance, which complicates robust patient classification and biomarker discovery. Recent benchmarking work in metabolomics biomedical data systematically evaluated feature extraction and feature selection pipelines and reported that supervised feature selection can improve downstream classification performance, providing practical guidance on when to apply these techniques [12]. In parallel, predictive biomedicine increasingly combines multiple data modalities (e.g., multi-omics measurements with imaging or genomic variation), further increasing the complexity of evidence synthesis and interpretation [13]. Existing biomedical literature-mining tools such as BioTextQuest v2.0 support term-driven clustering and named-entity extraction from PubMed abstracts, but remain primarily query-and-browse systems rather than evidence-grounded natural-language question answering interfaces [14].

2.2. Core LLM Components for Biomedical Applications

LLMs constitute the cornerstone of modern NLP, fundamentally built upon three core components: tokenization, embeddings, and the transformer-based architecture. A nuanced understanding of these components is vital for effectively deploying and customizing RAG systems, particularly within specialized contexts such as biomedical applications.

Tokenization is the foundational preprocessing step, converting raw textual data into discrete units, or tokens, that models can interpret computationally [15]. The primary goal of tokenization is to segment continuous text into manageable pieces, typically words, subwords, or characters. Subword tokenization methods, such as Byte Pair Encoding (BPE), WordPiece, and SentencePiece, have become predominant due to their ability to handle extensive vocabularies efficiently, mitigating the "out-of-vocabulary" (OOV) problem by decomposing words into meaningful fragments [16, 17]. In biomedical contexts, traditional tokenizers often inadequately handle complex terminologies involving gene names, chemical compounds, and clinical terms. This limitation has led to the development of domain-specific tokenizers designed explicitly for biomedical terminologies, thus enhancing entity recognition and semantic fidelity [18, 19]. For instance, custom tokenizers associated with BioBERT or PubMedBERT significantly outperform general-purpose tokenizers by accurately preserving semantic

integrity of biomedical terms [20, 21]. Effective tokenization directly influences the quality of embeddings generated, as inaccurate segmentation of specialized biomedical terms can lead to suboptimal semantic representations. Consequently, tokenization precision is a crucial prerequisite for producing high-quality embeddings tailored to biomedical applications.

Embeddings transform discrete tokens into dense, continuous vector representations that capture semantic and contextual meanings [22]. These representations enable models to quantify semantic similarities and differences between terms. In practice, embeddings are often precomputed during a model's training and optimized to encapsulate syntactic, semantic, and even domain-specific knowledge. In general-purpose LLMs (e.g., GPT-4, GPT-3.5, LLaMA), embeddings capture broad linguistic patterns useful for diverse applications. However, when fine-tuned or pretrained on specialized biomedical datasets, such as PubMed abstracts, clinical guidelines, and medical textbooks, these embeddings become highly enriched with domain-specific semantic structures [23, 24]. Biomedical embeddings from models like BioBERT, PubMedBERT, ClinicalBERT, and MedCPT better reflect nuanced relationships specific to medical terminology, drug interactions, and biological pathways, significantly enhancing performance on domain-specific tasks [15, 25]. These embedding representations are subsequently processed by transformer architectures, which leverage self-attention mechanisms to contextually interpret token relationships. Thus, embeddings serve as the critical bridge between tokenized input and the sophisticated understanding capabilities of transformer models.

Transformer architectures, driven by their self-attention mechanisms, form the foundation of nearly all modern LLMs [26]. These mechanisms enable the model to dynamically assess the importance of each token in a sequence relative to its context. While the intricacies of attention mechanisms underpin their functionality, transformers can be categorized into three types based on their architectural design and output: encoder-only models, which transform input sequences into contextual vector representations; decoder-only models, which generate text autoregressively; and encoder-decoder models, which integrate both components to map input sequences to distinct outputs. The evolution of transformers began with the encoder-decoder architecture, introduced in the original Transformer paper by Vaswani et al. [27], designed for sequence-to-sequence tasks like machine translation and summarization. This was followed by the rise of decoder-only models, such as the GPT series, which focus on autoregressive text generation and excel in tasks like text completion and conversational AI. Later, encoder-only models, such as BERT, emerged, leveraging bidirectional context for tasks requiring deep language understanding, such as classification and named entity recognition. Today, decoder-only models dominate generative applications due to their scalability with increasing parameters and vast datasets, enhancing their few-shot learning capabilities, while encoder-only models remain vital for understanding-focused tasks, particularly in specialized domains like biomedicine [1]. Encoder-only architectures (BioBERT, MedCPT) excel in contextual understanding critical for biomedical retrieval tasks.

1 Decoder-only models (e.g., GPT-4) dominate generative tasks
2 but often require extensive fine-tuning for precision in biomed-
3 ical contexts [1, 28].

4 The performance of LLMs in specialized tasks is pro-
5 foundly influenced by how effectively tokenization, embed-
6 dings, and transformer architectures are adapted and aligned.
7 This interplay becomes especially evident when comparing
8 general-purpose models against domain-specific counterparts
9 in biomedical contexts. Modern LLMs broadly categorize
10 into general-purpose and domain-specific models. General-
11 purpose LLMs, such as GPT-4, GPT-3.5, and LLaMA, come
12 equipped with integrated tokenizers, embedding models, and
13 transformer architectures, suitable for diverse language pro-
14 cessing tasks. However, they may struggle with the depth and
15 specificity of biomedical terminology without further adapta-
16 tion [17]. Conversely, domain-specific LLMs are either fine-
17 tuned or pretrained extensively on specialized corpora, adapt-
18 ing their tokenizers and embeddings to better capture domain-
19 specific language nuances. Biomedical domain-specific LLMs
20 exemplified by BioBERT, PubMedBERT, ClinicalBERT, and
21 MedCPT are trained explicitly on biomedical corpora, such
22 as PubMed abstracts, clinical question-answer datasets (e.g.,
23 MedQA, MedMCQA), electronic health records (EHRs), and
24 specialized medical textbooks [18, 19, 25]. For example,
25 BioBERT leverages extensive pretraining on PubMed and PMC
26 corpora to effectively handle biomedical named entity recog-
27 nition (NER) and semantic textual similarity tasks [21]. Pub-
28 MedBERT further specializes by exclusively pretraining on
29 PubMed abstracts, yielding superior performance in biomed-
30 ical text mining tasks [28]. ClinicalBERT utilizes patient-
31 centric clinical notes, enabling more accurate understanding of
32 clinical text [15]. Meanwhile, MedCPT introduces advanced
33 embedding methods based on user interactions from PubMed
34 searches, capturing relevance judgments at scale to significantly
35 enhance zero-shot semantic retrieval [19]. In summary, each
36 component—tokenization, embeddings, and transformer archi-
37 tecture—plays a critical role in defining LLM functionality.
38 Domain-specific adaptations of these components, particularly
39 within the biomedical field, substantially enhance the accuracy,
40 reliability, and contextual appropriateness of LLM-based appli-
41 cations, effectively bridging the gap between general linguistic
42 understanding and the highly specialized demands of biomed-
43 ical research and clinical practice. In biomedical NLP, these
44 modelling choices must also be evaluated through the lens of
45 transparency and interpretability. A recent scoping review in
46 healthcare NLP highlights persistent gaps in best practices and
47 systematic benchmarks for explainable and interpretable deep
48 learning, reinforcing the need for designs that support auditabil-
49 ity and evidence traceability in high-stakes settings [29].

53 2.3. Retrieval-Augmented Generation (RAG)

54 Retrieval-Augmented Generation (RAG) transforms natu-
55 ral language processing by integrating LLMs with real-time
56 retrieval from external knowledge bases. Unlike traditional
57 LLMs, which rely on static, potentially outdated knowledge,
58 RAG dynamically retrieves contextually relevant information,
59 making it ideal for biomedical research where precision and

up-to-date literature are critical [6]. By grounding responses
in retrieved evidence, RAG mitigates LLM hallucinations
and domain mismatches, enhancing accuracy in fields like
metabolomics [4, 18].

RAG’s core mechanism involves encoding a user
query—e.g., “What metabolites are associated with early-
stage Alzheimer’s disease?”—into a dense vector embedding
using models like MedCPT or BioBERT [19, 21]. This em-
bedding, capturing semantic intent, queries a vector database
of pre-processed documents (e.g., PubMed articles). Retrieved
text chunks augment the query, enabling the LLM to gener-
ate evidence-based responses [22]. Embeddings, built from
tokenized text (as discussed in the Background section), are
indexed in vector databases like FAISS or Qdrant, using algo-
rithms like HNSW (Hierarchical Navigable Small Worlds) for
efficient similarity searches [15, 23].

The retrieval pipeline begins with chunking, segmenting
documents into 512–1000-token units with overlaps to fit LLM
context windows (e.g., 4096 tokens for GPT-4) and preserve
biomedical coherence (e.g., keeping “glycolysis intermediates”
intact) [1, 16]. Chunks are embedded and indexed for rapid re-
trieval, with optimizations like quantization or term weighting
(e.g., “biomarker”) enhancing domain relevance [8, 25]. This
infrastructure supports RAG’s scalability and precision across
large datasets.

RAG implementations vary: *Classical RAG* uses simple co-
sine similarity for retrieval, as in BioBERT with FAISS, but
struggles with complex queries [15, 22]. *Custom/Advanced*
RAG combines lexical (BM25) and semantic (MedCPT) re-
trieval with re-ranking or iterative prompting, improving entity-
aware retrieval for metabolomics [16, 19]. *Graph-Based RAG*
leverages knowledge graphs, embedding nodes (e.g., metabo-
lites) and edges (relationships) for multi-hop reasoning, as
in LORE, excelling at uncovering latent biomedical connec-
tions [30, 31]. Each type uses embeddings differently, from
direct similarity to relational reasoning, addressing diverse re-
search needs. Knowledge-graph resources can further comple-
ment text retrieval by encoding metabolite–gene–disease re-
lationships and enabling multi-hop evidence traversal. However,
constructing and maintaining biomedical knowledge graphs at
scale remains challenging. BioKGrapher provides an exam-
ple of automated KG construction from PubMed IDs using
NER/NEL with UMLS normalization, concept weighting and
re-ranking, and ontology-driven relation building (e.g., via ter-
minologies such as SNOMED CT and NCI), illustrating a
practical pathway for producing condition-specific graphs that
could be coupled to graph-based retrieval in biomedical RAG
pipelines [32].

Evaluation metrics like Faithfulness, Answer Correct-
ness, and Context Precision and Recall assess RAG perfor-
mance. Biomedical QA datasets, especially for subfields like
metabolomics, are challenging to develop due to content diver-
sity, requiring tailored frameworks [7, 33]. RAG’s embedding-
driven approach, scalable across classical, custom, and graph-
based variants, significantly advances biomedical knowledge
extraction. Recent biomedical RAG work increasingly treats
evaluation as a retrieval-and-grounding problem rather than re-

lying only on lexical overlap metrics, particularly for short, technical answers. To address the limited scalability of expert-curated benchmarks, multiple strands of recent literature propose LLM-driven synthetic QA generation pipelines paired with automated filtering or retrievability checks to ensure generated questions remain answerable from the indexed corpus. In parallel, automated RAG evaluation toolkits (e.g., judge-based or prompt-based frameworks) operationalize claim-level faithfulness and context relevance, enabling larger-scale comparisons when human reference answers are unavailable. This motivates a two-tier evaluation framing in which a small expert-curated set supports reference-based validation, while larger retrievability-filtered synthetic sets support stress-testing of retrieval grounding and robustness at scale.

3. Methods

We implement *MedDiscover* as an open-source RAG evaluation pipeline and apply it to metabolomics/metabolic-disorder literature using a two-tier dataset (Gold and Silver). The primary comparisons fix the decoder (GPT-4o) to isolate retrieval effects, while decoder upgrades (GPT-4.1-nano/mini) are reported as sensitivity analyses. NotebookLM is included as a black-box comparator for answer style, with grounding metrics reported only where retrieved contexts are available.

The evaluation compares three primary strategies: (1) domain-specific embeddings (MedCPT) with a GPT-4o decoder, (2) general-purpose embeddings (OpenAI Ada-002) with a GPT-4o decoder, and (3) a non-RAG baseline (GPT-4o without retrieval). The choice of MedCPT, Ada, and GPT-4o was guided by an extensive literature review, which identified MedCPT as the only published, publicly available domain-specific embedding model with validated superior performance. Other alternatives (e.g., BioGPT, SciPhi) were either not available for standalone use or did not surpass MedCPT in comparative studies[19]. Ada and GPT-4o represent widely used general-purpose embedding and generation baselines, enabling a controlled comparison.

Detailed methodologies for data collection, preparation, the RAG pipeline, and evaluation metrics are described in the following subsections. Further experimental setups and extensive supplementary results can be found in the Supplementary Material, ensuring transparency, reproducibility (see Appendix A), and facilitating further exploration of our methodology.

3.1. Data Collection and Preparation

We assembled a metabolomics/metabolic-disorder corpus of 100 papers published between 2014 and 2024 from PubMed and Google Scholar, curated using ICD-10 metabolic disorder codes E70–E88. Text extraction and cleaning were performed by extracting raw text from PDFs using PyPDF2 and removing headers, footers, and non-informative formatting. Numeric and scientifically relevant data (metabolite concentrations, p-values) were preserved, and each processed article retained original paragraph sequences with metadata identifiers mapping back to the source PDF.

3.1.1. Gold QA set (expert-curated)

The Gold set consists of ten metabolomics papers focusing on Gaucher disease (Type 1), Nonalcoholic Fatty Liver Disease (NAFLD), and Type 1 diabetes, with 30 expert-curated questions and traceable reference answers. This set is used to validate answer quality against human references. The full Gold set contains 30 expert-curated QA pairs. For cross-system comparisons that require outputs from every evaluated configuration (including black-box comparators), we report results on the subset of 20 Gold questions for which all model outputs are available (evaluation subset; see Section 4).

3.1.2. Silver QA set (synthetic, retrievability-filtered)

The Silver set uses the 100-paper corpus and ~600 synthetic QA pairs generated with retrievability metadata (Ada n=300, MedCPT n=300). This set is used to stress-test retrieval grounding at scale rather than to provide human reference answers. We refer to this collection as a *Silver* set because the questions and reference answers are machine-generated and only partially filtered: each QA pair is retained based on lightweight quality controls (Section 3.2), including a retrievability check, but the content is not exhaustively expert-validated. Accordingly, Silver-set results are interpreted primarily as evidence about retrieval grounding and robustness at scale, rather than as definitive clinical correctness.

3.2. Synthetic QA generation with retrievability filtering (Silver set)

Synthetic QA pairs were generated from the Silver corpus using our QA-generation pipeline. The pipeline: (1) ingests PDF text, (2) chunks text into 500-token windows with 50-token overlap, (3) groups adjacent chunks to meet a minimum context length, (4) embeds contexts with Ada-002 or MedCPT and builds a FAISS index, (5) generates QA pairs with GPT-4.1-mini (Dec 14, 2025 runs; 3 questions per PDF), and (6) performs a retrievability check by re-embedding each generated question, querying the FAISS index, and recording whether the supporting passage was retrieved and its rank. The output records each question and answer, the source document and chunk identifiers, the retrieved context, retrievability indicators, disease category, and the embedding/LLM configuration. This retrievability logging provides a lightweight filtering signal inspired by synthetic QA pipelines in the literature [34]. Our use of machine-generated QA as a silver-standard resource is aligned with recent dataset efforts that pair medical questions with automatically generated answers and evidence-supported references, while emphasizing that expert validation remains essential for gold-standard claims [35]. In addition to retrievability logging, we applied lightweight QA-quality heuristics before constructing the final Silver benchmark:

- **De-duplication (question-level):** questions were normalized by lowercasing, stripping punctuation, and collapsing whitespace; exact duplicates after normalization were removed.

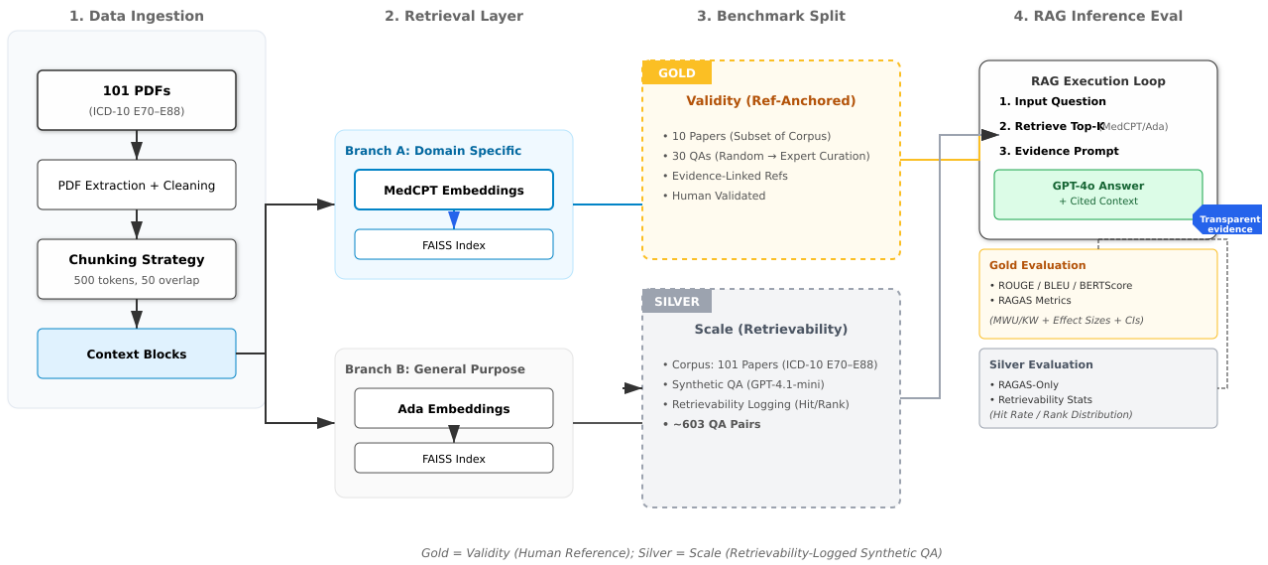


Figure 1: MedDiscover pipeline and benchmark design. A metabolomics/metabolic-disorder corpus (ICD-10 E70–E88) is ingested, cleaned, chunked (500 tokens with 50-token overlap), and indexed with two embedding configurations (MedCPT and Ada-002) using FAISS. Evaluation is organized into a Gold set (expert-curated QA with traceable references) and a Silver set (synthetic QA with retrievability metadata). RAG inference retrieves top- k contexts and uses an evidence-grounded prompt to produce answers with explicit supporting context; evaluation reports reference-based metrics on Gold and retrieval-grounding metrics plus retrievability statistics on Silver.

Table 1: Dataset summary (Gold vs Silver).

Dataset	Papers	Domains/diseases	QA pairs	Reference type	Primary use
Gold	10	Gaucher (Type 1), NAFLD, Type 1 diabetes	30	Human-curated	Answer quality vs references
Silver	100	ICD-10 E70–E88 metabolic disorders	~600 (Ada 300; MedCPT 300)	Synthetic + retrievability	Retrieval stress test

- **Minimum length:** we removed degenerate items using whitespace-token counts: $\text{question} \geq 8$ tokens and $\text{answer} \geq 6$ tokens.
- **Template filtering:** to reduce over-representation of trivial prompts, we filtered a small set of highly templated question openings (e.g., “what is”, “define”, “what does ... mean”) when they occurred disproportionately in the generated pool.

After QC filtering, we sampled 300 QA pairs per embedding configuration to keep the two Silver subsets balanced (Ada $n = 300$, MedCPT $n = 300$). These QC steps are implemented as a deterministic post-processing stage in the Silver QA pipeline (see Appendix A). Because Silver QA generation is stochastic and was executed separately for each embedding configuration, the resulting Ada and MedCPT QA sets are not strictly paired (Ada $n = 300$, MedCPT $n = 300$). We therefore treat Silver-set comparisons as independent samples and use Mann–Whitney U tests with effect sizes (Cliff’s δ). Reviewer feedback described early synthetic questions as “non-human,” which we interpret as overly surface-level, templated, or extractive prompts that can be answered by copying a single sentence. In this revision we mitigate this by (i) separating expert-curated Gold evalu-

ation from synthetic Silver benchmarking, (ii) explicitly documenting retrievability and minimal QA-quality controls, and (iii) treating Silver results as silver-standard evidence rather than expert-validated ground truth. Nevertheless, the Silver set remains machine-generated and may not fully reflect real investigator query distributions.

3.3. Text Chunking and Embedding (for MedDiscover RAG)

The chunking process uses overlapping chunks of 500 tokens with a 50-token overlap, a standard RAG setting selected based on prevailing practices in recent biomedical studies (e.g., RDGuru, AsthmaBot, EARA) and to preserve entity co-occurrence across chunk boundaries [16]. This chunk size is compatible with the context windows of current LLMs and balances retrieval precision with contextual integrity. For documents exceeding the LLM token limit, multiple overlapping chunks were generated and, during retrieval, the top-ranked segments were concatenated or batched as input context for the decoder LLM.

Text chunks were embedded using two distinct embedding models for the RAG pipelines:

- **MedCPT (Domain-specific):** 768-dimensional, pre-

Table 2: Qualitative illustration of question styles in Gold vs Silver. Examples are paraphrased for brevity.

Question type	Gold (expert-curated)	Silver (synthetic)
Biomarker association	Links a specific metabolite panel to a named disease stage and requests directionality (increase/decrease) with evidence.	Often focuses on a single metabolite–disease association extracted from one paragraph.
Study design / cohort	Asks about cohort characteristics, assay conditions, or comparison groups required to interpret results.	More frequently asks factoid questions about reported findings without design context.
Pathway / mechanism	Requests interpretation of how observed metabolite shifts map onto a pathway (e.g., lipid metabolism) with supporting text.	Can be partially inferential but may remain closer to extractive summaries.

Table 3: Silver QA post-processing quality controls.

Control	Operational definition
De-duplication	Normalize question text (lowercase, strip punctuation, collapse whitespace); drop exact duplicates.
Minimum length	Drop items with <code>question</code> < 8 whitespace tokens or <code>answer</code> < 6 whitespace tokens.
Template filtering	Reduce over-represented definition-only templates (e.g., “what is”, “define”) in the candidate pool prior to sampling.

trained on PubMed abstracts and biomedical corpora [19].

- **OpenAI text-embedding-ada-002 (General-purpose):** 1,536-dimensional, pretrained on extensive general-domain textual data.

Two parallel embedding databases were generated for direct comparative evaluation within the MedDiscover framework.

Indexing and vector storage involved indexing vector representations using FAISS. A flat index structure with cosine similarity was chosen for its balance of retrieval accuracy and computational efficiency in semantic search tasks across moderate-sized corpora. Metadata bindings linked each vector to its original text source, facilitating precise retrieval verification.

3.4. Retrieval-Augmented Generation Pipeline (MedDiscover)

The following pipeline was uniformly applied for both MedCPT and general-purpose (Ada) embeddings within MedDiscover, with GPT-4o as the decoder LLM. The output token limit for the GPT-4o decoder in MedDiscover was set to a maximum of 25 tokens during comparative experiments to encourage concise answers.

1. **Query Encoding:** User queries were embedded using the respective embedding model (MedCPT or Ada).
2. **Context Retrieval:** Top- k ($k = 3$) nearest-neighbor chunks were retrieved from the FAISS index using cosine similarity.
3. **Context Aggregation:** Retrieved chunks were concatenated into context passages, minimizing redundancy.

4. **Answer Generation:** Contextual prompts were fed into the GPT-4o decoder for generative response synthesis, instructed to reference provided evidence explicitly and adhere to the output token limit.

The non-RAG baseline utilized direct GPT-4o responses without external retrieval, also adhering to a similar concise output style for fair comparison where applicable.

3.5. Secondary Evaluation with NotebookLM

For a secondary evaluation, we utilized Google’s NotebookLM on the Gold set only (ten test publications, 30 queries). Responses were collected and evaluated using the same generation metrics where applicable. Since NotebookLM provides direct responses without exposing the underlying retrieved context, RAG-specific metrics such as Context Precision, Context Recall, and Faithfulness (which often relies on comparing generated statements to specific retrieved chunks) could not be fully evaluated in the same manner as for MedDiscover. Given that the MedDiscover GPT-4o decoder was constrained to a maximum of 25 output tokens for this experimentation phase, leading to very precise and minimal answers, initial comparisons with NotebookLM’s default output length were challenging. Therefore, to facilitate a more aligned comparison, a second set of responses was generated from NotebookLM by appending the instruction “Answer in Minimal Words” to each query. Both sets of NotebookLM responses (standard and minimal) were then evaluated.

3.6. Evaluation Metrics

Performance evaluation incorporated both generation-centric and retrieval-centric metrics, applied to MedDiscover

configurations and, where appropriate, to NotebookLM. Let $\phi(\cdot)$ denote the embedding function used for semantic similarity metrics.

Generation Metrics: Accuracy, Precision, Recall, and F1 Score quantitatively assessed the correctness and comprehensiveness of generated responses. ROUGE-1/2/L measure unigram overlap, bigram overlap, and longest-common-subsequence overlap between candidate and reference answers. BERTScore captures semantic similarity:

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (1)$$

$$\text{BERTScore} = \text{CosSim}(\phi(\text{generated}), \phi(\text{reference})) \quad (2)$$

Retrieval Metrics (Primarily for RAG systems like MedDiscover): Context Precision and Context Recall assessed the relevance and completeness of retrieved information, while Faithfulness measured factual grounding of generated content:

$$\text{Context Precision@K} = \frac{\text{Relevant Retrieved Contexts}}{K} \quad (3)$$

$$\text{Context Recall} = \frac{\text{Relevant Retrieved Contexts}}{\text{Total Relevant Contexts}} \quad (4)$$

$$\text{Faithfulness} = \frac{\text{Supported Claims in Generated Answer}}{\text{Total Claims in Generated Answer}} \quad (5)$$

Additionally, Answer Relevancy quantified semantic alignment between queries and responses:

$$\text{Answer Relevancy} = \frac{1}{N} \sum_{i=1}^N \cos(\phi(\text{answer}_i), \phi(\text{query}_i)) \quad (6)$$

We compute RAGAS metrics (faithfulness, answer correctness, context recall, context precision, answer relevancy) using the RAGAS framework [36] and report these where retrieved contexts are available. ROUGE and BLEU are reported only for the Gold set because the Silver set lacks human reference answers. For NotebookLM, only metrics not dependent on explicit retrieved context (like Answer Relevancy and generation metrics) are fully comparable.

Despite strong overall performance, MedDiscover displays several failure patterns. With a fixed $k = 3$ retrieval, relevant but lower-ranked metabolites may be omitted if they are distributed across the corpus, especially for diseases with multiple biomarkers. When queries refer to diseases or metabolites absent from the corpus, the system correctly returns that no evidence is available. If the corpus spans several diseases, queries about multiple diseases may yield incomplete answers, typically addressing only the best-matched case. For best results, we recommend applying MedDiscover to focused, single-disease corpora. These patterns suggest future improvements, including dynamic k adjustment and enhanced entity disambiguation.

4. Results

4.1. Results on the Gold set (expert curated)

We evaluated an expert-curated Gold set of 30 QA pairs; results reported here use the 20-question evaluation subset for which outputs are available across all compared configurations. Primary comparisons fix the decoder (GPT-4o) to isolate retrieval effects across the non-RAG baseline, Ada + GPT-4o, and MedCPT + GPT-4o. Supplementary Figure B.4 summarizes LLM-specific and retrieval-specific metrics for these models, along with NotebookLM and the MedCPT decoder sensitivity variants.

4.1.1. RAG vs non-RAG (fixed decoder)

With the decoder held constant (GPT-4o), retrieval augmentation improves grounding relative to the non-RAG baseline, particularly on faithfulness and context precision/recall. Differences between Ada and MedCPT under the same decoder are modest on the Gold set, and most pairwise contrasts do not reach statistical significance after correction.

4.1.2. NotebookLM comparison

NotebookLM shows strong lexical overlap, especially when prompted for minimal output, but it does not expose retrieved contexts. As a result, grounding metrics (context precision/recall, faithfulness) cannot be computed for NotebookLM, limiting its comparability in evidence-critical settings.

4.1.3. Sensitivity analysis (decoder upgrades)

MedCPT with GPT-4.1-nano/mini improves several lexical and answer-quality metrics on the Gold set; however, these variants are not evaluated for the Ada comparator. We therefore treat decoder-upgrade results as a supplementary sensitivity analysis rather than as evidence for embedding superiority.

4.2. Results on the Silver set (synthetic, 100-paper benchmark)

The Silver benchmark evaluates 600 synthetic QA pairs drawn from 100 papers (Ada $n=300$, MedCPT $n=300$) using RAGAS metrics only. Table 4 reports mean scores with bootstrap 95% confidence intervals (2,000 resamples), Mann-Whitney U p -values, and Cliff’s δ effect sizes (MedCPT vs Ada). MedCPT yields a statistically significant improvement in answer correctness ($p = 5.72 \times 10^{-11}$, $\delta = 0.303$), while other retrieval-centric metrics are comparable.

The effect size for answer correctness on the Silver benchmark (Cliff’s $\delta = 0.303$) corresponds to a small-to-moderate separation. Interpreted as a common-language effect size, this implies that a randomly selected MedCPT run has an estimated 65.1% probability of achieving a higher answer-correctness score than a randomly selected Ada run under this benchmark. Because the Silver benchmark uses synthetic reference answers that are identical to the responses in the evaluation files, answer-quality metrics should be interpreted as proxy measures of answerability and grounding rather than as expert-validated clinical correctness.

Primary Tests

Secondary / Sensitivity Tests

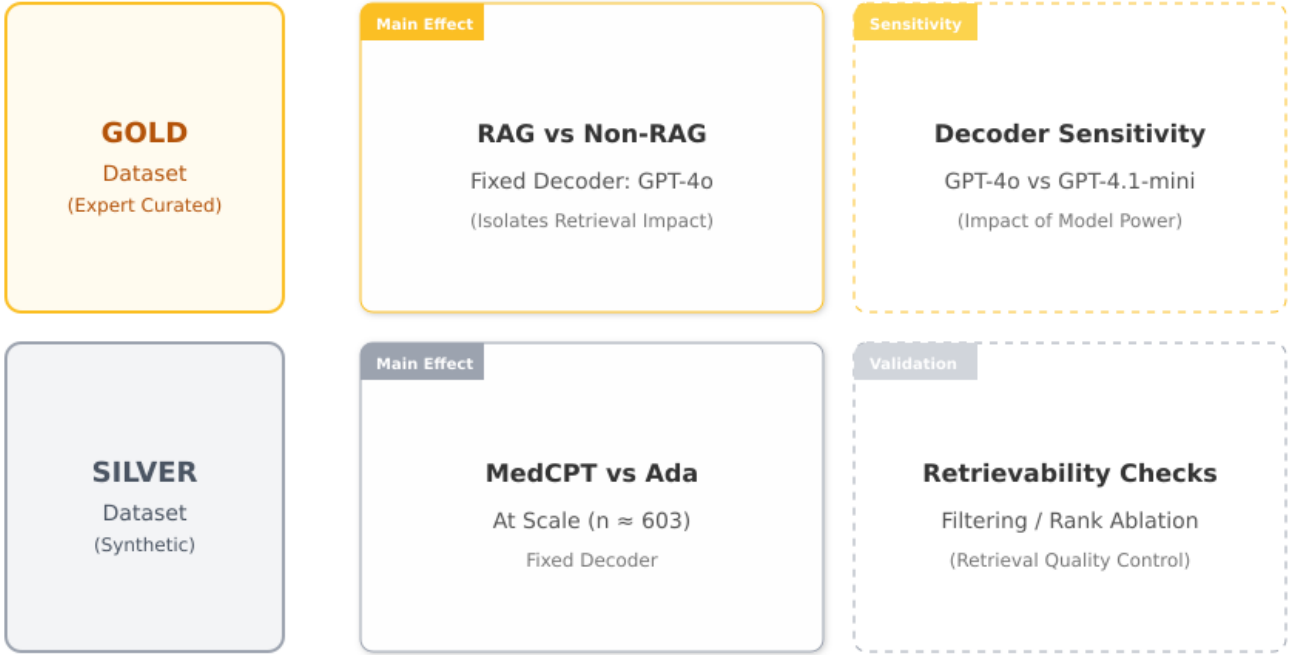


Figure 2: Experimental design matrix separating primary hypotheses from secondary sensitivity analyses.

Figure 2: Experimental design overview. Primary tests fix the decoder (GPT-4o) to isolate retrieval effects (RAG vs non-RAG on the Gold set; MedCPT vs Ada at scale on the Silver benchmark). Secondary analyses include decoder sensitivity (GPT-4o vs GPT-4.1 variants on Gold) and retrievability quality-control analyses on Silver (hit-rate and rank-based ablations).

4.3. Statistical analysis

Unless otherwise stated, we treat the retrieval-centric RAGAS metrics (faithfulness, answer correctness, context recall, context precision, answer relevancy) as a primary metric family and control family-wise error across this family using Holm–Bonferroni adjustment. Metrics outside this family are treated as secondary and interpreted cautiously. All model-by-metric score distributions were subjected to the Shapiro–Wilk test; almost every distribution departed from normality ($p < 0.05$). A subsequent Levene test revealed unequal variances for several evaluation metrics. In view of these results, groupwise comparisons were carried out with rank-based, non-parametric methods.

For the Gold set, a Kruskal–Wallis one-way ANOVA on ranks was computed across the model configurations, followed by Dunn’s post-hoc tests with Ada + GPT-4o as the reference group. Holm–Bonferroni correction was applied across the primary metric family (faithfulness, answer correctness, context recall, context precision, answer relevancy). Supplementary Figure B.4 reports adjusted contrasts.

For the Silver benchmark, we report Mann–Whitney U tests for Ada vs MedCPT on each RAGAS metric, along with Cliff’s δ effect sizes and bootstrap 95% confidence intervals for mean

scores (2,000 resamples; Table 4). This isolates embedding effects under a fixed decoder and provides a scale-appropriate estimate of practical impact.

5. Discussion

5.1. Principal Findings

This study evaluates MedDiscover as a reproducible RAG benchmark with two datasets: a Gold set (expert-curated) and a Silver set (retrievability-filtered synthetic) (see Appendix A). On the Gold set, retrieval augmentation improves grounding relative to a non-RAG baseline when the decoder is held constant. When the decoder is fixed at GPT-4o, differences between MedCPT and Ada are modest and not consistently significant across metrics.

On the Silver benchmark, embedding choice produces comparable retrieval-centric metrics, while MedCPT yields a statistically significant improvement in answer correctness (Table 4). This indicates that domain-specific embeddings can provide a targeted gain in answer quality, but not a universal advantage across all grounding metrics.

NotebookLM provides strong lexical alignment under minimal-output prompting, but its retrieval mechanism is

Table 4: Silver benchmark results (Ada vs MedCPT). Means with bootstrap 95% confidence intervals; Cliff’s δ for MedCPT vs Ada.

Metric	Ada mean [95% CI]	MedCPT mean [95% CI]	p	Cliff’s δ
Faithfulness	0.703 [0.667, 0.740]	0.687 [0.650, 0.725]	0.85	-0.009
Answer correctness	0.968 [0.960, 0.976]	0.980 [0.974, 0.985]	5.72×10^{-11}	0.303
Context recall	0.743 [0.702, 0.779]	0.722 [0.682, 0.762]	0.50	-0.029
Context precision	0.923 [0.890, 0.953]	0.908 [0.875, 0.937]	0.49	-0.016
Answer relevancy	0.800 [0.776, 0.823]	0.810 [0.788, 0.830]	0.82	0.011
Accuracy	0.997 [0.990, 1.000]	1.000 [1.000, 1.000]	0.32	0.003

opaque. Because retrieved contexts are not exposed, grounding metrics cannot be computed for NotebookLM, limiting its suitability for evidence-critical biomedical workflows.

All MedDiscover runs restricted GPT-4o to 25 generation tokens to control verbosity and encourage precision. This design choice likely depressed length-sensitive scores such as ROUGE and BLEU relative to longer responses, underscoring the need to interpret lexical overlap metrics cautiously in short-answer biomedical settings.

5.2. Interpretation in the Context of Previous Work

5.2.1. Evolution of Biomedical RAG

The question of how to generate accurate responses within a specialised knowledge domain has spurred several complementary approaches. Early efforts centred on domain-pre-trained encoders such as BioBERT [37], PubMedBERT [38], and SciBERT [39], each trained on large biomedical corpora. While effective at the time of training, these static models struggle to keep pace with the rapidly expanding literature. Regularly re-training a full language model is computationally prohibitive, and even frequent updates leave an inevitable gap between the model’s knowledge and the latest publications.

RAG frameworks were introduced to bridge this gap by injecting up-to-date evidence at inference time. Systems such as BIORAG [18], RAG², and Self-BioRAG have since added multi-stage retrieval pipelines, self-evaluation loops, and knowledge-graph conditioning to cope with the scale and dynamism of biomedical knowledge. MedDiscover follows this lineage but narrows the focus to metabolomics, a sub-discipline that has received comparatively little attention in prior RAG work. By anchoring retrieval to experimentally validated metabolite literature rather than broader biomedical text, our study illustrates how increased domain granularity can sharpen factual grounding.

5.2.2. Retrieval Architectures: Similarities and Divergences

Compared with earlier frameworks, MedDiscover adopts a deliberately lightweight retrieval architecture: a single-stage dense lookup over FAISS indices produced by MedCPT or Ada-002 embeddings. DRAGON and RAG², by contrast, implement multi-hop or multi-stage retrieval. DRAGON fuses knowledge graphs with text through a cross-modal encoder, while RAG² first rewrites the query into a rationale, then balances evidence across PubMed, PMC, textbooks and guidelines, and finally reranks with a cross-encoder. BioRAG ex-

tends the concept still further by coupling MeSH-filtered hierarchical look-ups with external search-engine calls and a self-evaluation module that decides when additional retrieval is needed.

Although these more elaborate pipelines improve performance on broad medical QA benchmarks, our results indicate that a streamlined, single-stage design can suffice for narrowly scoped, high-precision tasks. MedCPT-based retrieval delivers competitive context precision and faithfulness on the Gold set while maintaining lower latency and a smaller memory footprint. This finding implies that retrieval complexity should be matched to the breadth of the information need: dense, domain-specific embeddings may already capture enough local structure when the target corpus is well circumscribed.

5.2.3. Embedding Strategies and Domain Adaptation

A consistent theme across biomedical RAG studies is the advantage of domain-specialised vector representations. DRAGON initialises its language component with BioLinkBERT and supplements it with knowledge graphs; BIORAG uses a CLIP (Contrastive Language-Image Pretraining)-enhanced Pub-MedBERT [18]; Self-BioRAG and RAG² rely on MedCPT for top-k retrieval. Our head-to-head comparison between MedCPT (768 dimensions) and the larger, general-purpose Ada-002 (1536 dimensions) shows that the two retrievers achieve broadly comparable grounding metrics on the Silver benchmark, with a modest but significant improvement in answer correctness for MedCPT (Cliff’s $\delta = 0.303$; Table 4). These findings highlight a practical trade-off: domain-tuned embeddings reduce index size while offering targeted gains in answer quality, an important consideration for on-premise deployments constrained by GPU memory.

5.2.4. Comparative Performance Metrics

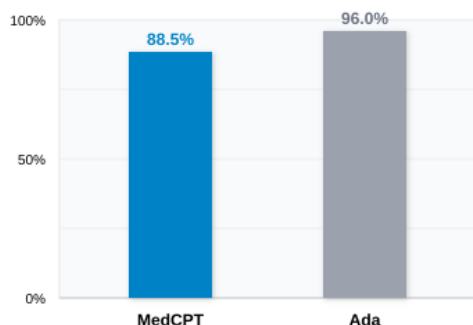
Direct numerical comparison across studies is complicated by differences in task formulation (multiple-choice MedQA versus factoid metabolite QA), as well as by the fact that the authors of DRAGON, BIORAG, Self-BioRAG and RAG² largely report accuracy. Nevertheless, several insights emerge. First, in the Gold set sensitivity analysis our Answer Correctness reaches 0.75 (with GPT-4.1-mini decoding), which matches the 0.73 accuracy reported for BIORAG on MedMCQA and exceeds the 0.55 reported for Self-BioRAG on PubMedQA, though being largely surpassed by 0.91 accuracy on MedMCQA. Second, our single-hop Answer Correctness of 0.39 (0.75

Retrievability Quality Control (Silver Dataset)

MedCPT Ada

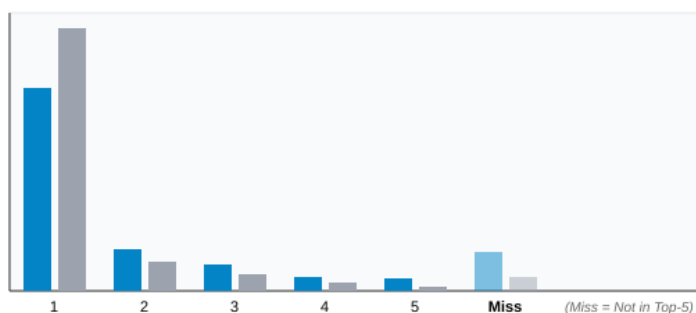
A. Retrieval Hit Rate

Percentage of generated Qs where answer chunk is in Top-5



B. Rank Distribution (Top-5)

Histogram of retrieved chunk rank (Top-5)



C. Ablation: Impact of Rank on RAGAS Metrics

Faithfulness Context Precision drop as retrieval rank increases

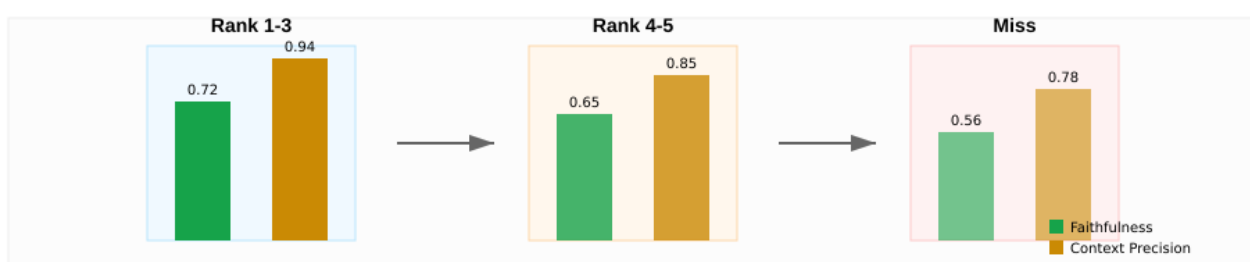


Figure 3: Retrievability quality control on the Silver benchmark (top-5 retrieval). (A) Hit rate: fraction of generated questions whose answer-supporting chunk is retrieved within the top-5. (B) Distribution of observed ranks for the answer-supporting chunk; “Miss” indicates not retrieved in the top-5. (C) Rank-based ablation showing that retrieval quality correlates with grounding metrics: faithfulness and context precision decrease as the answer-supporting chunk rank worsens.

with the stronger GPT-4.1-mini decoder) is broadly in line with DRAGON’s 0.475 accuracy on MedQA, but is attained with a far smaller corpus and without knowledge graph augmentation. These results indicate that metabolomics-specific retrieval can attain competitive grounding even in the absence of multi-hop reasoning.

Taken together, these comparisons position MedDiscover as a complementary addition to the biomedical RAG ecosystem. The framework highlights the value of transparent evaluation and suggests that domain-specific embeddings can yield targeted gains in answer quality, while simple dense retrieval can remain competitive in focused sub-domains.

5.3. Methodological considerations

The experimental design sought to balance realism with tractability, yet each methodological choice carried consequences that shape the interpretation of our results. We discuss here the principal evaluation metrics, the statistical power afforded by the dataset, the indexing and generation parameters, and the challenges inherent in benchmarking against a proprietary baseline.

We adopted a dual-lens strategy that combines generation-centric metrics (ROUGE, BLEU, BERTScore) with retrieval-centric ones (Context Precision, Context Recall, and Faithfulness). The latter group constitutes a clear strength of the study:

by explicitly measuring how many retrieved passages are relevant and how many answer claims are grounded in those passages, we directly probe the defining mechanism of a RAG system. Purely surface-level metrics alone cannot characterize grounding. At the same time, our reliance on ROUGE and BLEU exposes a known weakness: both metrics reward n-gram overlap and therefore under-value concise technical answers, precisely the style encouraged by our 25-token output cap. Moreover, neither metric is sensitive to subtle semantic shifts, such as confusing “decreased” with “unchanged,” that often make the difference in biomedical interpretation.

The Gold set contains only thirty expert-curated question-answer pairs mapped to ten articles, which limits statistical power for subtle effects. We therefore complement it with a larger Silver benchmark (100 papers; ~600 synthetic QA pairs) to increase power for retrieval-centric comparisons, while acknowledging that the corpus remains scoped to metabolic-disorder literature.

Manual chunking at 500 tokens with a 50-token overlap was chosen to fit comfortably inside the GPT-4o context window while preserving local biomedical coherence. Although effective, this heuristic could bias recall: over-long chunks dilute term specificity, whereas under-long chunks risk splitting meaningful entities or co-occurrences. An ablation study that systematically varies chunk size could clarify how sensitive retrieval performance is to this hyper-parameter. The contrast-

ing dimensionalities of the two embedding models, 768 for MedCPT versus 1 536 for Ada-002, introduced further trade-offs. Higher-dimensional vectors can improve representational fidelity but entail larger indices, slower similarity search, and higher memory footprints; future deployments will need to weigh these factors against marginal accuracy gains.

Capping GPT-4o outputs at 25 tokens served two purposes: it ensured that the comparison with NotebookLM and with the non-RAG baseline remained fair, and it made the Faithfulness assessment tractable by limiting the number of factual claims per answer. The downside is reduced explanatory depth; many clinically meaningful responses require context about study cohorts, assay conditions, or effect sizes that simply cannot fit within such a narrow bandwidth. Adaptive length policies, short by default but expandable on demand, may provide a better balance between brevity and informativeness.

Google’s NotebookLM offered a convenient external reference, yet its proprietary retrieval pipeline remains opaque. Because retrieved passages are not exposed, we could not compute Context Precision, Context Recall, or Faithfulness for that system. Consequently, the comparison is incomplete: NotebookLM’s excellent ROUGE scores in “minimal words” mode speak to lexical alignment, but say little about factual verifiability. Until commercial platforms provide richer telemetry or standardised RAG benchmarks emerge, evaluations will continue to suffer from this parity gap.

Taken together, these methodological reflections explain both the strengths and the blind spots of the current study. They highlight why improvements in retrieval grounding were detectable despite the small sample, but also why claims about domain generality, computational efficiency, and user-facing transparency must remain provisional pending more extensive and varied experimentation.

5.4. Limitations

This study demonstrates the potential of RAG frameworks for metabolomic biomarker discovery, but several important limitations must be acknowledged. The Gold set is limited to ten English articles spanning three diseases (Gaucher disease Type 1, NAFLD, and Type 1 diabetes), which constrains the strength of reference-based conclusions. The Silver set expands coverage to 100 papers, but the scope remains restricted to metabolic-disorder literature (ICD-10 E70–E88), limiting generalizability to other biomedical domains. A larger and more diverse corpus would provide a more representative evaluation of the framework’s scalability and applicability across a wider range of conditions. Gold and Silver sets differ in both scope and construction: the Gold set is expert-curated over three disease-focused topics, whereas the Silver set spans ICD-10 E70–E88 metabolic-disorder literature and uses machine-generated QA with partial filtering. This introduces selection and distribution shift between tiers; consequently, we do not interpret absolute metric values as directly comparable across Gold and Silver, and we use each tier to support different validity claims (reference-based validation vs retrieval-grounding stress testing).

Another limitation stems from the study’s focus on short, targeted queries. While these types of questions are well-suited to the initial objectives, they do not capture the complexity of real-world metabolomic research. In practice, researchers often need to tackle multi-faceted, interdisciplinary questions that require multi-hop reasoning and the synthesis of diverse data sources over time. The current framework’s capabilities in such scenarios remain unexplored. Additionally, the study relied exclusively on textual data, which, while valuable, ignores the multi-modal nature of metabolomic research. Data from spectrometry or molecular interaction networks are central to metabolomics and could offer richer context and insights, which the framework did not address.

The indexing and retrieval processes used in this study were static, with no mechanisms for continuous updates or adaptive refinement. Biomedical research is a rapidly evolving field, and the inability of the framework to incorporate new findings or adjust to user feedback limits its relevance and longevity. Furthermore, this study did not sufficiently explore real-world usability concerns, such as providing confidence scores, highlighting retrieved documents, or offering transparent explanations of how generated responses were synthesized. These features are critical for building trust and fostering adoption among clinicians and researchers who depend on reliability and interpretability.

From an engineering perspective, computational efficiency and cost were not rigorously profiled. The study relied on commercial GPT-4o decoding and two embedding models without accounting for the latency or financial burden that would accrue in large-scale or real-time deployments. For institutions with limited computational budgets or privacy-sensitive environments that cannot rely on third-party APIs, these cost and infrastructure questions are pivotal.

In addition to the previously discussed limitations, several further aspects warrant critical consideration. First, the current evaluation strategy relied exclusively on automated metrics (e.g., ROUGE, BLEU, BERTScore, Faithfulness), without incorporating structured human-in-the-loop assessments or inter-rater agreement. This absence of qualitative expert judgment may obscure important nuances in factual accuracy and biomedical relevance, especially in domains where minor terminological shifts (e.g., “reduced” vs. “unchanged”) can meaningfully alter interpretation. Second, all generative outputs were produced using variants of GPT-4 (GPT-4o or GPT-4.1-mini), thereby limiting insights into decoder variability. Given that different LLM architectures exhibit distinct generation behaviors, a broader decoder comparison, including open-source biomedical models, would be essential for generalizing the framework’s applicability across institutional settings. Third, the study did not develop a formal taxonomy of observed error types. While macro-level metrics suggested performance differences between domain-specific and general-purpose pipelines, it remains unclear whether these stemmed from hallucinations, factual omissions, semantic contradiction, or ambiguous phrasing. A structured error analysis would help isolate these causes and guide future improvements. Fourth, while the QA dataset was intentionally designed for clinical specificity and scientific rigor, it does not capture the breadth or ambiguity of real-

1 world information-seeking behavior by clinicians, biologists,
2 or translational researchers. Incorporating user-centric evalu-
3 ations, log analyses, or exploratory query types could improve
4 the system’s ecological validity. Lastly, although the framework
5 holds promise for biomedical support tasks, its ethical, regula-
6 tory, and operational readiness for clinical deployment remains
7 unassessed. Issues such as GDPR compliance, CE/FDA docu-
8 mentation traceability, and robust audit trails for generated con-
9 tent were beyond the current scope but are indispensable for
10 translation into regulated medical environments.

12 5.5. Future work

13 Although the present study demonstrates that retrieval aug-
14 mentation improves grounding and that MedCPT yields a mod-
15 est gain in answer correctness on the Silver benchmark, sev-
16 eral technical extensions are necessary before such a system
17 can serve as a comprehensive biomedical assistant.

18 A first priority is corpus expansion. While the Silver bench-
19 mark already spans 100 papers, the Gold set remains small and
20 the corpus is limited to metabolic disorders. Broadening cover-
21 age to additional disease classes and metabolomic phenotypes
22 would increase statistical power and support deeper domain
23 adaptation. Continuous ingestion of new publications from
24 PubMed, pre-print servers, and publicly available metabolomic
25 repositories, together with integration of additional clinical co-
26 horts, will allow iterative fine-tuning of both the retriever and
27 the decoder on an ever-growing biomedical corpus. Even real-
28 time, dynamic retrieval can be envisaged. Biomedical knowl-
29 edge grows too quickly for periodic, batch-style index rebuild-
30 ing to suffice. Implementing incremental or streaming indexing,
31 possibly via infrastructures such as Apache Kafka or Elastic-
32 search, would permit new studies to enter the search space on
33 the fly. Coupling this capability with iterative re-ranking strate-
34 gies and user-driven feedback loops would enable the system
35 to retire outdated passages and elevate novel, high-quality evi-
36 dence, ensuring that clinicians and researchers receive the most
37 current insights available.

38 The second avenue concerns the inherently multimodal
39 character of metabolomics. Beyond text, researchers rou-
40 tinely interpret numeric assay read-outs, LC-/GC-MS spectra,
41 molecular-interaction graphs, and, increasingly, imaging data.
42 Future iterations of MedDiscover should therefore move toward
43 multimodal embedding pipelines in which specialised encoders
44 e.g. vision transformers for images, graph neural networks for
45 interaction maps, and tabular encoders for quantitative mea-
46 surements) project their outputs into a shared latent space that is
47 jointly searchable with textual embeddings. Retrieval architec-
48 tures based on late fusion or cross-modal attention could then
49 merge heterogeneous evidence streams at query time, offering
50 a more holistic perspective on metabolic processes.

51 Transparency is key to gain researchers’ trust. Users should
52 be able to inspect why a particular passage was retrieved and
53 how it influenced the generated answer. Future work will there-
54 fore explore techniques for displaying source information with
55 similarity scores alongside the retrieved text. Visualising token-
56 level saliency could provide experts a deeper understanding of
57 the logic of both the retriever and the generator.

While the present work centres on metabolomics, the under-
lying methodology is readily transferable to adjacent biomed-
ical domains such as drug discovery, adverse-event surveil-
lance, or clinical decision support. These settings will re-
quire additional capabilities, including advanced query pars-
ing, multi-hop retrieval to chain together several pieces of evi-
dence, and domain-specific output formats, such as recom-
mended dosage ranges or structured adverse-event reports. Ex-
tending the framework to non-English literature through cross-
lingual embedding alignment will further enlarge its practical
reach, particularly in regions where locally generated biomed-
ical research is substantial yet under-indexed in English-centred
databases.

In addition to these planned extensions, several further re-
search avenues emerge. First, future iterations of MedDiscover
should integrate uncertainty quantification techniques, such as
token-level confidence scores, prediction intervals, or Bayesian
calibration layers, to better support user decision-making in
high-stakes biomedical settings. These additions could enable
users to distinguish between confidently grounded outputs and
potentially unreliable inferences, fostering appropriate trust cal-
ibration.

Second, while this study focused on system-level perfor-
mance, a structured taxonomy of model errors remains to be
developed. Categorizing answer-level failures into hallucina-
tions, omissions, factual inaccuracies, or semantic ambiguities
would yield actionable insights into the interaction between re-
trieval and generation components and help prioritize areas for
fine-tuning.

Third, user-centered evaluation is currently lacking. Future
work should conduct structured interaction studies involving
domain experts, such as clinical researchers and metabolomics
specialists, applying human-centered design methodologies
(e.g., cognitive walkthroughs, SUS scores, or semi-structured
interviews) to assess usability, transparency, and alignment with
real-world workflows.

Fourth, incorporating structured biomedical resources such
as MeSH, ChEBI, or SNOMED CT into the retrieval pipeline
may enhance semantic disambiguation and enable multi-hop
reasoning across ontological hierarchies. These resources could
be embedded alongside free text in hybrid indices to improve
retrieval granularity for entity-centric questions.

Finally, to promote open science and benchmarking, future
work will involve the release of a public QA benchmark for
metabolomics, including annotated queries, evidence passages,
and relevance judgments. This will support community-driven
evaluation of biomedical RAG systems and encourage shared
task initiatives within the BioNLP and biomedical IR commu-
nities.

Pursuing these directions of corpus expansion, multi-
modal retrieval, enhanced interpretability, and cross-domain
and cross-lingual generalisation, alongside user-centered eval-
uation, structured error analysis, uncertainty quantification, and
ontology-integrated retrieval, will transform MedDiscover from
a proof-of-concept into a versatile, up-to-date, and trustworthy
bridge between the ever-growing biomedical literature and the
actionable insights required for precision medicine.

6. Conclusion

This study presents MedDiscover as a reproducible RAG evaluation benchmark for metabolomics and metabolic-disorder literature (see Appendix A). Using a two-tier design (Gold expert-curated QA and Silver retrievability-filtered synthetic QA), we show that retrieval augmentation improves grounding relative to non-RAG baselines, and that MedCPT vs Ada yields comparable retrieval metrics with a modest but significant gain in answer correctness on the Silver benchmark. The open-source pipeline and document lists support reproducibility (see Appendix A) and provide a foundation for expansion into broader biomedical domains and modalities.

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Data and Code Availability

The MedDiscover framework is available as open-source software at <https://github.com/VatsalPatel18/MedDiscover>. The analysis pipeline, evaluation outputs, and figure-generation scripts used in this manuscript are available at <https://github.com/VatsalPatel18/MedDiscover-Analysis>. A live application demo can be tested at <https://huggingface.co/spaces/VatsalPatel18/MedDiscover-space>; the original GPT-based application requires OpenAI private API keys, which can be recreated from the GitHub code.

Due to copyright and licensing constraints, we do not redistribute publisher PDFs or extracted full-text derived from closed-access articles. Instead, we release the list of PubMed IDs/DOIs used to assemble the corpus and provide scripts to reproduce preprocessing and indexing from documents that the user can legally access (e.g., open-access sources) (see Appendix A). Released derived artifacts (Gold QA pairs, Silver QA pairs, retrieval metadata, and evaluation tables) are provided without redistributing publisher PDF files.

References

[1] T. B. Brown, B. Mann, N. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, S. Agarwal, A. Herbert-Voss, G. Krueger, T. Henighan, R. Child, A. Ramesh, D. M. Ziegler, J. Wu, C. Winter, C. Hesse, M. Chen, E. Sigler, M. Litwin, S. Gray, B. Chess, J. Clark, C. Berner, S. McCandlish, A. Radford, I. Sutskever, D. Amodei, Language models are few-shot learners, *Advances in Neural Information Processing Systems* 33 (2020). URL <https://arxiv.org/abs/2005.14165>

[2] S. Lista, R. González-Domínguez, S. López-Ortiz, et al., Integrative metabolomics science in alzheimer's disease: Relevance and future perspectives, *Ageing Research Reviews* 89 (2023) 101987. doi:10.1016/j.arr.2023.101987. URL <https://www.sciencedirect.com/science/article/pii/S1568163723001460>

[3] L. Brennan, Metabolomics: a tool to aid dietary assessment in nutrition, *Current Opinion in Food Science* 16 (2017) 96–99. doi:10.1016/j.cofs.2017.09.003. URL <https://www.sciencedirect.com/science/article/pii/S2214799317300966>

[4] L. Huang, W. Yu, W. Ma, W. Zhong, Z. Feng, H. Wang, Q. Chen, W. Peng, X. Feng, B. Qin, T. Liu, A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions (2023). arXiv:2311.05232. URL <https://arxiv.org/abs/2311.05232>

[5] C. Zhang, A. Le, Diabetes and cancer: The epidemiological and metabolic associations, *Advances in Experimental Medicine and Biology* 1311 (2021) 217–227. doi:10.1007/978-3-030-65768-0_16.

[6] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, S. Riedel, Retrieval-augmented generation for knowledge-intensive nlp tasks, arXiv preprint (2020). arXiv:arXiv:2005.11401.

[7] V. Karpukhin, B. Oğuz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, W. tau Yih, Dense passage retrieval for open-domain question answering (2020). arXiv:2004.04906. URL <https://arxiv.org/abs/2004.04906>

[8] D. S. Wishart, Emerging applications of metabolomics in drug discovery and precision medicine, *Nature Reviews Drug Discovery* 15 (2016) 473–484. doi:10.1038/nrd.2016.32. URL <https://doi.org/10.1038/nrd.2016.32>

[9] E. G. Armitage, C. Barbas, Metabolomics in cancer biomarker discovery: current trends and future perspectives, *Journal of Pharmaceutical and Biomedical Analysis* 87 (2014) 1–11. doi:10.1016/j.jpba.2013.08.041.

[10] C. H. Johnson, J. Ivanisevic, G. Siuzdak, Metabolomics: beyond biomarkers and towards mechanisms, *Nature Reviews Molecular Cell Biology* 17 (7) (2016) 451–459. doi:10.1038/nrm.2016.25. URL <https://doi.org/10.1038/nrm.2016.25>

[11] A. Rosato, L. Tenori, M. Cascante, et al., From correlation to causation: analysis of metabolomics data using systems biology approaches, *Metabolomics* 14 (2018) 37. doi:10.1007/s11306-018-1335-y. URL <https://doi.org/10.1007/s11306-018-1335-y>

[12] J. Labory, E. Njomgue-Fotso, S. Bottini, Benchmarking feature selection and feature extraction methods to improve the performances of machine-learning algorithms for patient classification using metabolomics biomedical data, *Computational and Structural Biotechnology Journal* 23 (2024) 1274–1287. doi:10.1016/j.csbj.2024.03.016.

[13] A. Qoku, N. Katsaouni, N. Flinner, F. Buettner, M. H. Schulz, Multimodal analysis methods in predictive biomedicine, *Computational and Structural Biotechnology Journal* 21 (2023) 5829–5838. doi:10.1016/j.csbj.2023.11.011.

[14] T. Theodosiou, K. Vrettos, I. Baltsavia, F. Baltoumas, N. Papanikolaou, A. N. Antonakis, D. Mossialos, C. A. Ouzounis, V. J. Promponas, M. Karagiani, E. Chatzaki, S. Brandau, G. A. Pavlopoulos, E. Andreakos, I. Iliopoulos, Biotextquest v2.0: An evolved tool for biomedical literature mining and concept discovery, *Computational and Structural Biotechnology Journal* 23 (2024) 3247–3253. doi:10.1016/j.csbj.2024.08.016.

[15] S. Murali, S. S., S. R, ReMAG-KR: Retrieval and Medically Assisted Generation with Knowledge Reduction for Medical Question Answering, in: X. Fu, E. Fleisig (Eds.), *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 4: Student Research Workshop)*, Association for Computational Linguistics, Bangkok, Thailand, 2024, pp. 140–145. doi:10.18653/v1/2024.acl-srw.13. URL <https://aclanthology.org/2024.acl-srw.13>

[16] Y. Xiong, X. Yang, L. Liu, K.-C. Wong, Q. Chen, Y. Xiang, B. Tang, EARA: Improving Biomedical Semantic Textual Similarity with Entity-Aligned Attention and Retrieval Augmentation, in: H. Bouamor, J. Pino, K. Bali (Eds.), *Findings of the Association for Computational Linguistics: EMNLP 2023*, Association for Computational Linguistics, Singapore, 2023, pp. 8760–8771. doi:10.18653/v1/2023.findings-emnlp.586. URL <https://aclanthology.org/2023.findings-emnlp.586>

[17] B. Bašaragin, A. Ljajić, D. Medvecki, L. Cassano, M. Košprdić, N. Milošević, How do you know that? Teaching Generative Language Models to Reference Answers to Biomedical Questions, arXiv:2407.05015 (Jul. 2024). doi:10.48550/arXiv.2407.05015.

- URL <http://arxiv.org/abs/2407.05015>
- [18] C. Wang, Q. Long, M. Xiao, X. Cai, C. Wu, Z. Meng, X. Wang, Y. Zhou, BioRAG: A RAG-LLM Framework for Biological Question Reasoning, *arXiv:2408.01107* (Aug. 2024). doi:10.48550/arXiv.2408.01107. URL <http://arxiv.org/abs/2408.01107>
- [19] Q. Jin, W. Kim, Q. Chen, D. C. Comeau, L. Yeganova, W. J. Wilbur, Z. Lu, Medcpt: Contrastive pre-trained transformers with large-scale pubmed search logs for zero-shot biomedical information retrieval, *Bioinformatics* 39 (11) (2023) btad651. doi:10.1093/bioinformatics/btad651. URL <http://dx.doi.org/10.1093/bioinformatics/btad651>
- [20] D. Soong, S. Sridhar, H. Si, J.-S. Wagner, A. C. C. Sá, C. Y. Yu, K. Karagoz, M. Guan, S. Kumar, H. Hamadeh, B. W. Higgs, Improving accuracy of GPT-3/4 results on biomedical data using a retrieval-augmented language model, *PLOS Digital Health* 3 (8) (2024) e0000568, publisher: Public Library of Science. doi:10.1371/journal.pdig.0000568. URL <https://journals.plos.org/digitalhealth/article?id=10.1371/journal.pdig.0000568>
- [21] J. Lee, W. Yoon, S. Kim, D. Kim, S. Kim, C. H. So, J. Kang, Biobert: a pre-trained biomedical language representation model for biomedical text mining, *Bioinformatics* 36 (4) (2019) 1234–1240. *arXiv:https://academic.oup.com/bioinformatics/article-pdf/36/4/1234/48983216/bioinformatics_36_4_1234.pdf*, doi:10.1093/bioinformatics/btz682. URL <https://doi.org/10.1093/bioinformatics/btz682>
- [22] A. R. Pelletier, J. Ramirez, I. Adam, S. Sankar, Y. Yan, D. Wang, D. Steinicke, W. Wang, P. Ping, Explainable Biomedical Hypothesis Generation via Retrieval Augmented Generation enabled Large Language Models, *arXiv:2407.12888* (Jul. 2024). doi:10.48550/arXiv.2407.12888. URL <http://arxiv.org/abs/2407.12888>
- [23] Y. Shi, S. Xu, T. Yang, Z. Liu, T. Liu, Q. Li, X. Li, N. Liu, MKRAG: Medical Knowledge Retrieval Augmented Generation for Medical Question Answering, *arXiv:2309.16035* (Aug. 2024). doi:10.48550/arXiv.2309.16035. URL <http://arxiv.org/abs/2309.16035>
- [24] J. Yang, L. Shu, H. Duan, H. Li, RDguru: A Conversational Intelligent Agent for Rare Diseases, *IEEE Journal of Biomedical and Health Informatics* (2024) 1–13Conference Name: IEEE Journal of Biomedical and Health Informatics. doi:10.1109/JBHI.2024.3464555. URL <https://ieeexplore.ieee.org/abstract/document/10684379>
- [25] Z. Yuan, Q. Jin, C. Tan, Z. Zhao, H. Yuan, F. Huang, S. Huang, RAMM: Retrieval-augmented Biomedical Visual Question Answering with Multimodal Pre-training, in: *Proceedings of the 31st ACM International Conference on Multimedia, MM '23*, Association for Computing Machinery, New York, NY, USA, 2023, pp. 547–556. doi:10.1145/3581783.3611830. URL <https://doi.org/10.1145/3581783.3611830>
- [26] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, I. Polosukhin, Attention is all you need (2023). *arXiv:1706.03762*. URL <https://arxiv.org/abs/1706.03762>
- [27] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, I. Polosukhin, Attention is all you need (2021). *arXiv:1706.03762*. URL <https://arxiv.org/abs/1706.03762>
- [28] C. Wang, Q. Long, M. Xiao, X. Cai, C. Wu, Z. Meng, X. Wang, Y. Zhou, Biorag: A rag-llm framework for biological question reasoning (2024). *arXiv:2408.01107*. URL <https://arxiv.org/abs/2408.01107>
- [29] G. Huang, Y. Li, S. Jameel, Y. Long, G. Papanastasiou, From explainable to interpretable deep learning for natural language processing in healthcare: How far from reality?, *Computational and Structural Biotechnology Journal* 24 (2024) 362–373. doi:10.1016/j.csbj.2024.05.004.
- [30] X. Lin, S. Ma, J. Shan, X. Zhang, S. X. Hu, T. Guo, S. Z. Li, K. Yu, BioKGBench: A Knowledge Graph Checking Benchmark of AI Agent for Biomedical Science, *arXiv:2407.00466* (Jun. 2024). doi:10.48550/arXiv.2407.00466. URL <http://arxiv.org/abs/2407.00466>
- [31] P.-H. Li, Y.-Y. Sun, H.-F. Juan, C.-Y. Chen, H.-K. Tsai, J.-H. Huang, LORE: A Literature Semantics Framework for Evidenced Disease-Gene Pathogenicity Prediction at Scale, pages: 2024.08.10.24311801 (Aug. 2024). doi:10.1101/2024.08.10.24311801. URL <https://www.medrxiv.org/content/10.1101/2024.08.10.24311801v1>
- [32] H. Schäfer, A. Idrissi-Yaghir, K. Arzideh, H. Damm, T. M. G. Pakull, C. S. Schmidt, M. Bahn, G. Lodde, E. Livingstone, D. Schadendorf, F. Nensa, P. A. Horn, C. M. Friedrich, Biokgrapher: Initial evaluation of automated knowledge graph construction from biomedical literature, *Computational and Structural Biotechnology Journal* 24 (2024) 639–660. doi:10.1016/j.csbj.2024.10.017.
- [33] M. Sud, E. Fahy, D. Cotter, K. Azam, I. Vadivelu, C. Burant, A. Edison, O. Fiehn, R. Higashi, K. S. Nair, S. Sumner, S. Subramaniam, Metabolomics workbench: An international repository for metabolomics data and metadata, metabolite standards, protocols, tutorials and training, and analysis tools, *Nucleic Acids Research* 44 (D1) (2016) D463–D470. doi:10.1093/nar/gkv1042. URL <https://doi.org/10.1093/nar/gkv1042>
- [34] P. Italiani, G. Moro, L. Ragazzi, Enhancing legal question answering with data generation and knowledge distillation from large language models, *Artificial Intelligence and Law* (2025). doi:10.1007/s10506-025-09463-9. URL <https://link.springer.com/article/10.1007/s10506-025-09463-9>
- [35] D. Gupta, D. Bartels, D. Demner-Fushman, A dataset of medical questions paired with automatically generated answers and evidence-supported references, *Scientific Data* 12 (1) (Jun. 2025). doi:10.1038/s41597-025-05233-z. URL <http://dx.doi.org/10.1038/s41597-025-05233-z>
- [36] Ragas: Automated evaluation of retrieval-augmented generation, *arXiv:2309.15217* (2023). URL <https://arxiv.org/abs/2309.15217>
- [37] J. Lee, W. Yoon, S. Kim, D. Kim, S. Kim, C. H. So, J. Kang, Biobert: a pre-trained biomedical language representation model for biomedical text mining, *Bioinformatics* 36 (4) (2019) 1234–1240. doi:10.1093/bioinformatics/btz682.
- [38] T. Almeida, R. A. A. Jonker, J. Reis, J. R. Almeida, S. Matos, Notebook for the BioASQ Lab at CLEF 2024, Conference paper (2023).
- [39] I. Beltagy, K. Lo, A. Cohan, Scibert: A pretrained language model for scientific text (2019). *arXiv:1903.10676*. URL <https://arxiv.org/abs/1903.10676>

1 Appendix A. Reproducibility checklist

2 Appendix B. Supplementary Gold-set results

3 Appendix C. Detailed Statistical Significance Tests

4 This appendix expands on the Gold set statistical analysis of
5 Section 4. The goal was to determine whether any of the seven
6 model pipelines differed reliably in their median metric scores.
7 The following steps were applied.

- 8 1. **Normality checks.** Every model–metric distribution was
9 examined with the Shapiro–Wilk test. All yielded $p < 0.05$, implying non-normal data.
- 10 2. **Variance homogeneity.** Levene’s test across models re-
11 vealed unequal variances for seven of the ten metrics, es-
12 pecially the retrieval-oriented ones (Faithfulness, Context
13 Recall, Context Precision) and BLEU.
- 14 3. **Omnibus comparison.** Because parametric assumptions
15 were violated, a Kruskal–Wallis test on ranks was carried
16 out for each metric across all model variants: the non-
17 RAG baseline, Ada + GPT-4o, three MedCPT configu-
18 rations (GPT-4, GPT-4.1-nano, GPT-4.1-mini), and two
19 NotebookLM variants.
- 20 4. **Pairwise contrasts.** Significant omnibus results were
21 followed up with Dunn’s test, applying a Bonferroni cor-
22 rection. The general-purpose Ada + GPT-4o pipeline
23 served as the reference.

24 Appendix C.1. Raw p -values

25 The adjusted Dunn p -values and corresponding Kruskal–
26 Wallis p for each metric are listed in Tables C.6 and C.7. The
27 numbers were imported directly from `../MedDiscover-Ana-`
28 `lysis/stats_ada_vs_others_basic.csv` and `../MedDi-`
29 `scover-Analysis/stats_ada_vs_others_contextual`
30 `.csv`.
31 Overall, the non-parametric analysis on the Gold set indi-
32 cates that retrieval augmentation improves grounding relative to
33 the non-RAG baseline, while decoder upgrades (GPT-4.1 vari-
34 ants) improve lexical and answer-quality metrics beyond fixed-
35 decoder comparisons.

36 Appendix D. Prompt templates

37 Appendix D.1. Evidence-grounded answering prompt (Med- 38 Discover RAG; GPT-4o)

39 [SYSTEM]

40 You are a biomedical assistant. Answer the question using ONLY the provided context.

41 If the context does not contain the answer, reply: "Insufficient evidence in the provided context."

42 [USER]

43 Question: {question}

44 Context:

45 {top_k_context_chunks}

46 Instructions:

- Provide a concise answer (max 25 tokens).
- Prefer specific entities (metabolites, diseases, study).
- Do not add unsupported claims.
- If multiple findings exist, report the one directly supported by the evidence.

47 Appendix D.2. Synthetic QA generation prompt (Silver set; 48 gpt-4.1-mini)

49 [SYSTEM]

50 You generate biomedical questions and short answers from

51 [USER]

52 Given the excerpt below, generate exactly 3 diverse ques-
53 tions from the excerpt. Prefer non-trivial questions when possible.

54 Excerpt:

55 {chunk_or_grouped_context}

56 Output format (one per line):

57 Q: ...

58 A: ...

59 Appendix D.3. NotebookLM minimal-output instruction

60 Answer in minimal words.

Table A.5: Reproducibility checklist (artifacts and where to find them).

Artifact	Location	Notes
Core RAG pipeline	https://github.com/VatsalPatel118/MedDiscover	Indexing, retrieval, prompting, and RAG inference implementation.
Analysis + figures	https://github.com/VatsalPatel118/MedDiscover-Analysis	Scripts to reproduce tables/figures and evaluation outputs.
Paper list (IDs)	MedDiscover-Analysis release package (evaluation_set.zip)	PubMed IDs / DOIs used to assemble the 100-paper Silver corpus.
Gold QA set	MedDiscover release package (qa_samples_gold.tsv)	30 expert-curated QA pairs + references; evaluation subset noted in Section 4.
Silver QA outputs	MedDiscover-Analysis evaluation outputs (qa_72_plus_29_ada_with_ragas.csv)	Synthetic QA + retrievability metadata; post-QC sampled to 300 per embedding.
Silver QA generator	MedDiscover QA-generation pipeline	GPT-4.1-mini; 500-token chunks, 50-token overlap.
QC post-processing	MedDiscover QA post-processing pipeline	De-duplication, min-length filters, and template filtering (Table 3).
Stats inputs (Gold)	MedDiscover-Analysis statistics tables	Dunn/KW tables used for Supplementary statistics appendix.
Environment	MedDiscover-Analysis requirements	Python requirements for reproducible analysis.

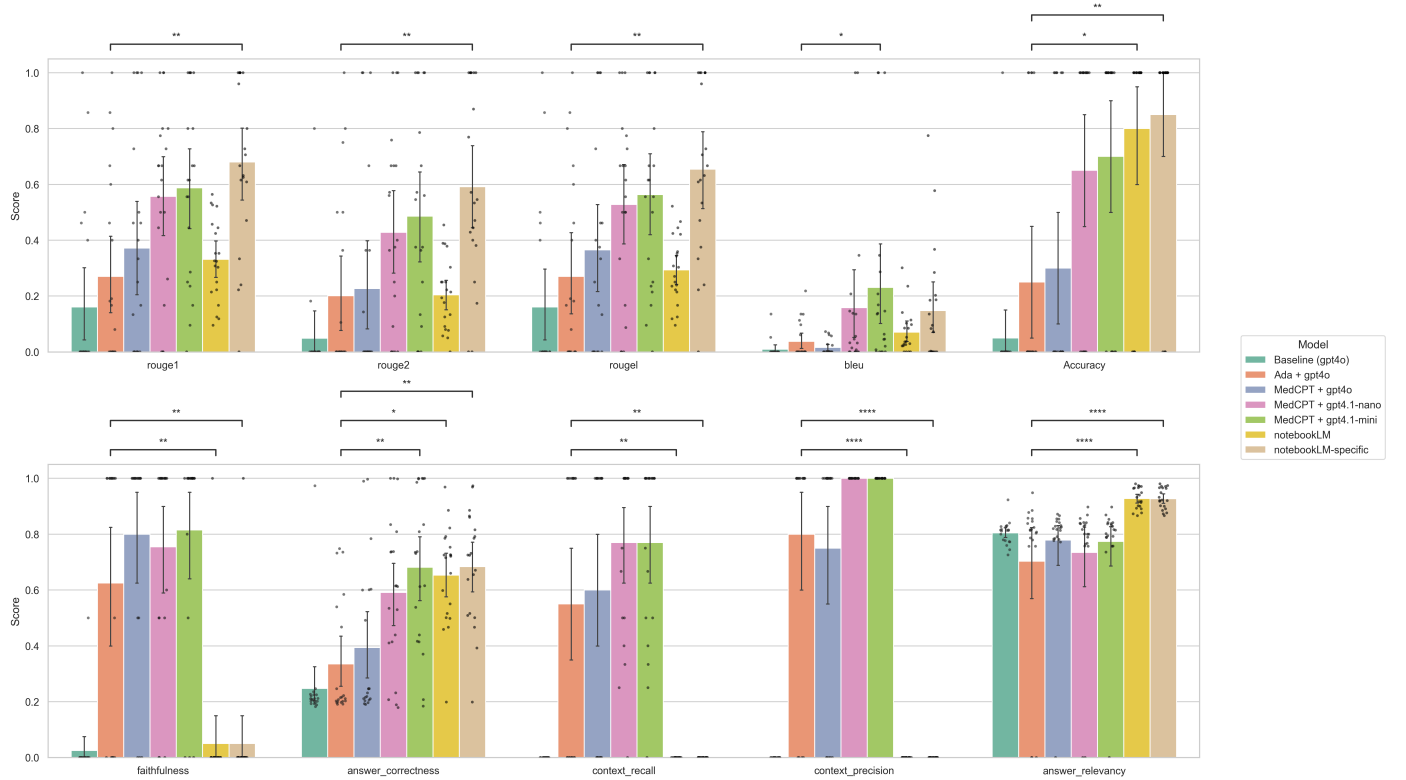


Figure B.4: Supplementary Gold-set results across additional configurations (decoder sensitivity variants and black-box comparator). This figure is provided for completeness; the primary comparisons in the main text restrict to fixed-decoder (GPT-4o) evaluations to avoid conflating retriever effects with decoder capacity. Note that NotebookLM does not expose retrieved contexts; context-dependent metrics are therefore not comparable and should be interpreted as not available rather than as zero performance.

Table C.6: Dunn’s test results (basic metrics).

Metric	Compared model	p_{adj}	p_{KW}
rouge1	MedCPT + gpt4o	1	2.003×10^{-6}
rouge1	MedCPT + gpt4.1-nano	0.172794	2.003×10^{-6}
rouge1	MedCPT + gpt4.1-mini	0.0729964	2.003×10^{-6}
rouge1	notebookLM	1	2.003×10^{-6}
rouge1	notebookLM-specific	0.00442385	2.003×10^{-6}
rouge2	MedCPT + gpt4o	1	1.521×10^{-7}
rouge2	MedCPT + gpt4.1-nano	0.314738	1.521×10^{-7}
rouge2	MedCPT + gpt4.1-mini	0.0923275	1.521×10^{-7}
rouge2	notebookLM	1	1.521×10^{-7}
rouge2	notebookLM-specific	0.00301024	1.521×10^{-7}
rougel	MedCPT + gpt4o	1	5.122×10^{-6}
rougel	MedCPT + gpt4.1-nano	0.261727	5.122×10^{-6}
rougel	MedCPT + gpt4.1-mini	0.104081	5.122×10^{-6}
rougel	notebookLM	1	5.122×10^{-6}
rougel	notebookLM-specific	0.00784415	5.122×10^{-6}
bleu	MedCPT + gpt4o	1	5.924×10^{-6}
bleu	MedCPT + gpt4.1-nano	0.683111	5.924×10^{-6}
bleu	MedCPT + gpt4.1-mini	0.034923	5.924×10^{-6}
bleu	notebookLM	0.44618	5.924×10^{-6}
bleu	notebookLM-specific	0.648565	5.924×10^{-6}
Accuracy	MedCPT + gpt4o	1	2.971×10^{-8}
Accuracy	MedCPT + gpt4.1-nano	0.245187	2.971×10^{-8}
Accuracy	MedCPT + gpt4.1-mini	0.0956256	2.971×10^{-8}
Accuracy	notebookLM	0.0110328	2.971×10^{-8}
Accuracy	notebookLM-specific	0.00325786	2.971×10^{-8}

Table C.7: Dunn’s test results (context metrics).

Metric	Compared model	p_{adj}	p_{KW}
faithfulness	MedCPT + gpt4o	1	2.240×10^{-14}
faithfulness	MedCPT + gpt4.1-nano	1	2.240×10^{-14}
faithfulness	MedCPT + gpt4.1-mini	1	2.240×10^{-14}
faithfulness	notebookLM	0.00310424	2.240×10^{-14}
faithfulness	notebookLM-specific	0.00310424	2.240×10^{-14}
answer_correctness	MedCPT + gpt4o	1	2.678×10^{-8}
answer_correctness	MedCPT + gpt4.1-nano	0.0929803	2.678×10^{-8}
answer_correctness	MedCPT + gpt4.1-mini	0.00302486	2.678×10^{-8}
answer_correctness	notebookLM	0.0110999	2.678×10^{-8}
answer_correctness	notebookLM-specific	0.00262388	2.678×10^{-8}
context_recall	MedCPT + gpt4o	1	2.072×10^{-15}
context_recall	MedCPT + gpt4.1-nano	1	2.072×10^{-15}
context_recall	MedCPT + gpt4.1-mini	1	2.072×10^{-15}
context_recall	notebookLM	0.00397699	2.072×10^{-15}
context_recall	notebookLM-specific	0.00397699	2.072×10^{-15}
context_precision	MedCPT + gpt4o	1	5.022×10^{-19}
context_precision	MedCPT + gpt4.1-nano	1	5.022×10^{-19}
context_precision	MedCPT + gpt4.1-mini	1	5.022×10^{-19}
context_precision	notebookLM	2.18907×10^{-8}	5.022×10^{-19}
context_precision	notebookLM-specific	2.18907×10^{-8}	5.022×10^{-19}
answer_relevancy	MedCPT + gpt4o	1	6.635×10^{-15}
answer_relevancy	MedCPT + gpt4.1-nano	1	6.635×10^{-15}
answer_relevancy	MedCPT + gpt4.1-mini	1	6.635×10^{-15}
answer_relevancy	notebookLM	6.44792×10^{-6}	6.635×10^{-15}
answer_relevancy	notebookLM-specific	6.05982×10^{-6}	6.635×10^{-15}

Leveraging MedDiscover: A Domain-Specific Retrieval-Augmented Generation with Large Language Models (RAG-LLMs) Framework for Biomedical Evidence-Grounded Knowledge Extraction in Metabolomics

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Abstract

Large Language Models (LLMs) combined with Retrieval-Augmented Generation (RAG) methods hold significant promise for enhancing biomedical knowledge extraction. In this study, we introduce *MedDiscover*, a specialized RAG-LLM framework optimized for metabolomics, which can improve biomedical question answering, employing but claims about domain-specific embeddings (MedCPT) integrated with GPT-based decoders. We rigorously evaluated *MedDiscover* against general-purpose RAG (OpenAI Ada-002), a non-RAG baseline (GPT-4o), and Google's NotebookLM, using a carefully curated metabolomics retrievers require transparent, reproducible evaluation. We present *MedDiscover*, an open-source RAG implementation and benchmark instantiated for metabolomics and metabolic-disorder literature, with a two-tier dataset design: a Gold set of 10 papers and 30 expert-curated question-answer dataset. Statistical analyses revealed that retrieval augmentation provided robust performance gains over pairs with traceable references, and a Silver set of 100 papers curated using ICD-10 metabolic disorder codes E70–E88 with ~600 synthetic QA pairs and retrievability metadata (Ada n=300, MedCPT n=300). On the Gold set, retrieval augmentation improves grounding compared with a non-RAG models (e.g., Faithfulness, Context Precision, and Context Recall, all $p < 0.001$). Notably, *MedDiscover* leveraging MedCPT embeddings paired with a GPT-4.1-mini decoder significantly outperformed general-purpose embeddings on Answer Correctness ($p = 0.003$), BLEU scores ($p = 0.035$), baseline. On the Silver benchmark, MedCPT and Ada show comparable retrieval-centric performance (faithfulness $p = 0.85$, context recall $p = 0.50$, context precision $p = 0.49$, answer relevancy $p = 0.82$), while MedCPT yields higher answer correctness (Mann–Whitney U $p = 5.72 \times 10^{-11}$, Cliff's $\delta = 0.303$; common-language effect = 0.651). We release code, evaluation scripts, and ROUGE metrics (ROUGE-1: $p = 0.073$, ROUGE-2: $p = 0.092$), underscoring the value of domain-specific embeddings and decoder capacity. Although NotebookLM achieved high lexical alignment (ROUGE-1: $p = 0.004$, ROUGE-2: $p = 0.003$), its opaque retrieval process limits its utility in scenarios requiring explicit evidence-based grounding. Overall, *MedDiscover* demonstrates superior retrieval precision and factual grounding in metabolomics knowledge extraction tasks, highlighting the critical synergy between domain adaptation and retrieval augmentation in biomedical research document lists to support reproducibility.

Keywords: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Biomedical Research, Metabolomics, Knowledge Extraction

1. Introduction

Large Language Models (LLMs), such as GPT-3 and GPT-4, have significantly transformed natural language processing (NLP), showcasing advanced capabilities in tasks including summarization, dialogue systems, reasoning, and text generation [1]. Their capacity to handle vast amounts of unstructured textual data, generating contextually relevant and coherent responses, presents significant opportunities in biomedical research, especially for navigating the extensive and complex biomedical literature available in repositories like PubMed [2]. Nonetheless, deploying these general-purpose LLMs in specialized biomedical subfields, such as metabolomics, presents substantial limitations, notably the occurrence of hallucinations—, outputs that seem plausible but are factually incorrect—, and difficulty in accurately interpreting complex biomedical terminologies [3–5]. Furthermore, traditional LLMs are limited

by static knowledge acquired during training, lacking dynamic integration of emerging, domain-specific data necessary in rapidly evolving fields like precision medicine and metabolomics.

Retrieval-Augmented Generation (RAG) methodologies have emerged to address these limitations by dynamically retrieving external knowledge and integrating it into the generative process. RAG significantly improves factual accuracy, transparency, and context-specificity of responses generated by LLMs [6]. Despite these advantages, optimizing RAG pipelines for domain-specific applications remains challenging. The quality and relevance of retrieved documents, efficiency of document ranking and filtering mechanisms, and the effective integration of retrieval signals into generative processes critically influence the overall performance of RAG-enhanced systems [7]. Moreover, evaluating the reliability and interpretability of responses generated by RAG systems remains essential, partic-

ularly in biomedical contexts where decision-making depends on accurate, reliable, and well-sourced information.

This paper introduces *MedDiscover*, a specialized RAG-LLM solution explicitly tailored for biomedical research, focusing on metabolomics an open-source RAG implementation and evaluation benchmark instantiated for metabolomics literature. Rather than proposing a new retrieval architecture, we focus on reproducible assessment of retrieval augmentation, embedding choice, and decoder sensitivity within a metabolomics/metabolic-disorder corpus (see Appendix Appendix A). Metabolomics, pivotal for biomarker discovery, disease diagnosis, and personalized medicine, faces significant literature management challenges due to its rapidly expanding knowledge base [8, 9]. We propose that a domain-specific RAG-LLM solution, such as MedDiscover, can substantially enhance information retrieval accuracy, interpretability,

We focus on metabolomics literature in the context of metabolic disorders (ICD-10 E70–E88) because these conditions are metabolite-centric and representative of biomarker-driven metabolomics workflows.

Our contributions are threefold:

- A two-tier benchmark: a Gold set with expert-curated QA pairs and a Silver set with retrievability-filtered synthetic QA pairs at scale.
- Transparent RAG evaluation with retrieval-centric metrics (RAGAS) and statistical testing that reports effect sizes and confidence intervals.
- An open-source reproducibility package (code, prompts, document lists, and evaluation pipeline) that enables independent re-analysis (see Appendix Appendix A).

2. Background: RAG-LLM RAG in Metabolomics

2.1. Metabolomics Research: Relevance and Challenges

Metabolomics connects genetic and environmental influences with phenotypic outcomes through metabolites, acting as substrates, intermediates, and end products within biological pathways, thereby reflecting immediate physiological states and adaptations [10, 11]. Researchers often struggle with the complexity, terminology, and rapidly expanding literature, complicating efficient knowledge extraction and practical hypothesis generation. Traditional metabolomics databases require structured queries, limiting intuitive exploration, especially for researchers with limited domain expertise. Beyond the rapid growth of the metabolomics literature, metabolomics datasets themselves are often high-dimensional and sparse (i.e., $p \gg n$), and can exhibit class- or feature-level imbalance, which complicates robust patient classification and biomarker discovery. Recent benchmarking work in metabolomics biomedical data systematically evaluated feature extraction and feature selection pipelines and reported that supervised feature selection can improve downstream classification performance, providing practical guidance on when to apply these techniques [12]. In parallel, predictive biomedicine increasingly combines multiple

data modalities (e.g., multi-omics measurements with imaging or genomic variation), further increasing the complexity of evidence synthesis and interpretation [13]. Existing biomedical literature-mining tools such as BioTextQuest v2.0 support term-driven clustering and named-entity extraction from PubMed abstracts, but remain primarily query-and-browse systems rather than evidence-grounded natural-language question answering interfaces [14].

2.2. Core LLM Components for Biomedical Applications

LLMs constitute the cornerstone of modern NLP, fundamentally built upon three core components: tokenization, embeddings, and the transformer-based architecture. A nuanced understanding of these components is vital for effectively deploying and customizing RAG-LLMs RAG systems, particularly within specialized contexts such as biomedical applications.

Tokenization is the foundational preprocessing step, converting raw textual data into discrete units, or tokens, that models can interpret computationally [15]. The primary goal of tokenization is to segment continuous text into manageable pieces, typically words, subwords, or characters. Subword tokenization methods, such as Byte Pair Encoding (BPE), WordPiece, and SentencePiece, have become predominant due to their ability to handle extensive vocabularies efficiently, mitigating the "out-of-vocabulary" (OOV) problem by decomposing words into meaningful fragments [16, 17]. In biomedical contexts, traditional tokenizers often inadequately handle complex terminologies involving gene names, chemical compounds, and clinical terms. This limitation has led to the development of domain-specific tokenizers designed explicitly for biomedical terminologies, thus enhancing entity recognition and semantic fidelity [18, 19]. For instance, custom tokenizers associated with BioBERT or PubMedBERT significantly outperform general-purpose tokenizers by accurately preserving semantic integrity of biomedical terms [20, 21]. Effective tokenization directly influences the quality of embeddings generated, as inaccurate segmentation of specialized biomedical terms can lead to suboptimal semantic representations. Consequently, tokenization precision is a crucial prerequisite for producing high-quality embeddings tailored to biomedical applications.

Embeddings transform discrete tokens into dense, continuous vector representations that capture semantic and contextual meanings [22]. These representations enable models to quantify semantic similarities and differences between terms. In practice, embeddings are often precomputed during a model's training and optimized to encapsulate syntactic, semantic, and even domain-specific knowledge. In general-purpose LLMs (e.g., GPT-4, GPT-3.5, LLaMA), embeddings capture broad linguistic patterns useful for diverse applications. However, when fine-tuned or pretrained on specialized biomedical datasets—, such as PubMed abstracts, clinical guidelines, and medical textbooks—, these embeddings become highly enriched with domain-specific semantic structures [23, 24]. Biomedical embeddings from models like BioBERT, PubMedBERT, ClinicalBERT, and MedCPT better reflect nuanced relationships specific to medical terminology, drug inter-

actions, and biological pathways, significantly enhancing performance on domain-specific tasks [15, 25]. These embedding representations are subsequently processed by transformer architectures, which leverage self-attention mechanisms to contextually interpret token relationships. Thus, embeddings serve as the critical bridge between tokenized input and the sophisticated understanding capabilities of transformer models.

Transformer architectures, driven by their self-attention mechanisms, form the foundation of nearly all modern LLMs [26]. These mechanisms enable the model to dynamically assess the importance of each token in a sequence relative to its context. While the intricacies of attention mechanisms underpin their functionality, transformers can be categorized into three types based on their architectural design and output: encoder-only models, which transform input sequences into contextual vector representations; decoder-only models, which generate text autoregressively; and encoder-decoder models, which integrate both components to map input sequences to distinct outputs. The evolution of transformers began with the encoder-decoder architecture, introduced in the original Transformer paper by Vaswani et al. [27], designed for sequence-to-sequence tasks like machine translation and summarization. This was followed by the rise of decoder-only models, such as the GPT series, which focus on autoregressive text generation and excel in tasks like text completion and conversational AI. Later, encoder-only models, such as BERT, emerged, leveraging bidirectional context for tasks requiring deep language understanding, such as classification and named entity recognition. Today, decoder-only models dominate generative applications due to their scalability with increasing parameters and vast datasets, enhancing their few-shot learning capabilities, while encoder-only models remain vital for understanding-focused tasks, particularly in specialized domains like biomedicine [1]. Encoder-only architectures (BioBERT, MedCPT) excel in contextual understanding critical for biomedical retrieval tasks. Decoder-only models (e.g., GPT-4) dominate generative tasks but often require extensive fine-tuning for precision in biomedical contexts [1, 28].

The performance of LLMs in specialized tasks is profoundly influenced by how effectively tokenization, embeddings, and transformer architectures are adapted and aligned. This interplay becomes especially evident when comparing general-purpose models against domain-specific counterparts in biomedical contexts. Modern LLMs broadly categorize into general-purpose and domain-specific models. General-purpose LLMs, such as GPT-4, GPT-3.5, and LLaMA, come equipped with integrated tokenizers, embedding models, and transformer architectures, suitable for diverse language processing tasks. However, they may struggle with the depth and specificity of biomedical terminology without further adaptation [17]. Conversely, domain-specific LLMs are either fine-tuned or pretrained extensively on specialized corpora, adapting their tokenizers and embeddings to better capture domain-specific language nuances. Biomedical domain-specific LLMs exemplified by BioBERT, PubMedBERT, ClinicalBERT, and MedCPT are trained explicitly on biomedical corpora, such as PubMed abstracts, clinical question-answer datasets (e.g.,

MedQA, MedMCQA), electronic health records (EHRs), and specialized medical textbooks [18, 19, 25]. For example, BioBERT leverages extensive pretraining on PubMed and PMC corpora to effectively handle biomedical named entity recognition (NER) and semantic textual similarity tasks [21]. PubMedBERT further specializes by exclusively pretraining on PubMed abstracts, yielding superior performance in biomedical text mining tasks [28]. ClinicalBERT utilizes patient-centric clinical notes, enabling more accurate understanding of clinical text [15]. Meanwhile, MedCPT introduces advanced embedding methods based on user interactions from PubMed searches, capturing relevance judgments at scale to significantly enhance zero-shot semantic retrieval [19]. In summary, each component—tokenization, embeddings, and transformer architecture—plays a critical role in defining LLM functionality. Domain-specific adaptations of these components, particularly within the biomedical field, substantially enhance the accuracy, reliability, and contextual appropriateness of LLM-based applications, effectively bridging the gap between general linguistic understanding and the highly specialized demands of biomedical research and clinical practice. [In biomedical NLP, these modelling choices must also be evaluated through the lens of transparency and interpretability. A recent scoping review in healthcare NLP highlights persistent gaps in best practices and systematic benchmarks for explainable and interpretable deep learning, reinforcing the need for designs that support auditability and evidence traceability in high-stakes settings \[29\].](#)

2.3. Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) transforms natural language processing by integrating LLMs with real-time retrieval from external knowledge bases. Unlike traditional LLMs, which rely on static, potentially outdated knowledge, RAG dynamically retrieves contextually relevant information, making it ideal for biomedical research where precision and up-to-date literature are critical [6]. By grounding responses in retrieved evidence, RAG mitigates LLM hallucinations and domain mismatches, enhancing accuracy in fields like metabolomics [4, 18].

RAG’s core mechanism involves encoding a user query—e.g., “What metabolites are associated with early-stage Alzheimer’s disease?”—into a dense vector embedding using models like MedCPT or BioBERT [19, 21]. This embedding, capturing semantic intent, queries a vector database of pre-processed documents (e.g., PubMed articles). Retrieved text chunks augment the query, enabling the LLM to generate evidence-based responses [22]. Embeddings, built from tokenized text (as discussed in the Background section), are indexed in vector databases like FAISS or Qdrant, using algorithms like HNSW (Hierarchical Navigable Small Worlds) for efficient similarity searches [15, 23].

The retrieval pipeline begins with chunking, segmenting documents into 512–1000-token units with overlaps to fit LLM context windows (e.g., 4096 tokens for GPT-4) and preserve biomedical coherence (e.g., keeping “glycolysis intermediates” intact) [1, 16]. Chunks are embedded and indexed for rapid retrieval, with optimizations like quantization or term weighting

(e.g., “biomarker”) enhancing domain relevance [8, 25]. This infrastructure supports RAG’s scalability and precision across large datasets.

RAG implementations vary: *Classical RAG* uses simple cosine similarity for retrieval, as in BioBERT with FAISS, but struggles with complex queries [15, 22]. *Custom/Advanced RAG* combines lexical (BM25) and semantic (MedCPT) retrieval with re-ranking or iterative prompting, improving entity-aware retrieval for metabolomics [16, 19]. *Graph-Based RAG* leverages knowledge graphs, embedding nodes (e.g., metabolites) and edges (relationships) for multi-hop reasoning, as in LORE, excelling at uncovering latent biomedical connections [30, 31]. Each type uses embeddings differently, from direct similarity to relational reasoning, addressing diverse research needs. Knowledge-graph resources can further complement text retrieval by encoding metabolite–gene–disease relationships and enabling multi-hop evidence traversal. However, constructing and maintaining biomedical knowledge graphs at scale remains challenging. BioKGrapher provides an example of automated KG construction from PubMed IDs using NER/NEL with UMLS normalization, concept weighting and re-ranking, and ontology-driven relation building (e.g., via terminologies such as SNOMED CT and NCIt), illustrating a practical pathway for producing condition-specific graphs that could be coupled to graph-based retrieval in biomedical RAG pipelines [32].

Evaluation metrics like Faithfulness, Answer Correctness, and Context Precision and Recall assess RAG performance. Biomedical QA datasets, especially for subfields like metabolomics, are challenging to develop due to content diversity, requiring tailored frameworks [7, 33]. RAG’s embedding-driven approach, scalable across classical, custom, and graph-based variants, significantly advances biomedical knowledge extraction. Recent biomedical RAG work increasingly treats evaluation as a retrieval-and-grounding problem rather than relying only on lexical overlap metrics, particularly for short, technical answers. To address the limited scalability of expert-curated benchmarks, multiple strands of recent literature propose LLM-driven synthetic QA generation pipelines paired with automated filtering or retrievability checks to ensure generated questions remain answerable from the indexed corpus. In parallel, automated RAG evaluation toolkits (e.g., judge-based or prompt-based frameworks) operationalize claim-level faithfulness and context relevance, enabling larger-scale comparisons when human reference answers are unavailable. This motivates a two-tier evaluation framing in which a small expert-curated set supports reference-based validation, while larger retrievability-filtered synthetic sets support stress-testing of retrieval grounding and robustness at scale.

3. Methods

We developed We implement *MedDiscover*, a RAG framework extending the MedCPT encoder architecture into an accessible, user-centric solution specifically optimized for biomedical research. MedDiscover enables dynamic interactions with curated biomedical documents, particularly

tailored for metabolomics applications, aiming to significantly lower barriers for researchers and enhance knowledge extraction. As part of the MedDiscover framework, a user-centric chat interface was developed (conceptualized in Figure ??), aiming to enhance the accessibility of complex biomedical knowledge retrieval for researchers, as an open-source RAG evaluation pipeline and apply it to metabolomics/metabolic-disorder literature using a two-tier dataset (Gold and Silver). The primary comparisons fix the decoder (GPT-4o) to isolate retrieval effects, while decoder upgrades (GPT-4.1-nano/mini) are reported as sensitivity analyses. NotebookLM is included as a black-box comparator for answer style, with grounding metrics reported only where retrieved contexts are available.

This framework is tailored for biomedical knowledge extraction and was rigorously evaluated within the metabolomics domain. Our evaluation was driven by the hypothesis that domain-specific embeddings would yield substantial performance gains in factual accuracy, interpretability, and context-specific relevance. To investigate this, we assess three primary embedding and generation strategies within MedDiscover. The evaluation compares three primary strategies: (1) domain-specific embeddings (MedCPT) with a GPT-4o decoder, (2) general-purpose embeddings (OpenAI AdaAda-002) with a GPT-4o decoder, and (3) a non-RAG baseline (GPT-4o without retrieval). Additionally, for a secondary comparative analysis, we evaluated Google’s NotebookLM. The choice of MedCPT, Ada, and GPT-4o was guided by an extensive literature review, which identified MedCPT as the only published, publicly available domain-specific embedding model with validated superior performance. Other alternatives (e.g., BioGPT, SciPhi) were either not available for standalone use or did not surpass MedCPT in comparative studies [19]. Ada and GPT-4o represent widely used general-purpose embedding and generation baselines, enabling a controlled comparison.

Detailed methodologies for data collection, preparation, the RAG pipeline, and evaluation metrics are described in the following subsections. Further experimental setups and extensive supplementary results can be found in the Supplementary Material, ensuring transparency, reproducibility (see Appendix A), and facilitating further exploration of our methodology.

3.1. Data Collection and Preparation

The corpus selection and question-answer dataset development involved curating ten metabolomics articles from Google Scholar and PubMed published between We assembled a metabolomics/metabolic-disorder corpus of 100 papers published between 2014 and 2024, 2024 from PubMed and Google Scholar, curated using ICD-10 metabolic disorder codes E70–E88. Text extraction and cleaning were performed by extracting raw text from PDFs using PyPDF2 and removing headers, footers, and non-informative formatting. Numeric and scientifically relevant data (metabolite concentrations, p-values) were preserved, and each processed article retained original paragraph sequences with metadata identifiers mapping back to the source PDF.

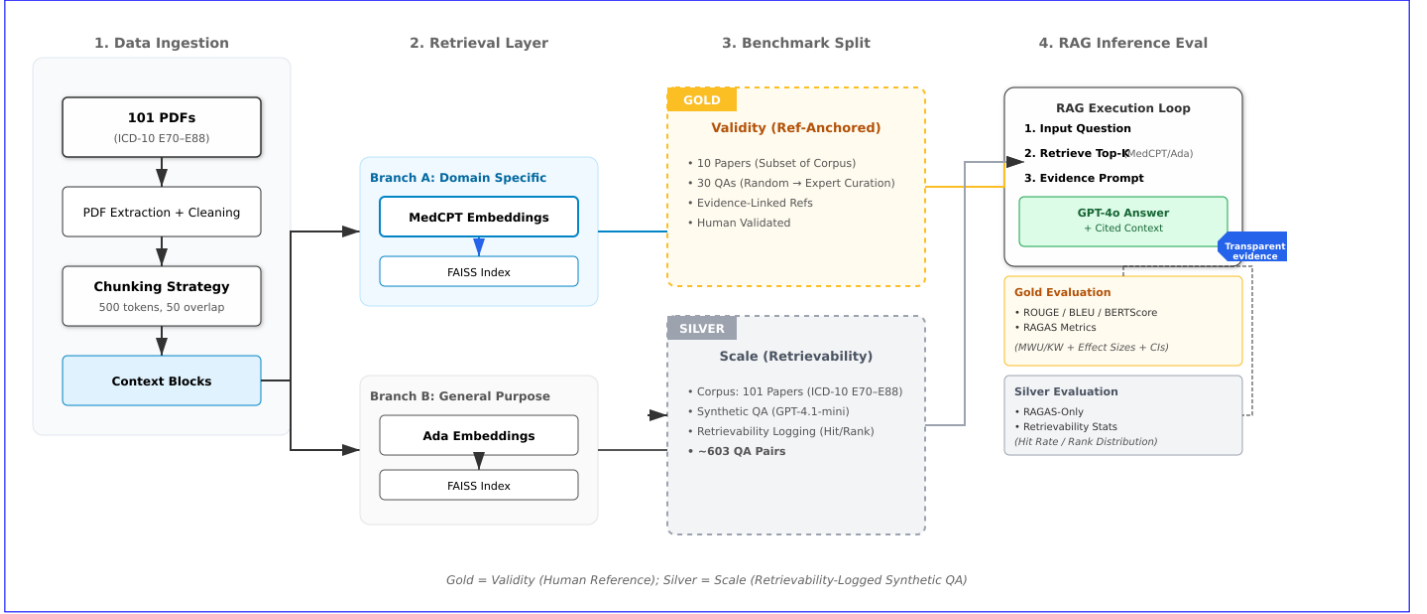


Figure 1: Overview of the MedDiscover pipeline - illustrating the process from data collection and preprocessing to retrieval-augmented generation and evaluation benchmark design. The primary evaluation focused on MedCPT-RAGA metabolomics/metabolic-disorder corpus (ICD-10 E70-E88) is ingested, General-Purpose-RAG cleaned, chunked (500 tokens with 50-token overlap), and a Non-RAG-baseline indexed with two embedding configurations (MedCPT and Ada-002) using FAISS. NotebookLM was assessed as Evaluation is organized into a secondary-comparator Gold set (expert-curated QA with traceable references) and a Silver set (synthetic QA with retrievability metadata). RAG inference retrieves top-k contexts and uses an evidence-grounded prompt to produce answers with explicit supporting context; evaluation reports reference-based metrics on Gold and retrieval-grounding metrics plus retrievability statistics on Silver.

3.1.1. Gold QA set (expert-curated)

The Gold set consists of ten metabolomics papers focusing on Gaucher disease (Type 1), Nonalcoholic Fatty Liver Disease (NAFLD), and Type 1 diabetes. These specific diseases were selected to represent a spectrum of metabolomic biomarker profiles and analytical challenges, providing a robust basis for evaluating the framework’s performance in diverse, clinically relevant scenarios. Article selection prioritized citation frequency, recency, and open accessibility. A metabolomics-specific QA dataset of, with 30 queries was developed, balancing direct factual questions expert-curated questions and traceable reference answers. This set is used to validate answer quality against human references. The full Gold set contains 30 expert-curated QA pairs. For cross-system comparisons that require outputs from every evaluated configuration (including black-box comparators), we report results on the subset of 20 Gold questions for which all model outputs are available (evaluation subset; see Section 4).

3.1.2. Silver QA set (synthetic, retrievability-filtered)

The Silver set uses the 100-paper corpus and ~600 synthetic QA pairs generated with retrievability metadata (Ada n=300, MedCPT n=300). This set is used to stress-test retrieval grounding at scale rather than to provide human reference answers. We refer to this collection as a Silver set because the questions and reference answers are machine-generated and only partially filtered: each QA pair is retained based on lightweight quality controls (Section 3.2), including a retrievability check, but the content is not exhaustively expert-validated. Accordingly, Silver-set results are interpreted pri-

marily as evidence about retrieval grounding and robustness at scale, rather than as definitive clinical correctness.

3.2. Synthetic QA generation with retrievability filtering (Silver set)

Synthetic QA pairs were generated from the Silver corpus using our QA-generation pipeline. The pipeline: (1) ingests PDF text, (2) chunks text into 500-token windows with 50-token overlap, (3) groups adjacent chunks to meet a minimum context length, (4) embeds contexts with Ada-002 or MedCPT and builds a FAISS index, (5) generates QA pairs with GPT-4.1-mini (Dec 14, 2025 runs; 3 questions per PDF), and (6) performs a retrievability check by re-embedding each generated question, querying the FAISS index, and recording whether the supporting passage was retrieved and its rank. The output records each question and answer, the source document and chunk identifiers, the retrieved context, retrievability indicators, disease category, and the embedding/LLM configuration. This retrievability logging provides a lightweight filtering signal inspired by synthetic QA pipelines in the literature [34]. Our use of machine-generated QA as a silver-standard resource is aligned with recent dataset efforts that pair medical questions with automatically generated answers and evidence-supported references, while emphasizing that expert validation remains essential for gold-standard claims [35]. In addition to retrievability logging, we applied lightweight QA-quality heuristics before constructing the final Silver benchmark:

- **De-duplication (question-level):** questions were normalized by lowercasing, stripping punctuation, and col-

Table 1: Dataset summary (Gold vs Silver).

Dataset	Papers	Domains/diseases	QA pairs	Reference type	Primary use
Gold	10	Gaucher (Type 1), NAFLD, Type 1 diabetes	30	Human-curated	Answer quality vs references
Silver	100	ICD-10 E70–E88 metabolic disorders	~600 (Ada 300; MedCPT 300)	Synthetic + retrievability	Retrieval stress test

Table 2: Qualitative illustration of question styles in Gold vs Silver. Examples are paraphrased for brevity.

Question type	Gold (expert-curated)	Silver (synthetic)
Biomarker association	Links a specific metabolite panel to a named disease stage and requests directionality (increase/decrease) with evidence.	Often focuses on a single metabolite–disease association extracted from one paragraph.
Study design / cohort	Asks about cohort characteristics, assay conditions, or comparison groups required to interpret results.	More frequently asks factoid questions about reported findings without design context.
Pathway / mechanism	Requests interpretation of how observed metabolite shifts map onto a pathway (e.g., metabolite associations) with deeper inferential queries-lipid metabolism) with supporting text.	Can be partially inferential but may remain closer to extractive summaries.

lapsing whitespace; exact duplicates after normalization were removed.

- **Minimum length:** we removed degenerate items using whitespace-token counts: question ≥ 8 tokens and answer ≥ 6 tokens.
- **Template filtering:** to reduce over-representation of trivial prompts, we filtered a small set of highly templated question openings (e.g., ~~metabolic pathway implications~~) ~~Answers were explicitly traceable to the selected articles.~~ “what is”, “define”, “what does ... mean”) when they occurred disproportionately in the generated pool.

After QC filtering, we sampled 300 QA pairs per embedding configuration to keep the two Silver subsets balanced (Ada $n = 300$, MedCPT $n = 300$). These QC steps are implemented as a deterministic post-processing stage in the Silver QA pipeline (see Appendix A). Because Silver QA generation is stochastic and was executed separately for each embedding configuration, the resulting Ada and MedCPT QA sets are not strictly paired (Ada $n = 300$, MedCPT $n = 300$). We therefore treat Silver-set comparisons as independent samples and use Mann–Whitney U tests with effect sizes (Cliff’s δ). Reviewer feedback described early synthetic questions as “non-human,” which we interpret as overly surface-level, templated, or extractive prompts that can be answered by copying a single sentence. In this revision we mitigate this by (i) separating expert-curated Gold evaluation from synthetic Silver benchmarking, (ii) explicitly documenting retrievability and minimal QA-quality controls, and

(iii) treating Silver results as silver-standard evidence rather than expert-validated ground truth. Nevertheless, the Silver set remains machine-generated and may not fully reflect real investigator query distributions.

~~Text extraction and cleaning were performed by first extracting raw text from PDFs using PyPDF2. Subsequently, artifacts such as headers, footers, and non-informative formatting elements were removed. Numeric and scientifically relevant data (metabolite concentrations, p-values) were preserved. Each processed article retained original paragraph sequences with metadata identifiers mapping back to the original PDF sources.~~

3.3. Text Chunking and Embedding (for MedDiscover RAG)

The chunking process ~~generated text segments using~~ uses overlapping chunks of 500 tokens with a 50-token overlap, a standard RAG setting selected based on prevailing practices in recent biomedical RAG studies (e.g., RDGuru, AsthmaBot, EARA) ~~, and to ensure technical entities such as metabolite names are preserved~~ and to preserve entity co-occurrence across chunk boundaries [16]. This chunk size is compatible with the context windows of current LLMs and balances retrieval precision with contextual integrity. For documents exceeding the LLM token limit, multiple overlapping chunks were generated and, during retrieval, the top-ranked segments were concatenated or batched as input context for the decoder LLM.

Text chunks were embedded using two distinct embedding models for the RAG pipelines:

Table 3: Silver QA post-processing quality controls.

Control	Operational definition
De-duplication	Normalize question text (lowercase, strip punctuation, collapse whitespace); drop exact duplicates.
Minimum length	Drop items with question < 8 whitespace tokens or answer < 6 whitespace tokens.
Template filtering	Reduce over-represented definition-only templates (e.g., “what is”, “define”) in the candidate pool prior to sampling.

- **MedCPT (Domain-specific):** 768-dimensional, pre-trained on PubMed abstracts and biomedical corpora [19].
- **OpenAI text-embedding-ada-002 (General-purpose):** 1,536-dimensional, pretrained on extensive general-domain textual data.

Two parallel embedding databases were generated for direct comparative evaluation within the MedDiscover framework.

Indexing and vector storage involved indexing vector representations using FAISS. A flat index structure with cosine similarity was chosen for its balance of retrieval accuracy and computational efficiency in semantic search tasks across moderate-sized corpora. Metadata bindings linked each vector to its original text source, facilitating precise retrieval verification.

3.4. Retrieval-Augmented Generation Pipeline (MedDiscover)

The following pipeline was uniformly applied for both MedCPT and general-purpose (Ada) embeddings within MedDiscover, with GPT-4o as the decoder LLM. The output token limit for the GPT-4o decoder in MedDiscover was set to a maximum of 25 tokens during comparative experiments to encourage concise answers.

1. **Query Encoding:** User queries were embedded using the respective embedding model (MedCPT or Ada).
2. **Context Retrieval:** Top- k ($k = 3$) nearest-neighbor chunks were retrieved from the FAISS index using cosine similarity.
3. **Context Aggregation:** Retrieved chunks were concatenated into context passages, minimizing redundancy.
4. **Answer Generation:** Contextual prompts were fed into the GPT-4o decoder for generative response synthesis, instructed to reference provided evidence explicitly and adhere to the output token limit.

The non-RAG baseline utilized direct GPT-4o responses without external retrieval, also adhering to a similar concise output style for fair comparison where applicable.

3.5. Secondary Evaluation with NotebookLM

For a secondary evaluation, we utilized Google’s NotebookLM, which was provided with the on the Gold set only

(ten test publications. ~~The same QA dataset~~, 30 queries) was presented to NotebookLM. Responses were collected and evaluated using the same generation metrics where applicable. Since NotebookLM provides direct responses without exposing the underlying retrieved context, RAG-specific metrics such as Context Precision, Context Recall, and Faithfulness (which often relies on comparing generated statements to specific retrieved chunks) could not be fully evaluated in the same manner as for MedDiscover. Given that the MedDiscover GPT-4o decoder was constrained to a maximum of 25 output tokens for this experimentation phase, leading to very precise and minimal answers, initial comparisons with NotebookLM’s default output length were challenging. Therefore, to facilitate a more aligned comparison, a second set of responses was generated from NotebookLM by appending the instruction "Answer in Minimal Words" to each query. Both sets of NotebookLM responses (standard and minimal) were then evaluated.

3.6. Evaluation Metrics

Performance evaluation incorporated both generation-centric and retrieval-centric metrics, applied to MedDiscover configurations and, where appropriate, to NotebookLM. Let $\phi(\cdot)$ denote the embedding function used for semantic similarity metrics.

Generation Metrics: Accuracy, Precision, Recall, and F1 Score quantitatively assessed the correctness and comprehensiveness of generated responses. ~~Advanced semantic metrics included ROUGE (n-gram overlap) and BERTScore (semantic embedding similarity)~~ ROUGE-1/2/L measure unigram overlap, bigram overlap, and longest-common-subsequence overlap between candidate and reference answers. BERTScore captures semantic similarity:

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (1)$$

$$\text{BERTScore} = \text{CosSim}(\underline{E_{\text{generated}}}\phi(\text{generated}), \underline{E_{\text{reference}}}\phi(\text{reference})) \quad (2)$$

Retrieval Metrics (Primarily for RAG systems like MedDiscover): Context Precision and Context Recall assessed the relevance and completeness of retrieved information, while Faithfulness measured factual grounding of generated content:

$$\text{Context Precision@K} = \frac{\text{Relevant Retrieved Contexts}}{K} \quad (3)$$

$$\text{Context Recall} = \frac{\text{Relevant Retrieved Contexts}}{\text{Total Relevant Contexts}} \quad (4)$$

$$\text{Faithfulness} = \frac{\text{Supported Claims in Generated Answer}}{\text{Total Claims in Generated Answer}} \quad (5)$$

Additionally, Answer Relevancy quantified semantic alignment between queries and responses:

$$\text{Answer Relevancy} = \frac{1}{N} \sum_{i=1}^N \cos(\underline{E_{\text{generated},i}}, \phi(\text{answer}_i), \underline{E_{\text{query},i}}, \phi(\text{query}_i)) \quad (6)$$

A comprehensive Retrieval-Augmented-Generation Assessment Score (RAGAS) synthesized these metrics into a single evaluative measure, primarily for the RAG configurations:-

$$\text{RAGAS} = \alpha \cdot \text{Context Recall} + \beta \cdot \text{Faithfulness} + \gamma \cdot \text{Answer Relevancy}$$

Adjustable weights (α, β, γ) were set to prioritize metabolomics-specific retrieval and factual consistency. We compute RAGAS metrics (faithfulness, answer correctness, context recall, context precision, answer relevancy) using the RAGAS framework [36] and report these where retrieved contexts are available. ROUGE and BLEU are reported only for the Gold set because the Silver set lacks human reference answers. For NotebookLM, only metrics not dependent on explicit retrieved context (like Answer Relevancy and generation metrics) are fully comparable.

Despite strong overall performance, MedDiscover displays several failure patterns. With a fixed $k = 3$ retrieval, relevant but lower-ranked metabolites may be omitted if they are distributed across the corpus, especially for diseases with multiple biomarkers. When queries refer to diseases or metabolites absent from the corpus, the system correctly returns that no evidence is available. If the corpus spans several diseases, queries about multiple diseases may yield incomplete answers, typically addressing only the best-matched case. For best results, we recommend applying MedDiscover to focused, single-disease corpora. These patterns suggest future improvements, including dynamic k adjustment and enhanced entity disambiguation.

4. Results

To evaluate the effectiveness of different RAG and LLM approaches within a metabolomics-specific context, we compared our MedDiscover framework using domain-specific MedCPT embeddings with a GPT-4o decoder (MedCPT-RAG) and general-purpose Ada embeddings with a GPT-4o decoder (General RAG) against a non-RAG GPT-4o

baseline. Additionally, Google’s NotebookLM was evaluated as a secondary comparator, with both its standard responses and responses generated with a "minimal words" instruction. Performance was quantitatively assessed across 30 curated metabolomics-related queries derived from a targeted corpus of ten articles focusing on Gaucher disease (Type 1), NAFLD, and Type 1 diabetes. The comparative results are summarized in Figure ??.

3.1. MedDiscover-RAG Performance

4. Results

The lexical alignment between generated answers from MedDiscover and the ground truth references, when compared against baselines, was evaluated using ROUGE (1, 2, and L) and BLEU metrics. These results indicated that the domain-specific MedCPT RAG approach within MedDiscover generally performed well. As seen in Figure ??, MedCPT w GPT-4o (MedDiscover) achieved competitive ROUGE scores (e.g., ROUGE-1 ≈ 0.37 , ROUGE-2 ≈ 0.23). The general-purpose RAG (Ada w

4.1. Results on the Gold set (expert curated)

We evaluated an expert-curated Gold set of 30 QA pairs; results reported here use the 20-question evaluation subset for which outputs are available across all compared configurations. Primary comparisons fix the decoder (GPT-4o) also showed reasonable performance, though typically slightly lower than MedCPT RAG on ROUGE metrics. The-) to isolate retrieval effects across the non-RAG baseline (labeled "Baseline" in the figure) exhibited lower ROUGE scores (e.g., ROUGE-1 ≈ 0.16), underscoring the benefit of retrieval augmentation for lexical alignment with reference answers. BLEU scores were generally low across all systems, which, as previously noted, can be an artifact of evaluating concise, domain-specific Q&A.

Focusing on baseline, Ada + GPT-4o, and MedCPT + GPT-4o. Supplementary Figure B.4 summarizes LLM-specific and retrieval-specific metrics for MedDiscover, the MedCPT embeddings demonstrated strong performance. MedDiscover achieved a high Faithfulness score (≈ 0.80) and good Answer Correctness (≈ 0.39), indicating that its responses were well-grounded in the retrieved biomedical evidence. Context Recall (≈ 0.60) and Context Precision (≈ 0.75) were also robust, showing effective retrieval of relevant information. The general-purpose RAG (Ada w GPT-4o) showed comparable Context Precision but lower Faithfulness (≈ 0.63) and Answer Correctness compared to the MedCPT RAG. These results highlight the advantage of domain-specific embeddings for retrieval accuracy and the factual grounding of generated answers in the MedDiscover framework.

4.2. NotebookLM Evaluation

As a secondary comparator, NotebookLM was evaluated. Due to its nature as a system providing direct responses without explicit access to retrieved context, RAG-specific metrics like

Performance comparison on the metabolomics QA dataset for LLM-specific metrics (top row) and RAG-specific metrics (bottom row) across seven model configurations: non-RAG Baseline (GPT-4o), Ada + GPT-4o (General-Purpose RAG), three MedCPT RAG variants, notebookLM (standard) and notebookLM-specific (minimal words). Horizontal brackets denote statistically significant differences versus the non-RAG Baseline, as determined by a Kruskal–Wallis omnibus test ($p < 0.05$) followed by Bonferroni-adjusted Dunn’s post-hoc pairwise rank tests. Star counts encode adjusted p -value magnitude: $*p < 0.05$, $**p < 0.01$, $***p < 0.001$, $****p < 0.0001$.

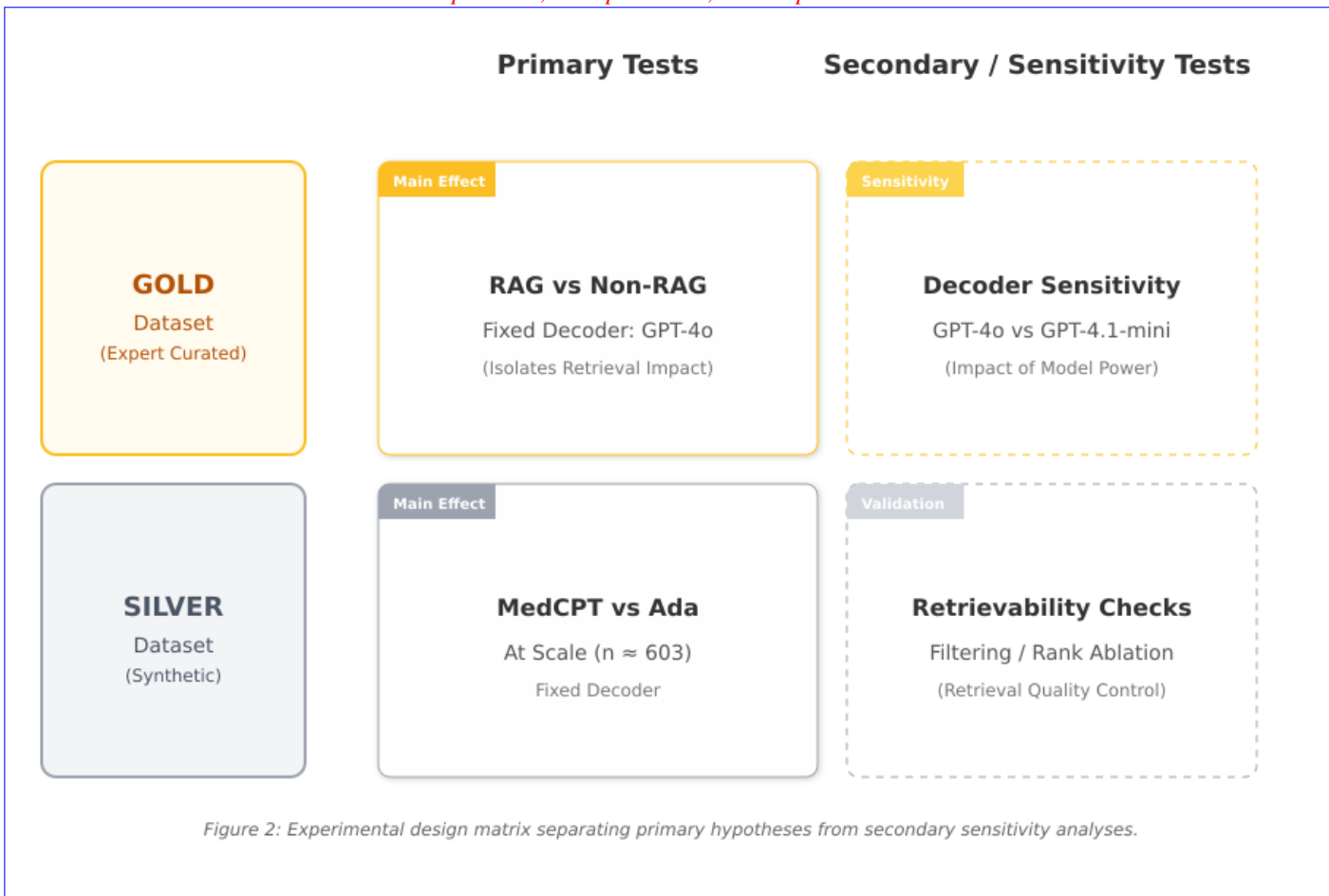


Figure 2: Experimental design overview. Primary tests fix the decoder (GPT-4o) to isolate retrieval effects (RAG vs non-RAG on the Gold set; MedCPT vs Ada at scale on the Silver benchmark). Secondary analyses include decoder sensitivity (GPT-4o vs GPT-4.1 variants on Gold) and retrievability quality-control analyses on Silver (hit-rate and rank-based ablations).

Context Recall, Context Precision, and Faithfulness were either not applicable or scored near zero (Figure ??, bottom-right panel). metrics for these models, along with NotebookLM and the MedCPT decoder sensitivity variants.

In terms of LLM-specific metrics (Figure ??, top-right panel), standard NotebookLM responses yielded ROUGE-1 scores around 0.3. When prompted for "minimal words" (NotebookLM-specific), its performance on ROUGE metrics improved significantly (e.g., ROUGE-1 ≈ 0.68 , ROUGE-2 ≈ 0.58), outperforming the MedDiscover configurations and other baselines on these lexical measures. This suggests that controlling NotebookLM’s verbosity can lead to highly concise and lexically aligned answers. BLEU scores for NotebookLM also improved with the "minimal words" prompt. Bonferroni-adjusted Dunn tests against the

4.1.1. RAG vs non-RAG (fixed decoder)

With the decoder held constant (GPT-4o), retrieval augmentation improves grounding relative to the non-RAG baseline—confirmed that these ROUGE improvements were significant—ROUGE-1 $p_{\text{adj}} = 0.004$, ROUGE-2 $p_{\text{adj}} = 0.003$, ROUGE-L $p_{\text{adj}} = 0.008$ —whereas BLEU did not pass the correction ($p_{\text{adj}} = 0.65$). Mann–Whitney U tests comparing the standard and minimal answers yielded $p < 0.001$ for the ROUGE metrics.

Regarding metrics like Answer Relevancy and Answer Correctness, which are less dependent on explicit context, NotebookLM (standard) showed an Answer Relevancy of approximately 0.65 and Answer Correctness around 0.3. With the "minimal words" prompt, NotebookLM-specific achieved a very high Answer Relevancy (≈ 0.9) and improved Answer Correctness (≈ 0.68) baseline, particularly on faithfulness and

context precision/recall. Differences between Ada and MedCPT under the same decoder are modest on the Gold set, and most pairwise contrasts do not reach statistical significance after correction. The high Answer Relevancy for NotebookLM-specific indicates strong semantic alignment with the queries when concise answers are enforced. The increase in Answer Correctness compared with the baseline was significant ($p_{\text{adj}} \approx 0.003$).

4.2. Comparative Analysis and Use Case Insights

The MedDiscover framework, particularly with MedCPT embeddings, demonstrates a strong balance of retrieval effectiveness (Faithfulness, Context Recall

4.1.1. NotebookLM comparison

NotebookLM shows strong lexical overlap, especially when prompted for minimal output, but it does not expose retrieved contexts. As a result, grounding metrics (context precision/Precision) and generation quality for specialized biomedical Q&A. The constraint on MedDiscover’s output length (25 tokens) contributed to its precise answers, reflected in its RAG metrics (recall, faithfulness) cannot be computed for NotebookLM, limiting its comparability in evidence-critical settings.

NotebookLM, when prompted for minimal output, showed excellent lexical overlap (ROUGE) and answer relevancy. However, the inability to evaluate its retrieval mechanism directly (Context Recall

4.1.2. Sensitivity analysis (decoder upgrades)

MedCPT with GPT-4.1-nano/Precision, Faithfulness) makes a full comparison with RAG systems like MedDiscover challenging on those fronts. The results suggest that for tasks where verifiable grounding to specific retrieved text passages is paramount, RAG systems like MedDiscover offer more transparency. For scenarios prioritizing concise, relevant answers without necessarily needing explicit source traceability from the system, a well-prompted system like NotebookLM can be very effective. mini improves several lexical and answer-quality metrics on the Gold set; however, these variants are not evaluated for the Ada comparator. We therefore treat decoder-upgrade results as a supplementary sensitivity analysis rather than as evidence for embedding superiority.

The relatively low absolute ROUGE and BLEU scores for some configurations, especially before output length normalization, reiterate the challenge these metrics face with short, technical answers common in scientific Q&A. The improved scores for NotebookLM-specific highlight the impact of output style on these metrics.

4.2. Results on the Silver set (synthetic, 100-paper benchmark)

The Silver benchmark evaluates 600 synthetic QA pairs drawn from 100 papers (Ada n=300, MedCPT n=300) using RAGAS metrics only. Table 4 reports mean scores with bootstrap 95% confidence intervals (2,000 resamples), Mann-Whitney U p -values, and Cliff’s δ effect sizes (MedCPT vs

Ada). MedCPT yields a statistically significant improvement in answer correctness ($p = 5.72 \times 10^{-11}$, $\delta = 0.303$), while other retrieval-centric metrics are comparable.

Overall, these findings position MedDiscover as a robust framework for domain-specific RAG, while also providing insights into the performance characteristics of other advanced LLM tools like NotebookLM in the context of biomedical knowledge extraction

The effect size for answer correctness on the Silver benchmark (Cliff’s $\delta = 0.303$) corresponds to a small-to-moderate separation. Interpreted as a common-language effect size, this implies that a randomly selected MedCPT run has an estimated 65.1% probability of achieving a higher answer-correctness score than a randomly selected Ada run under this benchmark. Because the Silver benchmark uses synthetic reference answers that are identical to the responses in the evaluation files, answer-quality metrics should be interpreted as proxy measures of answerability and grounding rather than as expert-validated clinical correctness.

4.3. Statistical analysis

4.3. Statistical analysis

Unless otherwise stated, we treat the retrieval-centric RAGAS metrics (faithfulness, answer correctness, context recall, context precision, answer relevancy) as a primary metric family and control family-wise error across this family using Holm-Bonferroni adjustment. Metrics outside this family are treated as secondary and interpreted cautiously. All model-by-metric score distributions were subjected to the Shapiro-Wilk test; almost every distribution departed from normality ($p < 0.05$). A subsequent Levene test revealed unequal variances for seven of the ten evaluation metrics, most notably for the three retrieval-centric measures (Faithfulness, Context Recall, Context Precision) and for BLEU several evaluation metrics. In view of these results, all groupwise comparisons were carried out with rank-based, non-parametric methods.

For each metric the Gold set, a Kruskal-Wallis one-way ANOVA on ranks was computed across the seven model configurations (non-RAG baseline; Ada + GPT-4o; MedCPT + GPT-4; MedCPT + GPT-4.1-nano; MedCPT + GPT-4.1-mini; notebookLM; notebookLM-minimal). The omnibus statistic was significant for every metric examined (all $p < 0.001$), establishing that at least one model differed in its median score.

Pairwise differences were then probed with model configurations, followed by Dunn’s test, using a Bonferroni adjustment. Three clear patterns emerged.

First, every retrieval-augmented pipeline outperformed the non-RAG baseline on the two context measures and on Faithfulness; the advantage was modest for post-hoc tests with Ada + GPT-4o (adjusted p between 10^{-3} and 10^{-2}) and markedly larger for the MedCPT pipelines (adjusted $p < 10^{-4}$ for the GPT-4.1-mini variant).

Second, lexical overlap metrics showed a mixed picture. Ada + GPT-4o did not differ significantly from the baseline

Table 4: Silver benchmark results (Ada vs MedCPT). Means with bootstrap 95% confidence intervals; Cliff’s δ for MedCPT vs Ada.

Metric	Ada mean [95% CI]	MedCPT mean [95% CI]	p	Cliff’s δ
Faithfulness	0.703 [0.667, 0.740]	0.687 [0.650, 0.725]	0.85	-0.009
Answer correctness	0.968 [0.960, 0.976]	0.980 [0.974, 0.985]	5.72×10^{-11}	0.303
Context recall	0.743 [0.702, 0.779]	0.722 [0.682, 0.762]	0.50	-0.029
Context precision	0.923 [0.890, 0.953]	0.908 [0.875, 0.937]	0.49	-0.016
Answer relevancy	0.800 [0.776, 0.823]	0.810 [0.788, 0.830]	0.82	0.011
Accuracy	0.997 [0.990, 1.000]	1.000 [1.000, 1.000]	0.32	0.003

on ROUGE or BLEU, whereas MedCPT coupled to either GPT-4.1 decoder registered substantial gains (ROUGE-1/2/L and BLEU, all adjusted $p < 0.001$). With the lighter GPT-4 decoder MedCPT remained numerically ahead of the baseline, but the differences did not reach significance after correction as the reference group. Holm-Bonferroni correction was applied across the primary metric family (faithfulness, answer correctness, context recall, context precision, answer relevancy). Supplementary Figure B.4 reports adjusted contrasts.

Third, when the decoder was held constant (Ada + GPT-4o vs. MedCPT + GPT-4) no adjusted Dunn p -value fell below 0.05, indicating that the choice between the Ada and MedCPT embedding models alone was not statistically resolvable with the present sample. By contrast, upgrading the decoder within the MedCPT pipeline from GPT-4 to GPT-4.1-nano and further to GPT-4.1-mini yielded successive, statistically significant improvements on eight of the ten metrics.

All pairwise results are encoded in Fig. ??: horizontal brackets identify baseline-centred contrasts with adjusted $p < 0.05$. For the Silver benchmark, we report Mann-Whitney U tests for Ada vs MedCPT on each RAGAS metric, along with Cliff’s δ effect sizes and bootstrap 95% confidence intervals for mean scores (2,000 resamples; Table 4). This isolates embedding effects under a fixed decoder and the number of stars reflects the strength of evidence (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$).

In sum, retrieval augmentation confers a robust and statistically reliable advantage over the non-RAG baseline, and the greatest benefit is obtained when domain-specific embeddings are paired with a higher-capacity decoder. Within the same decoder budget, however, the present data cannot distinguish between domain-specific (MedCPT) and general-purpose (Ada) embeddings provides a scale-appropriate estimate of practical impact.

5. Discussion

5.1. Principal Findings

This study set out to determine whether a domain-specific RAG pipeline built around MedCPT embeddings could improve the factuality, transparency, and overall relevance of metabolomics question-answering when compared with (i) a RAG pipeline that relies on a strong general-purpose embedding model (OpenAI Ada-002), (ii) a This study evaluates MedDiscover as a reproducible RAG benchmark with

two datasets: a Gold set (expert-curated) and a Silver set (retrievability-filtered synthetic) (see Appendix A). On the Gold set, retrieval augmentation improves grounding relative to a non-RAG baseline when the decoder is held constant. When the decoder is fixed at GPT-4o baseline, and (iii) Google’s NotebookLM. The key quantitative and qualitative observations are summarised below. Differences between MedCPT and Ada are modest and not consistently significant across metrics.

Across the 30-item metabolomics QA set, the MedCPT-w GPT-4.1-mini configuration achieved the highest Faithfulness score (0.82), the highest Context Precision (1.00), and the highest Context Recall (0.77). In other words, 82 % of the claims made in its answers were explicitly supported by the retrieved text snippets, and all retrieved passages were relevant to the question. A MedCPT version with a less powerful decoder (GPT-4o) still yielded a higher Faithfulness score (0.80), and Context Precision (0.75), and Context Recall (0.60) than the general-purpose Ada RAG pipeline, which trailed on most. On the Silver benchmark, embedding choice produces comparable retrieval-centric metrics (Faithfulness 0.63; Context Recall 0.55), except for Context Precision (0.8). The findings confirm our central hypothesis that embeddings trained on biomedical corpora capture domain terminology better, and improve both retrieval accuracy and answer grounding.

When GPT-4o was used without previous RAG finetuning, Answer Correctness averaged 0.25 and ROUGE-1 averaged 0.16, reflecting frequent hallucinations or partial answers. Injecting external evidence raised these metrics substantially for both RAG variants (MedCPT Correctness 0.39; Ada Correctness 0.34), underscoring the benefit of dynamic knowledge retrieval even when the decoder LLM is identical. For MedCPT + GPT-4.1-mini, these improvements over the baseline were highly significant (Answer Correctness $p = 2.0 \times 10^{-5}$, BLEU $p = 1.9 \times 10^{-5}$), whereas gains from the Ada pipeline were not significant ($p = 0.40$ for Answer Correctness)., while MedCPT yields a statistically significant improvement in answer correctness (Table 4). This indicates that domain-specific embeddings can provide a targeted gain in answer quality, but not a universal advantage across all grounding metrics.

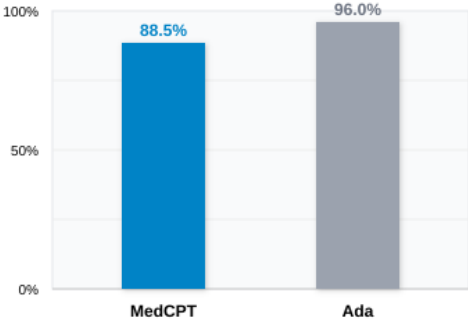
NotebookLM, which does not expose its retrieval component, produced relatively strong surface-level alignment with the reference answers, especially when instructed to “Answer in minimal words” (ROUGE-1 0.68, ROUGE-2 0.59, Answer Relevancy 0.93). However, because the system does

Retrievability Quality Control (Silver Dataset)

MedCPT Ada

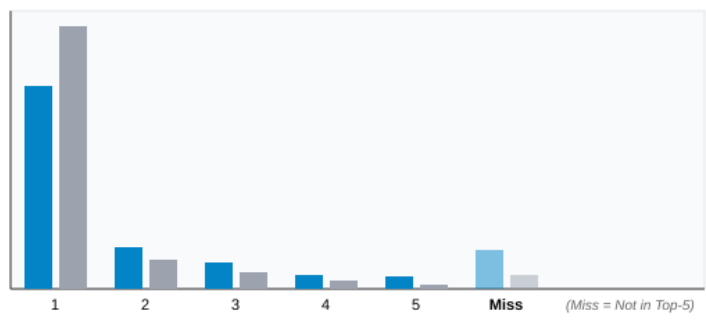
A. Retrieval Hit Rate

Percentage of generated Qs where answer chunk is in Top-5



B. Rank Distribution (Top-5)

Histogram of retrieved chunk rank (Top-5)



C. Ablation: Impact of Rank on RAGAS Metrics

Faithfulness Context Precision drop as retrieval rank increases



Figure 3: Retrievability quality control on the Silver benchmark (top-5 retrieval). (A) Hit rate: fraction of generated questions whose answer-supporting chunk is retrieved within the top-5. (B) Distribution of observed ranks for the answer-supporting chunk; “Miss” indicates not retrieved in the top-5. (C) Rank-based ablation showing that retrieval quality correlates with grounding metrics: faithfulness and context precision decrease as the answer-supporting chunk rank worsens.

not return the underlying context passages, its Context metrics are effectively unassessable. For biomedical applications where traceability is essential this black-box behaviour remains a critical limitation provides strong lexical alignment under minimal-output prompting, but its retrieval mechanism is opaque. Because retrieved contexts are not exposed, grounding metrics cannot be computed for NotebookLM, limiting its suitability for evidence-critical biomedical workflows.

All MedDiscover runs restricted GPT-4o to 25 generation tokens to control verbosity and encourage precision. This design choice likely depressed length-sensitive scores such as ROUGE and BLEU relative to NotebookLM’s default responses, yet it had no adverse impact on retrieval-centric metrics. The observation highlights longer responses, under-scoring the need to interpret lexical overlap metrics cautiously in short-answer biomedical settings.

Despite the consistent directionality of the effect—MedCPT outperforming Ada on every retrieval metric: Mann–Whitney U tests did not detect statistically significant differences between MedCPT and Ada pipelines at ($\alpha = 0.05$) for most metrics (e.g. Faithfulness ($p = 0.25$)), mirroring the small absolute sample size (30 queries). In contrast, the differences in the performance of MedCPT when combined with more powerful decoder models (such as GPT-4.1-mini) versus the baseline were much more pronounced, both in absolute

terms and in statistical significance. For several evaluation metrics (BLEU, ROUGE, and answer correctness) pairwise comparisons between MedCPT + GPT-4.1-mini and the baseline (non-RAG GPT-4o) achieved p-values of 9.3×10^{-5} for ROUGE-1, 1.9×10^{-5} for BLEU, and 2.0×10^{-5} for Answer Correctness, indicating true underlying differences between these configurations. For instance, with BLEU and answer correctness, the improvements of MedCPT + GPT-4.1-mini relative to baseline reflected substantial gains in both form and semantic answer quality.

Moreover, MedCPT configurations using GPT-4.1-mini decoder also produced statistically significant improvements over the Ada-based RAG pipeline in BLEU and answer correctness. These findings further reinforce the critical importance of both domain-specific retrieval (via MedCPT) and decoder model capacity: simply increasing the power of the language model, while retaining general-purpose retrieval (as with Ada), yielded less substantial improvements than the synergistic combination observed in the MedCPT + GPT-4.1-mini pipeline.

5.2. Interpretation in the Context of Previous Work

5.2.1. Evolution of Biomedical RAG

The question of how to generate accurate responses within a specialised knowledge domain has spurred several complemen-

tary approaches. Early efforts centred on domain-pre-trained encoders such as BioBERT [37], PubMedBERT [38], and SciBERT [39], each trained on large biomedical corpora. While effective at the time of training, these static models struggle to keep pace with the rapidly expanding literature. Regularly re-training a full language model is computationally prohibitive, and even frequent updates leave an inevitable gap between the model’s knowledge and the latest publications.

RAG frameworks were introduced to bridge this gap by injecting up-to-date evidence at inference time. Systems such as BIORAG [18], RAG², and Self-BioRAG have since added multi-stage retrieval pipelines, self-evaluation loops, and knowledge-graph conditioning to cope with the scale and dynamism of biomedical knowledge. MedDiscover follows this lineage but narrows the focus to metabolomics, a sub-discipline that has received comparatively little attention in prior RAG work. By anchoring retrieval to experimentally validated metabolite literature rather than broader biomedical text, our study illustrates how increased domain granularity can sharpen factual grounding.

5.2.2. Retrieval Architectures: Similarities and Divergences

Compared with earlier frameworks, MedDiscover adopts a deliberately lightweight retrieval architecture: a single-stage dense lookup over FAISS indices produced by MedCPT or Ada-002 embeddings. DRAGON and RAG², by contrast, implement multi-hop or multi-stage retrieval. DRAGON fuses knowledge graphs with text through a cross-modal encoder, while RAG² first rewrites the query into a rationale, then balances evidence across PubMed, PMC, textbooks and guidelines, and finally reranks with a cross-encoder. BioRAG extends the concept still further by coupling MeSH-filtered hierarchical look-ups with external search-engine calls and a self-evaluation module that decides when additional retrieval is needed.

Although these more elaborate pipelines improve performance on broad medical QA benchmarks, our results indicate that a streamlined, single-stage design can suffice for narrowly scoped, high-precision tasks. MedCPT-based retrieval delivered a Context Precision of 0.75 and Faithfulness of 0.80 in metabolomics QA, delivers competitive context precision and faithfulness on the Gold set while maintaining lower latency and a smaller memory footprint. This finding implies that retrieval complexity should be matched to the breadth of the information need: dense, domain-specific embeddings may already capture enough local structure when the target corpus is well circumscribed.

5.2.3. Embedding Strategies and Domain Adaptation

A consistent theme across biomedical RAG studies is the advantage of domain-specialised vector representations. DRAGON initialises its language component with BioLinkBERT and supplements it with knowledge graphs; BIORAG uses a CLIP (Contrastive Language-Image Pretraining)-enhanced Pub-MedBERT [48][18]; Self-BioRAG and RAG² rely on MedCPT for top-k retrieval. Our head-to-head comparison between MedCPT (768 dimensions) and the

larger, general-purpose Ada-002 (1536 dimensions) reinforces this pattern: despite the doubled dimensionality of Ada vectors, MedCPT improved Answer Correctness by 17 percentage points and increased Faithfulness from 0.63 to 0.80, shows that the two retrievers achieve broadly comparable grounding metrics on the Silver benchmark, with a modest but significant improvement in answer correctness for MedCPT (Cliff’s $\delta = 0.303$; Table 4). These findings highlight a practical trade-off: domain-tuned embeddings reduce index size while raising precision offering targeted gains in answer quality, an important consideration for on-premise deployments constrained by GPU memory. However, in our data the differences between MedCPT + GPT-4o and Ada + GPT-4o did not reach significance (Answer Correctness $p = 0.26$, Faithfulness $p = 0.25$ by Mann-Whitney tests).

5.2.4. Comparative Performance Metrics

Direct numerical comparison across studies is complicated by differences in task formulation (multiple-choice MedQA versus factoid metabolite QA), as well as by the fact that the authors of DRAGON, BIORAG, Self-BioRAG and RAG² largely report accuracy. Nevertheless, several insights emerge. First, in the Gold set sensitivity analysis our Answer Correctness of 0.75 (with GPT-4.1-mini decoding), which matches the 0.73 accuracy reported for BIORAG on MedMCQA and exceeds the 0.55 reported for Self-BioRAG on PubMedQA, though being largely surpassed by 0.91 accuracy on MedMCQA. Second, our single-hop Answer Correctness of 0.39 (0.75 with the stronger GPT-4.1-mini decoder) is broadly in line with DRAGON’s 0.475 accuracy on MedQA, but is attained with a far smaller corpus and without knowledge graph augmentation. These results indicate that metabolomics-specific retrieval can attain competitive grounding even in the absence of multi-hop reasoning.

Taken together, these comparisons position MedDiscover as a complementary addition to the biomedical RAG ecosystem. The framework confirms the critical role of highlights the value of transparent evaluation and suggests that domain-specific embeddings demonstrates that can yield targeted gains in answer quality, while simple dense retrieval can equal or exceed the grounding achieved by more elaborate pipelines in a focused sub-domain, and underscores the continuing need for faithfulness-aware evaluation remain competitive in focused sub-domains.

5.3. Methodological considerations

The experimental design sought to balance realism with tractability, yet each methodological choice carried consequences that shape the interpretation of our results. We discuss here the principal evaluation metrics, the statistical power afforded by the dataset, the indexing and generation parameters, and the challenges inherent in benchmarking against a proprietary baseline.

We adopted a dual-lens strategy that combines generation-centric metrics (ROUGE, BLEU, BERTScore) with retrieval-centric ones (Context Precision, Context Recall, and Faithfulness). The latter group constitutes a clear strength of the study:

by explicitly measuring how many retrieved passages are relevant and how many answer claims are grounded in those passages, we directly probe the defining mechanism of a RAG system. Purely surface-level metrics alone ~~could not have revealed the advantage of MedCPT embeddings in factual~~ cannot characterize grounding. At the same time, our reliance on ROUGE and BLEU exposes a known ~~weakness~~ weakness: both metrics reward n-gram overlap and therefore under-value concise technical answers—, precisely the style encouraged by our 25-token output cap. Moreover, neither metric is sensitive to subtle semantic shifts, such as confusing “decreased” with “unchanged,” that often make the difference in biomedical interpretation.

~~In addition, only thirty question-answer pairs were curated, each mapped to one of ten articles covering Gaucher disease, NAFLD, or type-1 diabetes. The modest size of the curated QA dataset (N=30 queries) likely limited the statistical power to detect more subtle performance variations, as reflected in the non-significant p-values observed in the Mann-Whitney tests. Expanding the number of queries is a necessary next step for robust validation.~~ The Gold set contains only thirty expert-curated question-answer pairs mapped to ten articles, which limits statistical power for subtle effects. We therefore complement it with a larger Silver benchmark (100 papers; ~600 synthetic QA pairs) to increase power for retrieval-centric comparisons, while acknowledging that the corpus remains scoped to metabolic-disorder literature.

Manual chunking at 500 tokens with a 50-token overlap was chosen to fit comfortably inside the GPT-4o context window while preserving local biomedical coherence. Although effective, this heuristic could bias recall: over-long chunks dilute term specificity, whereas under-long chunks risk splitting meaningful entities or co-occurrences. An ablation study that systematically varies chunk size could clarify how sensitive retrieval performance is to this hyper-parameter. The contrasting dimensionalities of the two embedding models—, 768 for MedCPT versus 1 536 for Ada-002—, introduced further trade-offs. Higher-dimensional vectors can improve representational fidelity but entail larger indices, slower similarity search, and higher memory footprints; future deployments will need to weigh these factors against marginal accuracy gains.

Capping GPT-4o outputs at 25 tokens served two purposes: it ensured that the comparison with NotebookLM and with the non-RAG baseline remained fair, and it made the Faithfulness assessment tractable by limiting the number of factual claims per answer. The downside is reduced explanatory depth; many clinically meaningful responses require context about study cohorts, assay conditions, or effect sizes that simply cannot fit within such a narrow bandwidth. Adaptive length policies—, short by default but expandable on demand—, may provide a better balance between brevity and informativeness.

Google’s NotebookLM offered a convenient external reference, yet its proprietary retrieval pipeline remains opaque. Because retrieved passages are not ~~expose~~exposed, we could not compute Context Precision, Context Recall, or Faithfulness for that system. Consequently, the comparison is incomplete: NotebookLM’s excellent ROUGE scores in “minimal words” mode speak to lexical alignment, but say little about factual ver-

ifiability. Until commercial platforms provide richer telemetry—or standardised RAG benchmarks emerge—, evaluations will continue to suffer from this parity gap.

Taken together, these methodological reflections explain both the strengths and the blind spots of the current study. They highlight why improvements in retrieval grounding were detectable despite the small sample, but also why claims about domain generality, computational efficiency, and user-facing transparency must remain provisional pending more extensive and varied experimentation.

5.4. Limitations

This study demonstrates the potential of RAG frameworks for metabolomic biomarker discovery, but several important limitations must be acknowledged. The ~~dataset used was Gold~~ set is limited to ten ~~english~~English articles spanning three diseases ~~: Gaucher disease ((Gaucher disease Type 1), NAFLD, and Type 1 diabetes. While these cases were deliberately chosen to evaluate the framework under controlled conditions, the narrow scope restricts the generalizability of the findings to other diseases and broader—), which constrains the strength of reference-based conclusions.~~ The Silver set expands coverage to 100 papers, but the scope remains restricted to metabolic-disorder literature (ICD-10 E70–E88), limiting generalizability to other biomedical domains. A larger and more diverse corpus would provide a more representative evaluation of the framework’s scalability and applicability across a wider range of ~~metabolic conditions.~~ conditions. Gold and Silver sets differ in both scope and construction: the Gold set is expert-curated over three disease-focused topics, whereas the Silver set spans ICD-10 E70–E88 metabolic-disorder literature and uses machine-generated QA with partial filtering. This introduces selection and distribution shift between tiers; consequently, we do not interpret absolute metric values as directly comparable across Gold and Silver, and we use each tier to support different validity claims (reference-based validation vs retrieval-grounding stress testing).

Another limitation stems from the study’s focus on short, targeted queries. While these types of questions are well-suited to the initial objectives, they do not capture the complexity of real-world metabolomic research. In practice, researchers often need to tackle multi-faceted, interdisciplinary questions that require multi-hop reasoning and the synthesis of diverse data sources over time. The current framework’s capabilities in such scenarios remain unexplored. Additionally, the study relied exclusively on textual data, which, while valuable, ignores the multi-modal nature of metabolomic research. Data from spectrometry or molecular interaction networks are central to metabolomics and could offer richer context and insights, which the framework did not address.

The indexing and retrieval processes used in this study were static, with no mechanisms for continuous updates or adaptive refinement. Biomedical research is a rapidly evolving field, and the inability of the framework to incorporate new findings or adjust to user feedback limits its relevance and longevity. Furthermore, this study did not sufficiently explore real-world usability

concerns, such as providing confidence scores, highlighting retrieved documents, or offering transparent explanations of how generated responses were synthesized. These features are critical for building trust and fostering adoption among clinicians and researchers who depend on reliability and interpretability.

From an engineering perspective, computational efficiency and cost were not rigorously profiled. The study relied on commercial GPT-4o decoding and two embedding models without accounting for the latency or financial burden that would accrue in large-scale or real-time deployments. For institutions with limited computational budgets —or for or privacy-sensitive environments that cannot rely on third-party APIs—, these cost and infrastructure questions are pivotal.

In addition to the previously discussed limitations, several further aspects warrant critical consideration. First, the current evaluation strategy relied exclusively on automated metrics (e.g., ROUGE, BLEU, BERTScore, Faithfulness), without incorporating structured human-in-the-loop assessments or inter-rater agreement. This absence of qualitative expert judgment may obscure important nuances in factual accuracy and biomedical relevance—, especially in domains where minor terminological shifts (e.g., “reduced” vs. “unchanged”) can meaningfully alter interpretation. Second, all generative outputs were produced using variants of GPT-4 (GPT-4o or GPT-4.1-mini), thereby limiting insights into decoder variability. Given that different LLM architectures exhibit distinct generation behaviors, a broader decoder comparison—, including open-source biomedical models—, would be essential for generalizing the framework’s applicability across institutional settings. Third, the study did not develop a formal taxonomy of observed error types. While macro-level metrics suggested performance differences between domain-specific and general-purpose pipelines, it remains unclear whether these stemmed from hallucinations, factual omissions, semantic contradiction, or ambiguous phrasing. A structured error analysis would help isolate these causes and guide future improvements. Fourth, while the QA dataset was intentionally designed for clinical specificity and scientific rigor, it does not capture the breadth or ambiguity of real-world information-seeking behavior by ~~clinician~~clinicians, biologists, or translational researchers. Incorporating user-centric evaluations, log analyses, or exploratory query types could improve the system’s ecological validity. Lastly, although the framework holds promise for biomedical support ~~task~~tasks, its ethical, regulatory, and operational readiness for clinical deployment remains unassessed. Issues such as GDPR ~~compliance~~compliance, CE/FDA documentation traceability, and robust audit trails for generated content were beyond the current scope but are indispensable for translation into regulated medical environments.

5.5. Future work

Although the present study demonstrates that ~~a domain-specific RAG pipeline built around MedCPT embeddings improves the factual grounding of metabolomics question-answering~~retrieval augmentation improves grounding and that MedCPT yields a modest gain in answer correctness on the Silver benchmark, several technical extensions are

necessary before such a system can serve as a comprehensive biomedical assistant.

A first priority is corpus expansion. ~~Our current evaluation relies on ten articles that focus on Gaucher disease, NAFLD, and type-1 diabetes. Broadening the corpus to encompass a wider spectrum of metabolic phenotypes and biomarker classes would not only~~While the Silver benchmark already spans 100 papers, the Gold set remains small and the corpus is limited to metabolic disorders. Broadening coverage to additional disease classes and metabolomic phenotypes would increase statistical power ~~but also provide the diversity required for and support~~ deeper domain adaptation. Continuous ingestion of new publications from PubMed, pre-print servers, and publicly available metabolomic repositories, together with integration of additional clinical ~~cohort~~cohorts, will allow iterative fine-tuning of both the retriever and the decoder on an ever-growing biomedical ~~corpora~~corpus. Even real-time, dynamic retrieval can be ~~envisioned~~envisioned. Biomedical knowledge grows too quickly for periodic, batch-style index rebuilding to suffice. Implementing incremental or streaming indexing, possibly via infrastructures such as Apache Kafka or Elasticsearch, would permit new studies to enter the search space on the fly. Coupling this capability with iterative re-ranking strategies and user-driven feedback loops would enable the system to retire outdated passages and elevate ~~novel~~novel, high-quality evidence, ensuring that clinicians and researchers receive the most current insights available.

The second avenue concerns the inherently multimodal character of metabolomics. Beyond text, researchers routinely interpret numeric assay read-outs, LC-/GC-MS spectra, molecular-interaction graphs, and, increasingly, imaging data. Future iterations of MedDiscover should therefore move toward multimodal embedding pipelines in which specialised encoders e.g. vision transformers for images, graph neural networks for interaction maps, and tabular encoders for quantitative measurements) project their outputs into a shared latent space that is jointly searchable with textual embeddings. Retrieval architectures based on late fusion or cross-modal attention could then merge heterogeneous evidence streams at query time, offering a more holistic perspective on metabolic processes.

Transparency is key to gain researchers’ ~~trust~~trust. Users should be able to inspect why a particular passage was retrieved and how it influenced the generated answer. Future work will therefore explore techniques for displaying source information with similarity scores alongside the retrieved text. Visualising token-level saliency could provide experts a deeper understanding of the logic of both the retriever and the generator.

While the present work centres on ~~metabolomic~~metabolomics, the underlying methodology is readily transferable to adjacent biomedical domains such as drug discovery, adverse-event surveillance, or clinical decision support. These settings will require additional capabilities, including advanced query parsing, multi-hop retrieval to chain together several pieces of evidence, and domain-specific output formats, such as recommended dosage ranges or structured adverse-event reports. Extending the framework to non-English literature through cross-lingual embedding alignment

will further enlarge its practical reach, particularly in regions where locally generated biomedical research is substantial yet under-indexed in English-centred databases.

In addition to these planned extensions, several further research avenues emerge. First, future iterations of MedDiscover should integrate uncertainty quantification techniques—, such as token-level confidence scores, prediction intervals, or Bayesian calibration layers—, to better support user decision-making in high-stakes biomedical settings. These additions could enable users to distinguish between confidently grounded outputs and potentially unreliable inferences, fostering appropriate trust calibration.

Second, while this study focused on system-level performance, a structured taxonomy of model errors remains to be developed. Categorizing answer-level failures into hallucinations, omissions, factual inaccuracies, or semantic ambiguities would yield actionable insights into the interaction between retrieval and generation components and help prioritize areas for fine-tuning.

Third, user-centered evaluation is currently lacking. Future work should conduct structured interaction studies involving domain experts—, such as clinical researchers and metabolomics specialists—, applying human-centered design methodologies (e.g., cognitive walkthroughs, SUS scores, or semi-structured interviews) to assess usability, transparency, and alignment with real-world workflows.

Fourth, incorporating structured biomedical resources such as MeSH, ChEBI, or SNOMED CT into the retrieval pipeline may enhance semantic disambiguation and enable multi-hop reasoning across ontological hierarchies. These resources could be embedded alongside free text in hybrid indices to improve retrieval granularity for entity-centric questions.

Finally, to promote open science and benchmarking, future work will involve the release of a public QA benchmark for metabolomics, including annotated queries, evidence passages, and relevance judgments. This will support community-driven evaluation of biomedical RAG systems and encourage shared task initiatives within the BioNLP and biomedical IR communities.

Pursuing these directions of corpus expansion, multimodal retrieval, enhanced interpretability, and cross-domain and cross-lingual generalisation—, alongside user-centered evaluation, structured error analysis, uncertainty quantification, and ontology-integrated retrieval—, will transform MedDiscover from a proof-of-concept into a versatile, up-to-date, and trustworthy bridge between the ever-growing biomedical literature and the actionable insights required for precision medicine.

6. Conclusion

This study presents MedDiscover ~~, a domain-specific RAG framework optimized for metabolomics research. Through systematic comparison with general-purpose and non-retrieval baselines, we demonstrate that domain-adapted embeddings significantly improve retrieval precision, factual grounding,~~ as a reproducible RAG evaluation benchmark

for metabolomics and answer correctness in biomedical question answering. The findings underscore the critical role of specialized representations and transparent evidence integration in high-stakes domains like precision medicine. While MedDiscover is currently tailored to metabolomics, its architecture offers a scalable metabolic-disorder literature (see Appendix A). Using a two-tier design (Gold expert-curated QA and Silver retrievability-filtered synthetic QA), we show that retrieval augmentation improves grounding relative to non-RAG baselines, and that MedCPT vs Ada yields comparable retrieval metrics with a modest but significant gain in answer correctness on the Silver benchmark. The open-source pipeline and document lists support reproducibility (see Appendix A) and provide a foundation for expansion into ~~multimodal, multilingual, and clinically regulated contexts. As biomedical knowledge continues to grow at unprecedented speed, domain-specific RAG systems like MedDiscover represent a promising pathway to bridge the gap between unstructured literature and actionable insight. Statistical testing showed that MedCPT with GPT-4.1-mini achieved markedly better BLEU, ROUGE, and Answer Correctness than both the non-RAG GPT-4o baseline and the Ada-based pipeline (all $p < 0.01$)~~ broader biomedical domains and modalities.

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Data and Code Availability

The MedDiscover framework is available as ~~independent, open-source software at <https://github.com/VatsalPatel18/MedDiscover>. All datasets, analysis scripts, and instructions necessary to reproduce the results and figures in this manuscript—including data preprocessing, embedding generation, retrieval pipelines, evaluation metrics, and the curated metabolomics QA dataset—are available at~~ The analysis pipeline, evaluation outputs, and figure-generation scripts used in this manuscript are available at <https://github.com/VatsalPatel18/MedDiscover-Analysis>. A live application demo can be tested at <https://huggingface.co/spaces/VatsalPatel18/MedDiscover-space>; the original GPT-based application requires OpenAI private API keys, which can be recreated from the GitHub code.

Due to copyright and licensing constraints, we do not redistribute publisher PDFs or extracted full-text derived from closed-access articles. Instead, we release the list of PubMed IDs/DOIs used to assemble the corpus and provide scripts to reproduce preprocessing and indexing from documents that the user can legally access (e.g., open-access sources) (see Appendix A). Released derived artifacts (Gold QA pairs, Silver QA pairs, retrieval metadata, and evaluation tables) are provided without redistributing publisher PDF files.

References

- [1] T. B. Brown, B. Mann, N. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, S. Agarwal, A. Herbert-Voss, G. Krueger, T. Henighan, R. Child, A. Ramesh, D. M. Ziegler, J. Wu, C. Winter, C. Hesse, M. Chen, E. Sigler, M. Litwin, S. Gray, B. Chess, J. Clark, C. Berner, S. McCandlish, A. Radford, I. Sutskever, D. Amodei, Language models are few-shot learners, *Advances in Neural Information Processing Systems* 33 (2020). URL <https://arxiv.org/abs/2005.14165>
- [2] S. Lista, R. González-Domínguez, S. López-Ortiz, et al., Integrative metabolomics science in alzheimer's disease: Relevance and future perspectives, *Ageing Research Reviews* 89 (2023) 101987. doi:10.1016/j.arr.2023.101987. URL <https://www.sciencedirect.com/science/article/pii/S1568163723001460>
- [3] L. Brennan, Metabolomics: a tool to aid dietary assessment in nutrition, *Current Opinion in Food Science* 16 (2017) 96–99. doi:10.1016/j.cofs.2017.09.003. URL <https://www.sciencedirect.com/science/article/pii/S2214799317300966>
- [4] L. Huang, W. Yu, W. Ma, W. Zhong, Z. Feng, H. Wang, Q. Chen, W. Peng, X. Feng, B. Qin, T. Liu, A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions (2023). arXiv:2311.05232. URL <https://arxiv.org/abs/2311.05232>
- [5] C. Zhang, A. Le, Diabetes and cancer: The epidemiological and metabolic associations, *Advances in Experimental Medicine and Biology* 1311 (2021) 217–227. doi:10.1007/978-3-030-65768-0_16.
- [6] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, S. Riedel, Retrieval-augmented generation for knowledge-intensive nlp tasks, arXiv preprint (2020). arXiv:arXiv:2005.11401.
- [7] V. Karpukhin, B. Oğuz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, W. tau Yih, Dense passage retrieval for open-domain question answering (2020). arXiv:2004.04906. URL <https://arxiv.org/abs/2004.04906>
- [8] D. S. Wishart, Emerging applications of metabolomics in drug discovery and precision medicine, *Nature Reviews Drug Discovery* 15 (2016) 473–484. doi:10.1038/nrd.2016.32. URL <https://doi.org/10.1038/nrd.2016.32>
- [9] E. G. Armitage, C. Barbas, Metabolomics in cancer biomarker discovery: current trends and future perspectives, *Journal of Pharmaceutical and Biomedical Analysis* 87 (2014) 1–11. doi:10.1016/j.jpba.2013.08.041.
- [10] C. H. Johnson, J. Ivanisevic, G. Siuzdak, Metabolomics: beyond biomarkers and towards mechanisms, *Nature Reviews Molecular Cell Biology* 17 (7) (2016) 451–459. doi:10.1038/nrm.2016.25. URL <https://doi.org/10.1038/nrm.2016.25>
- [11] A. Rosato, L. Tenori, M. Cascante, et al., From correlation to causation: analysis of metabolomics data using systems biology approaches, *Metabolomics* 14 (2018) 37. doi:10.1007/s11306-018-1335-y. URL <https://doi.org/10.1007/s11306-018-1335-y>
- [12] J. Labory, E. Njomgue-Fotso, S. Bottini, Benchmarking feature selection and feature extraction methods to improve the performances of machine-learning algorithms for patient classification using metabolomics biomedical data, *Computational and Structural Biotechnology Journal* 23 (2024) 1274–1287. doi:10.1016/j.csbj.2024.03.016.
- [13] A. Qoku, N. Katsaouni, N. Flinner, F. Buettner, M. H. Schulz, Multi-modal analysis methods in predictive biomedicine, *Computational and Structural Biotechnology Journal* 21 (2023) 5829–5838. doi:10.1016/j.csbj.2023.11.011.
- [14] T. Theodosiou, K. Vrettos, I. Baltavia, F. Baltoumas, N. Papanikolaou, A. N. Antonakis, D. Mossialos, C. A. Ouzounis, V. J. Promponas, M. Karaglani, E. Chatzaki, S. Brandau, G. A. Pavlopoulos, E. Andreaskos, I. Iliopoulos, Biotextquest v2.0: An evolved tool for biomedical literature mining and concept discovery, *Computational and Structural Biotechnology Journal* 23 (2024) 3247–3253. doi:10.1016/j.csbj.2024.08.016.
- [15] S. Murali, S. S., S. R. ReMAG-KR: Retrieval and Medically Assisted Generation with Knowledge Reduction for Medical Question Answering, in: X. Fu, E. Fleisig (Eds.), *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 4: Student Research Workshop)*, Association for Computational Linguistics, Bangkok, Thailand, 2024, pp. 140–145. doi:10.18653/v1/2024.acl-srw.13. URL <https://aclanthology.org/2024.acl-srw.13>
- [16] Y. Xiong, X. Yang, L. Liu, K.-C. Wong, Q. Chen, Y. Xiang, B. Tang, EARA: Improving Biomedical Semantic Textual Similarity with Entity-Aligned Attention and Retrieval Augmentation, in: H. Bouamor, J. Pino, K. Bali (Eds.), *Findings of the Association for Computational Linguistics: EMNLP 2023*, Association for Computational Linguistics, Singapore, 2023, pp. 8760–8771. doi:10.18653/v1/2023.findings-emnlp.586. URL <https://aclanthology.org/2023.findings-emnlp.586>
- [17] B. Bašaragin, A. Ljajić, D. Medvečki, L. Cassano, M. Košprdić, N. Milošević, How do you know that? Teaching Generative Language Models to Reference Answers to Biomedical Questions, arXiv:2407.05015 (Jul. 2024). doi:10.48550/arXiv.2407.05015. URL <http://arxiv.org/abs/2407.05015>
- [18] C. Wang, Q. Long, M. Xiao, X. Cai, C. Wu, Z. Meng, X. Wang, Y. Zhou, BioRAG: A RAG-LLM Framework for Biological Question Reasoning, arXiv:2408.01107 (Aug. 2024). doi:10.48550/arXiv.2408.01107. URL <http://arxiv.org/abs/2408.01107>
- [19] Q. Jin, W. Kim, Q. Chen, D. C. Comeau, L. Yeganova, W. J. Wilbur, Z. Lu, Medcpt: Contrastive pre-trained transformers with large-scale pubmed search logs for zero-shot biomedical information retrieval, *Bioinformatics* 39 (11) (2023) btad651. doi:10.1093/bioinformatics/btad651. URL <http://dx.doi.org/10.1093/bioinformatics/btad651>
- [20] D. Soong, S. Sridhar, H. Si, J.-S. Wagner, A. C. C. Sá, C. Y. Yu, K. Karagoz, M. Guan, S. Kumar, H. Hamadeh, B. W. Higgs, Improving accuracy of GPT-3/4 results on biomedical data using a retrieval-augmented language model, *PLOS Digital Health* 3 (8) (2024) e0000568, publisher: Public Library of Science. doi:10.1371/journal.pdig.0000568. URL <https://journals.plos.org/digitalhealth/article?id=10.1371/journal.pdig.0000568>
- [21] J. Lee, W. Yoon, S. Kim, D. Kim, S. Kim, C. H. So, J. Kang, Biobert: a pre-trained biomedical language representation model for biomedical text mining, *Bioinformatics* 36 (4) (2019) 1234–1240. arXiv:https://academic.oup.com/bioinformatics/article-pdf/36/4/1234/48983216/bioinformatics\36_4_1234.pdf, doi:10.1093/bioinformatics/btz682. URL <https://doi.org/10.1093/bioinformatics/btz682>
- [22] A. R. Pelletier, J. Ramirez, I. Adam, S. Sankar, Y. Yan, D. Wang, D. Steinicke, W. Wang, P. Ping, Explainable Biomedical Hypothesis Generation via Retrieval Augmented Generation enabled Large Language Models, arXiv:2407.12888 (Jul. 2024). doi:10.48550/arXiv.2407.12888. URL <http://arxiv.org/abs/2407.12888>
- [23] Y. Shi, S. Xu, T. Yang, Z. Liu, T. Liu, Q. Li, X. Li, N. Liu, MKRAG: Medical Knowledge Retrieval Augmented Generation for Medical Question Answering, arXiv:2309.16035 (Aug. 2024). doi:10.48550/arXiv.2309.16035. URL <https://arxiv.org/abs/2309.16035>
- [24] J. Yang, L. Shu, H. Duan, H. Li, RDguru: A Conversational Intelligent Agent for Rare Diseases, *IEEE Journal of Biomedical and Health Informatics* (2024) 1–13Conference Name: IEEE Journal of Biomedical and Health Informatics. doi:10.1109/JBHI.2024.3464555. URL <https://ieeexplore.ieee.org/abstract/document/10684379>
- [25] Z. Yuan, Q. Jin, C. Tan, Z. Zhao, H. Yuan, F. Huang, S. Huang, RAMM: Retrieval-augmented Biomedical Visual Question Answering with Multi-modal Pre-training, in: *Proceedings of the 31st ACM International Conference on Multimedia, MM '23*, Association for Computing Machinery, New York, NY, USA, 2023, pp. 547–556. doi:10.1145/3581783.3611830. URL <https://doi.org/10.1145/3581783.3611830>
- [26] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, I. Polosukhin, Attention is all you need (2023). arXiv:1706.03762. URL <https://arxiv.org/abs/1706.03762>
- [27] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, I. Polosukhin, Attention is all you need (2021). arXiv:1706.03762. URL <https://arxiv.org/abs/1706.03762>

- [28] C. Wang, Q. Long, M. Xiao, X. Cai, C. Wu, Z. Meng, X. Wang, Y. Zhou, Biorag: A rag-llm framework for biological question reasoning (2024). arXiv:2408.01107.
URL <https://arxiv.org/abs/2408.01107>
- [29] G. Huang, Y. Li, S. Jameel, Y. Long, G. Papanastasiou, From explainable to interpretable deep learning for natural language processing in healthcare: How far from reality?, Computational and Structural Biotechnology Journal 24 (2024) 362–373. doi:10.1016/j.csbj.2024.05.004.
- [30] X. Lin, S. Ma, J. Shan, X. Zhang, S. X. Hu, T. Guo, S. Z. Li, K. Yu, BioKGBench: A Knowledge Graph Checking Benchmark of AI Agent for Biomedical Science, arXiv:2407.00466 (Jun. 2024). doi:10.48550/arXiv.2407.00466.
URL <http://arxiv.org/abs/2407.00466>
- [31] P.-H. Li, Y.-Y. Sun, H.-F. Juan, C.-Y. Chen, H.-K. Tsai, J.-H. Huang, LORE: A Literature Semantics Framework for Evidenced Disease-Gene Pathogenicity Prediction at Scale, pages: 2024.08.10.24311801 (Aug. 2024). doi:10.1101/2024.08.10.24311801.
URL <https://www.medrxiv.org/content/10.1101/2024.08.10.24311801v1>
- [32] H. Schäfer, A. Idrissi-Yaghir, K. Arzideh, H. Damm, T. M. G. Pakull, C. S. Schmidt, M. Bahn, G. Lodde, E. Livingstone, D. Schadendorf, F. Nensa, P. A. Horn, C. M. Friedrich, Biokgrapher: Initial evaluation of automated knowledge graph construction from biomedical literature, Computational and Structural Biotechnology Journal 24 (2024) 639–660. doi:10.1016/j.csbj.2024.10.017.
- [33] M. Sud, E. Fahy, D. Cotter, K. Azam, I. Vadivelu, C. Burant, A. Edison, O. Fiehn, R. Higashi, K. S. Nair, S. Sumner, S. Subramaniam, Metabolomics workbench: An international repository for metabolomics data and metadata, metabolite standards, protocols, tutorials and training, and analysis tools, Nucleic Acids Research 44 (D1) (2016) D463–D470. doi:10.1093/nar/gkv1042.
URL <https://doi.org/10.1093/nar/gkv1042>
- [34] P. Italiani, G. Moro, L. Ragazzi, Enhancing legal question answering with data generation and knowledge distillation from large language models, Artificial Intelligence and Law (2025). doi:10.1007/s10506-025-09463-9.
URL <https://link.springer.com/article/10.1007/s10506-025-09463-9>
- [35] D. Gupta, D. Bartels, D. Demner-Fushman, A dataset of medical questions paired with automatically generated answers and evidence-supported references, Scientific Data 12 (1) (Jun. 2025). doi:10.1038/s41597-025-05233-z.
URL <http://dx.doi.org/10.1038/s41597-025-05233-z>
- [36] Ragas: Automated evaluation of retrieval-augmented generation, arXiv:2309.15217 (2023).
URL <https://arxiv.org/abs/2309.15217>
- [37] J. Lee, W. Yoon, S. Kim, D. Kim, S. Kim, C. H. So, J. Kang, Biobert: a pre-trained biomedical language representation model for biomedical text mining, Bioinformatics 36 (4) (2019) 1234–1240. doi:10.1093/bioinformatics/btz682.
- [38] T. Almeida, R. A. A. Jonker, J. Reis, J. R. Almeida, S. Matos, Notebook for the BioASQ Lab at CLEF 2024, Conference paper (2023).
- [39] I. Beltagy, K. Lo, A. Cohan, Scibert: A pretrained language model for scientific text (2019). arXiv:1903.10676.
URL <https://arxiv.org/abs/1903.10676>

Appendix A. Reproducibility checklist

Appendix B. Supplementary Gold-set results

Appendix C. Detailed Statistical Significance Tests

This appendix expands on the [Gold set](#) statistical analysis of Section 4. The goal was to determine whether any of the seven model pipelines differed reliably in their median metric scores. The following steps were applied.

- Normality checks.** Every model–metric distribution was examined with the Shapiro–Wilk test. All yielded $p < 0.05$, implying non-normal data.
- Variance homogeneity.** Levene’s test across models revealed unequal variances for seven of the ten metrics, especially the retrieval-oriented ones (Faithfulness, Context Recall, Context Precision) and BLEU.
- Omnibus comparison.** Because parametric assumptions were violated, a Kruskal–Wallis test on ranks was carried out for each metric across all model variants: the non-RAG baseline, Ada + GPT-4o, three MedCPT configurations (GPT-4, GPT-4.1-nano, GPT-4.1-mini), and two NotebookLM variants.
- Pairwise contrasts.** Significant omnibus results were followed up with Dunn’s test, applying a Bonferroni correction. The general-purpose Ada + GPT-4o pipeline served as the reference.

Appendix C.1. Raw p -values

The adjusted Dunn p -values and corresponding Kruskal–Wallis p for each metric are listed in Tables C.6 and C.7. The numbers were imported directly from [results/code/fresults/stats_ada_vs_others_basic.csv](#) and [results/code/fresults/stats_ada_vs_others_contextual/MedDiscover-Analysis/stats_ada_vs_others_contextual.csv](#) and [../MedDiscover-Analysis/stats_ada_vs_others_contextual.csv](#).

Overall, the non-parametric analysis ~~corroborates the main findings: retrieval-augmented systems consistently outperform on the Gold set indicates that retrieval augmentation improves grounding relative to the non-RAG baseline, and pairing MedCPT embeddings with the while decoder upgrades (GPT-4.1 -mini decoder yields the strongest gains, variants) improve lexical and answer-quality metrics beyond fixed-decoder comparisons.~~

Appendix D. Prompt templates

Appendix D.1. Evidence-grounded answering prompt (MedDiscover RAG; GPT-4o)

```
[SYSTEM]
You are a biomedical assistant. Answer the question using the context.
If the context does not contain the answer, reply: "Insufficient information."
[DIF] >
[USER]
Question: {question}
```


Table A.5: Reproducibility checklist (artifacts and where to find them).

Artifact	Location	Notes
Core RAG pipeline	https://github.com/VatsalPatel118/MedDiscover	Indexing, retrieval, prompting, and RAG inference implementation.
Analysis + figures	https://github.com/VatsalPatel118/MedDiscover-Analysis	Scripts to reproduce tables/figures and evaluation outputs.
Paper list (IDs)	MedDiscover-Analysis release package (evaluation_set.zip)	PubMed IDs / DOIs used to assemble the 100-paper Silver corpus.
Gold QA set	MedDiscover release package (qa_samples_gold.tsv)	30 expert-curated QA pairs + references; evaluation subset noted in Section 4.
Silver QA outputs	MedDiscover-Analysis evaluation outputs (qa_72_plus_29_ada_with_ragas.csv)	Synthetic QA + retrievability metadata; post-QC sampled to 300 per embedding.
Silver QA generator	MedDiscover QA-generation pipeline	GPT-4.1-mini; 500-token chunks, 50-token overlap.
QC post-processing	MedDiscover QA post-processing pipeline	De-duplication, min-length filters, and template filtering (Table 3).
Stats inputs (Gold)	MedDiscover-Analysis statistics tables	Dunn/KW tables used for Supplementary statistics appendix.
Environment	MedDiscover-Analysis requirements	Python requirements for reproducible analysis.

Appendix D.3. NotebookLM minimal-output instruction

Answer in minimal words.

```
%DIF >
Context:
{top_k_context_chunks}
%DIF >
Instructions:
- Provide a concise answer (max 25 tokens).
- Prefer specific entities (metabolites, diseases, study descriptors).
- Do not add unsupported claims.
- If multiple findings exist, report the one directly supported by the context.
Appendix D.2. Synthetic QA generation prompt (Silver set;
gpt-4.1-mini)

[SYSTEM]
You generate biomedical questions and short answers from scientific text excerpts.
%DIF >
[USER]
Given the excerpt below, generate exactly 3 diverse questions that are answerable
from the excerpt. Prefer non-trivial questions when possible.
%DIF >
Excerpt:
{chunk_or_grouped_context}
%DIF >
Output format (one per line):
Q: ...
A: ...
```

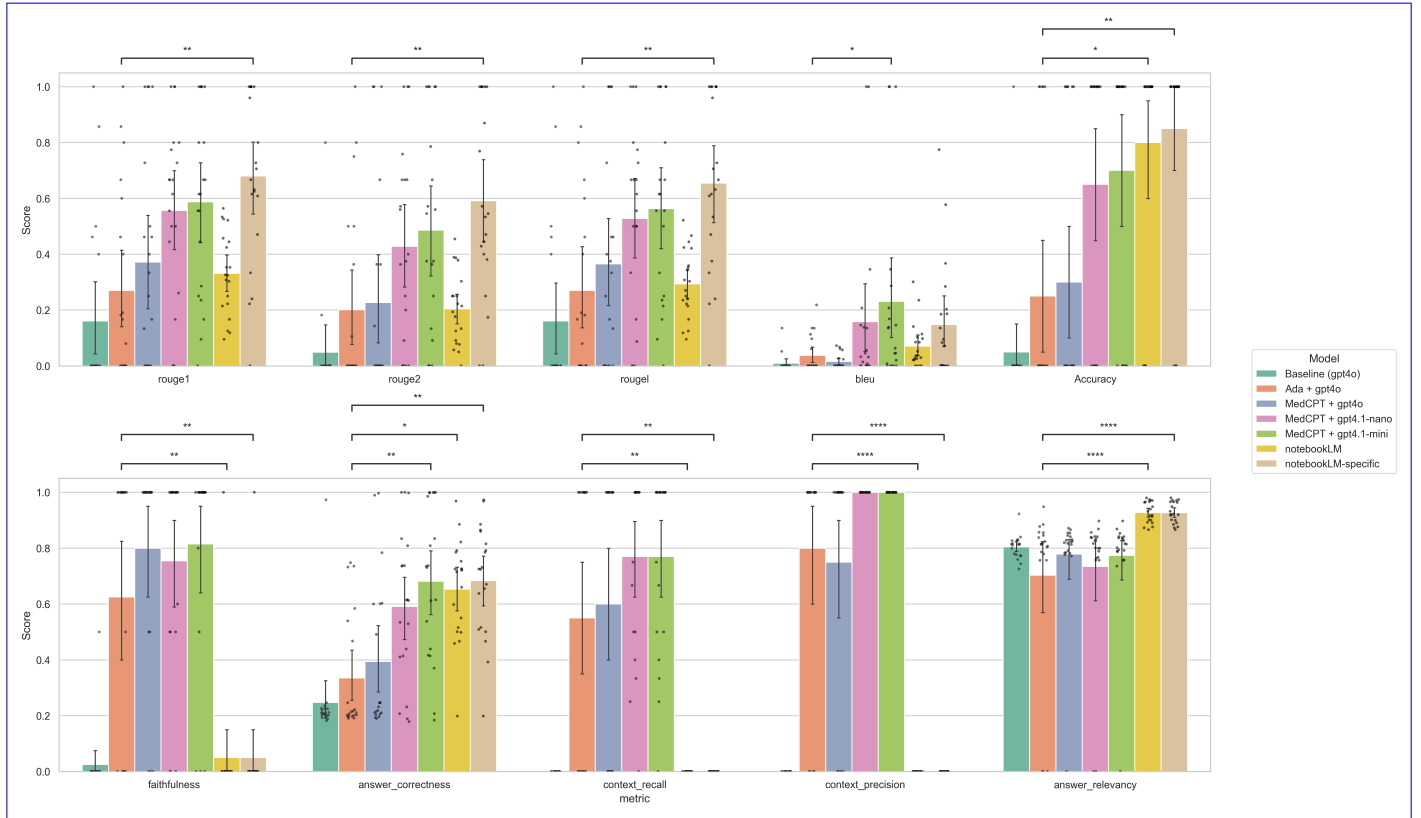


Figure B.4: Supplementary Gold-set results across additional configurations (decoder sensitivity variants and black-box comparator). This figure is provided for completeness; the primary comparisons in the main text restrict to fixed-decoder (GPT-4o) evaluations to avoid conflating retriever effects with decoder capacity. Note that NotebookLM does not expose retrieved contexts; context-dependent metrics are therefore not comparable and should be interpreted as not available rather than as zero performance.

Table C.6: Dunn’s test results (basic metrics).

Metric	Compared model	p_{adj}	p_{KW}
rouge1	MedCPT + gpt4o	1	2.003×10^{-6}
rouge1	MedCPT + gpt4.1-nano	0.172794	2.003×10^{-6}
rouge1	MedCPT + gpt4.1-mini	0.0729964	2.003×10^{-6}
rouge1	notebookLM	1	2.003×10^{-6}
rouge1	notebookLM-specific	0.00442385	2.003×10^{-6}
rouge2	MedCPT + gpt4o	1	1.521×10^{-7}
rouge2	MedCPT + gpt4.1-nano	0.314738	1.521×10^{-7}
rouge2	MedCPT + gpt4.1-mini	0.0923275	1.521×10^{-7}
rouge2	notebookLM	1	1.521×10^{-7}
rouge2	notebookLM-specific	0.00301024	1.521×10^{-7}
rougel	MedCPT + gpt4o	1	5.122×10^{-6}
rougel	MedCPT + gpt4.1-nano	0.261727	5.122×10^{-6}
rougel	MedCPT + gpt4.1-mini	0.104081	5.122×10^{-6}
rougel	notebookLM	1	5.122×10^{-6}
rougel	notebookLM-specific	0.00784415	5.122×10^{-6}
bleu	MedCPT + gpt4o	1	5.924×10^{-6}
bleu	MedCPT + gpt4.1-nano	0.683111	5.924×10^{-6}
bleu	MedCPT + gpt4.1-mini	0.034923	5.924×10^{-6}
bleu	notebookLM	0.44618	5.924×10^{-6}
bleu	notebookLM-specific	0.648565	5.924×10^{-6}
Accuracy	MedCPT + gpt4o	1	2.971×10^{-8}
Accuracy	MedCPT + gpt4.1-nano	0.245187	2.971×10^{-8}
Accuracy	MedCPT + gpt4.1-mini	0.0956256	2.971×10^{-8}
Accuracy	notebookLM	0.0110328	2.971×10^{-8}
Accuracy	notebookLM-specific	0.00325786	2.971×10^{-8}

Table C.7: Dunn’s test results (context metrics).

Metric	Compared model	p_{adj}	p_{KW}
faithfulness	MedCPT + gpt4o	1	2.240×10^{-14}
faithfulness	MedCPT + gpt4.1-nano	1	2.240×10^{-14}
faithfulness	MedCPT + gpt4.1-mini	1	2.240×10^{-14}
faithfulness	notebookLM	0.00310424	2.240×10^{-14}
faithfulness	notebookLM-specific	0.00310424	2.240×10^{-14}
answer_correctness	MedCPT + gpt4o	1	2.678×10^{-8}
answer_correctness	MedCPT + gpt4.1-nano	0.0929803	2.678×10^{-8}
answer_correctness	MedCPT + gpt4.1-mini	0.00302486	2.678×10^{-8}
answer_correctness	notebookLM	0.0110999	2.678×10^{-8}
answer_correctness	notebookLM-specific	0.00262388	2.678×10^{-8}
context_recall	MedCPT + gpt4o	1	2.072×10^{-15}
context_recall	MedCPT + gpt4.1-nano	1	2.072×10^{-15}
context_recall	MedCPT + gpt4.1-mini	1	2.072×10^{-15}
context_recall	notebookLM	0.00397699	2.072×10^{-15}
context_recall	notebookLM-specific	0.00397699	2.072×10^{-15}
context_precision	MedCPT + gpt4o	1	5.022×10^{-19}
context_precision	MedCPT + gpt4.1-nano	1	5.022×10^{-19}
context_precision	MedCPT + gpt4.1-mini	1	5.022×10^{-19}
context_precision	notebookLM	2.18907×10^{-8}	5.022×10^{-19}
context_precision	notebookLM-specific	2.18907×10^{-8}	5.022×10^{-19}
answer_relevancy	MedCPT + gpt4o	1	6.635×10^{-15}
answer_relevancy	MedCPT + gpt4.1-nano	1	6.635×10^{-15}
answer_relevancy	MedCPT + gpt4.1-mini	1	6.635×10^{-15}
answer_relevancy	notebookLM	6.44792×10^{-6}	6.635×10^{-15}
answer_relevancy	notebookLM-specific	6.05982×10^{-6}	6.635×10^{-15}

CRedit author statement

Vatsal Patel: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data Curation, Writing - Original Draft, Visualization. **Anne Schwert:** Writing - Review & Editing, Writing - Original Draft, Supervision, Project administration, Conceptualization. **Elena Jolkver:** Writing - Review & Editing, Writing - Original Draft.

Term	Definition
Conceptualization	Ideas; formulation or evolution of overarching research goals and aims
Methodology	Development or design of methodology; creation of models
Software	Programming, software development; designing computer programs; implementation of the computer code and supporting algorithms; testing of existing code components
Validation	Verification, whether as a part of the activity or separate, of the overall replication/ reproducibility of results/experiments and other research outputs
Formal analysis	Application of statistical, mathematical, computational, or other formal techniques to analyze or synthesize study data
Investigation	Conducting a research and investigation process, specifically performing the experiments, or data/evidence collection
Resources	Provision of study materials, reagents, materials, patients, laboratory samples, animals, instrumentation, computing resources, or other analysis tools
Data Curation	Management activities to annotate (produce metadata), scrub data and maintain research data (including software code, where it is

Term	Definition
	necessary for interpreting the data itself) for initial use and later reuse
Writing - Original Draft	Preparation, creation and/or presentation of the published work, specifically writing the initial draft (including substantive translation)
Writing - Review & Editing	Preparation, creation and/or presentation of the published work by those from the original research group, specifically critical review, commentary or revision – including pre-or postpublication stages
Visualization	Preparation, creation and/or presentation of the published work, specifically visualization/ data presentation
Supervision	Oversight and leadership responsibility for the research activity planning and execution, including mentorship external to the core team
Project administration	Management and coordination responsibility for the research activity planning and execution
Funding acquisition	Acquisition of the financial support for the project leading to this publication